Revision Highlights

Revision 31 to D631A002.8JP dated 11/20/2023

The purpose of this revision is to provide RSAT capability for aircraft serial numbers 42831, 42834, 42836 and 64992.

The following changes comprise this revision:

Chapter - Front Matter

Revision Approval (Revised)

This section uniquely identifies the approval authority and the reference number for this revision.

Log of Sections (Revised)

Revised to reflect sections changed in this revision.

Chapter - Certificate Limitations

Electronics - Ground Proximity Warning System (GPWS) - Runway Situation Awareness Tool (RSAT) (ASN Note)

Revised applicability note to add airplane serial numbers 42831, 42834, 42836 and 64992.

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Title **A163052**

737-8

FAA APPROVED

AIRPLANE FLIGHT MANUAL



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Revision Approval A232367

This Revision Approval may only be used in conjunction with a Log of Sections that refers to Revision Approval Number A232367 in the revision approval number column for the Revision Approval section.

Approved by:



Flight Analyst E-UM, ODA-300064-NM, for

11/20/2023

Manager, Flight Test & Human Factors Branch, AIR-710
Federal Aviation Administration

Approved Date

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Log of Sections A232367

D631A002.8JP Revision 31 Date: 11/20/2023

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*	Log of Sections	A232367	11/17/2023
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* = Revised Sections @ = Airplane Serial Number Note Change Only

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Airplane Serial Number & Appendix Effectivity A232474

The Airplane Flight Manual (AFM) is applicable to the following Airplane Serial Numbers (ASNs):

42825-42836, 43361, 62897, 62903, 62904, 62907, 62911, 62922, 62923, 62926, 62928, 62931, 63564, 63970-63973, 64992, 64993, 66350

The Appendices contained in this AFM are applicable to the following ASNs:

Appendix	ASNs	Notes
D631A002-CDL	All	
D631A002-LGE	All	
D631A002-EASA	All	
D631A002-OZONE	42825-42830	1
D631A002-AP3GIE	All	
D631A002-L1B27E	All	

Note(s):

1. Not applicable upon incorporation of Boeing Service Bulletin 737-21-1220.

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Appendix Applicability A222136

The following appendices are applicable to this AFM. The appendices in the first column may only be used in conjunction with the appendices in the Applicable Appendices column:

Appendix	Title	Applicable Appendices
D631A002-AP3GIE	EASA Autopilot Flight Director System: CATIIIa, GLS, IAN	All
D631A002-CDL	Configuration Deviation List	All
D631A002-EASA	EASA Appendix	All
D631A002-L1B27E	737-8 LEAP-1B27 EASA	All
D631A002-LGE	Landing Gear Extended	All
D631A002-OZONE	Ozone Exposure Appendix	All

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Preface A163052

This Airplane Flight Manual (AFM) complies with the U.S. Federal Aviation Administration (FAA) Title 14 of the Code of Federal Regulations (CFR) and is approved by the FAA.

This AFM may contain appendices or supplements that add, amend, replace and/or supersede information found in the FAA approved AFM.

The engineering developed and the FAA certification provided by the revisions to the AFM are applicable and valid only for the airplane serial numbers listed in the Airplane Serial Number & Appendix Effectivity Section, in their original Boeing delivery configuration, or as modified by the incorporation of approved Boeing Service Bulletins. With respect to any other modifications, it shall be the responsibility of the operator to obtain appropriate regulatory agency approval for application of the data provided by this AFM.

Each revision to this AFM includes a new Log of Sections listing the effective sections comprising the AFM. Revisions to text material are indicated by vertical bars in the margins.

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General A180643

The basic Airplane Flight Manual (AFM) for the Model 737 airplane is comprised of Front Matter, Certificate Limitations, Operating Procedures, Operational Data, and Performance.

Appendices will be provided for specific configurations and regulatory requirements.

Weight Limitations and Center of Gravity Limits A232320

Applicable to ASN(s) : 42825, 42826, 42827, 42828, 42829, 42830, 42831, 42832, 42833, 42834, 42835, 42836, 43361, 62897, 62903, 62904, 62907, 62911, 62922, 62923, 62926, 62928, 62931, 63564, 63970, 63971, 63972, 63973, 64992, 64993, 66350

	Pounds	Kilograms
Maximum Taxi Weight (MTW)	181,700	82,417
Maximum Takeoff Weight (MTOW)	181,200	82,190
Maximum Landing Weight (MLW)	152,800	69,308
Maximum Zero Fuel Weight (MZFW)	145,400	65,952
Minimum Flight Weight (MFW)	94,000	42,638

Notes:
The maximum weight limits may be less as limited by center of gravity, fuel density and fuel loading limits, as given in this section. Refer to the Weight and Balance Manual for additional specific airplane loading limitations.

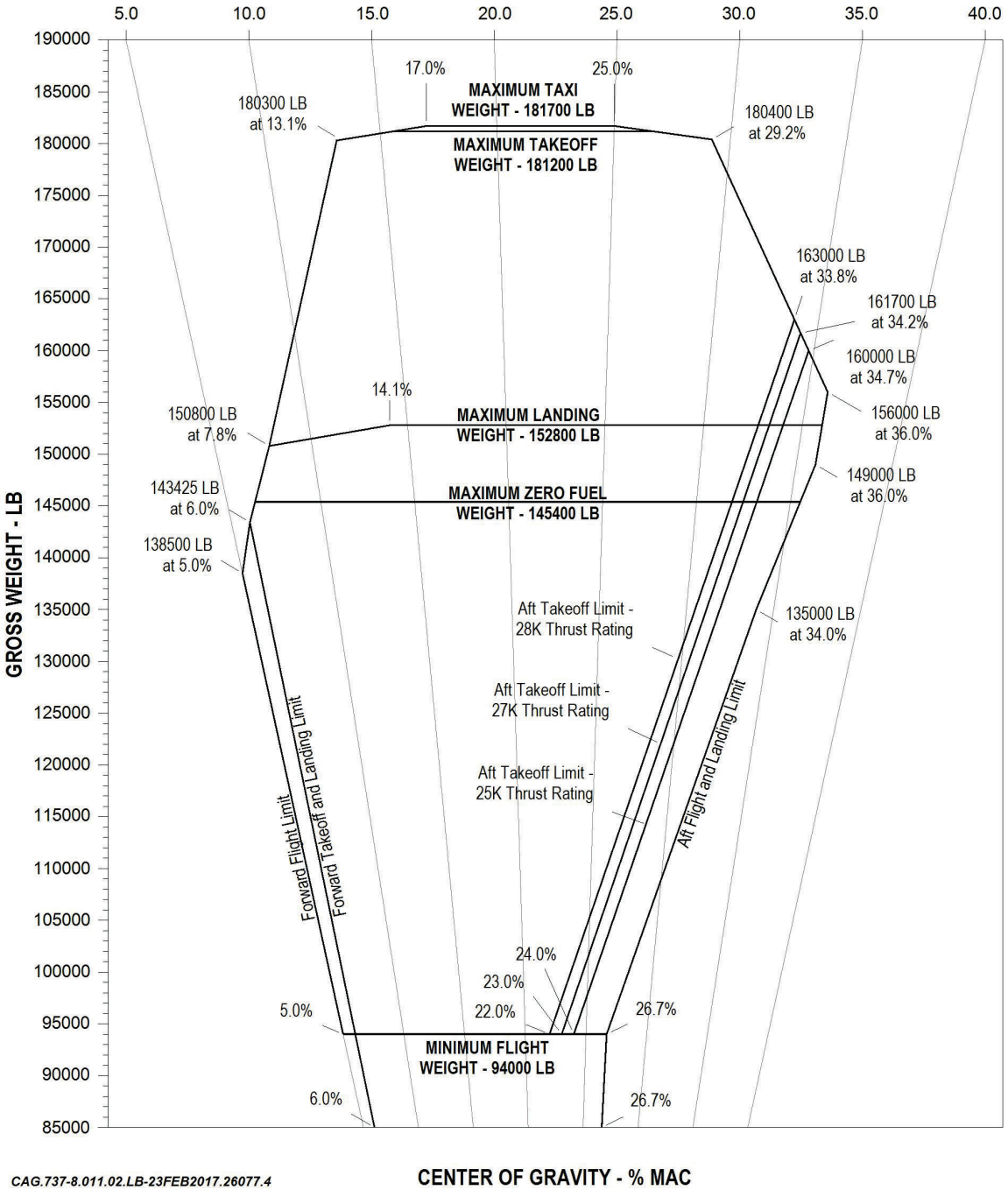
Gaugeable fuel must not be used to comply with Minimum Flight Weight.

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Refer to the applicable Weight and Balance Manual for identification of center of gravity coordinates and specific loading/operating restrictions.

Model 737-8 weights in pounds



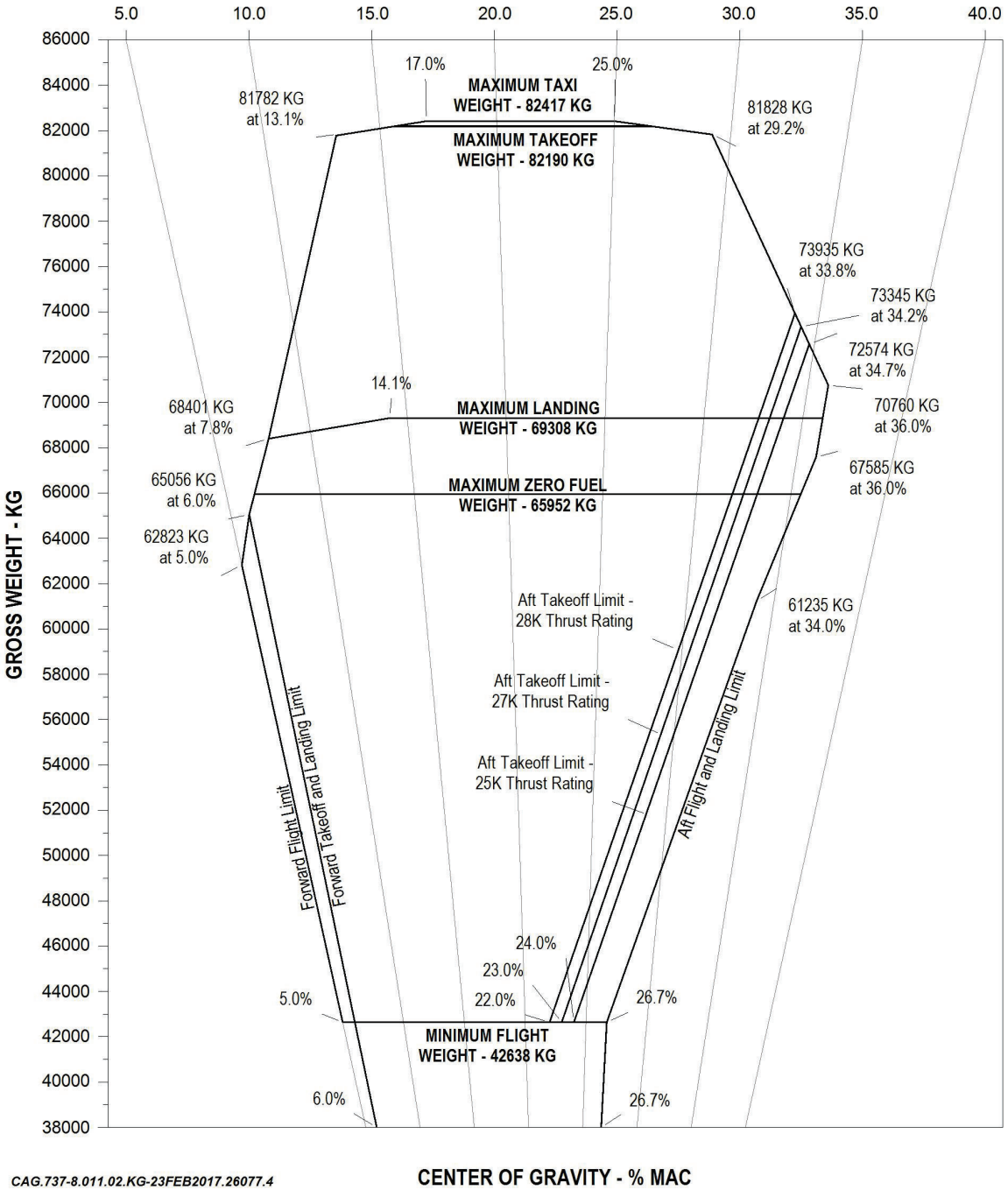
Refer to Airplane Flight Manual - Digital Performance Information (AFM-DPI) for operations with alternate forward center of gravity limits for takeoff. The alternate forward center of gravity limits for takeoff performance data in the AFM-DPI are only for use with the selected limits shown.

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Refer to the applicable Weight and Balance Manual for identification of center of gravity coordinates and specific loading/operating restrictions.

Model 737-8 weights in kilograms



Refer to Airplane Flight Manual - Digital Performance Information (AFM-DPI) for operations with alternate forward center of gravity limits for takeoff. The alternate forward center of gravity limits for takeoff performance data in the AFM-DPI are only for use with the selected limits shown.

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Section Weight Limitations and Center of Gravity Limits, continued from previous page

Autobrakes A163052

Autobrakes are not allowed with antiskid inoperative.

Auxiliary Power Unit (APU) Limitations A163052

Do not use APU, in flight, for both bleed and electrical load above 10,000 feet.

Do not use APU bleed above 17,000 feet.

APU may be used, on ground only, for both bleed and electrical loads to a maximum altitude of 15,000 feet.

APU operation has been demonstrated to 41,000 feet.

Cabin - Cabin Pressurization Limitations A170242

System Safety Relief Differential Pressure. Maximum 9.10 psi

Maximum allowable cabin pressure differential for takeoff and landing is 0.125 psi (236 feet below airport pressure altitude).

Cabin - Emergency Lighting (Photoluminescent) A163052

Photoluminescent Floor Emergency Lighting (if installed) must be charged in accordance with approved procedures.

Cabin - Passenger Evacuation A163052

Installation of handle covers on the overwing exits must be verified prior to departure whenever passengers are carried.

Cabin phones and/or handsets (if installed) must be stowed, and the cabin prepared to satisfy direct view requirements by lowering the passenger seat headrests and opening the viewing window (where applicable) prior to taxi, takeoff, and landing whenever passengers are carried.

Certification Status A163052

This airplane is certificated in the Transport Category, Title 14 of the Code of Federal Regulations (CFR) Part 25 and Part 36.

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Cold-Soaked Fuel Frost A163052

Takeoff with cold-soaked fuel frost on the wing tank upper surfaces is not permitted if any of the following conditions are present:

1. The ambient air temperature is below +4°C, +39°F.
2. The tank fuel temperature is below -16°C, +3°F.
3. There is cold-soaked fuel frost on the wing tank upper surfaces that is beyond the lines defining the permissible cold-soaked fuel frost area.
4. There is precipitation or visible moisture (rain, snow, drizzle, or fog with less than 1 mile visibility).

Deice and Anti-Ice Fluids A170119

Use of any other but the following airplane deicing and anti-icing fluids is prohibited: AMS1424/1 Type I and AMS1428/1 Type II, Type III, and Type IV Airplane Deicing and Anti-Icing Fluids. Use of these fluid types is prohibited at ambient temperatures below the Lowest Operational Use Temperature specified for the fluid.

Electronics - Aircraft Communications Addressing and Reporting System (ACARS) A163052

Applicable to ASN(s) : 42831, 42832, 42833, 42835, 62897, 62903, 62904, 62907, 62911, 62922, 62923, 62926, 62928, 62931, 63971, 63972, 63973, 64992, 66350

Protective Mode – Controller Pilot Datalink Communications (PM-CPDLC)

The PM-CPDLC installation is only intended to be used in cruise flight phase and for non-critical messaging.

The following limitation is not applicable if Rockwell Collins Communication Management Unit (CMU) part number 822-1239-151 or Honeywell CMU part number 965-0758-006, Protective Mode – Controller Pilot Datalink Communications (PM-CPDLC), is installed:

The following is applicable when using the ACARS Controller Pilot Datalink Communications (CPDLC):

The aircrew shall read back via voice every ATC clearance for altitude, heading, speed, route and rate of climb message received via CPDLC datalink. The aircrew shall obtain confirmation via voice of the accuracy of the clearance from the controlling authority before executing the clearance.

The aircrew shall independently verify any altimeter setting received via CPDLC.

Electronics - Flight Deck Auxiliary Power (if installed) A163052

The use of Flight Deck Auxiliary Power outlets in the flight deck requires operational regulatory approval.

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Electronics - Flight Management System - Barometric Altitude Temperature Compensation A201609

The following is not applicable upon installation of FMCS Software update U14 and later versions with BARO VNAV option enabled:

When the destination's reported temperature is below the minimum published temperature for the approach, instrument approach operations are prohibited unless manual temperature compensated barometric altitude corrections are applied in accordance with operationally approved procedures.

If no minimum temperature is published for the approach, manual temperature compensated barometric altitude corrections must be applied in accordance with applicable operationally approved procedures, as specified by the operator's regulatory authority or by the authority of the country in which the operation is conducted (including Air Traffic Control).

Warning: Use of published procedures without consideration of cold temperature effects on barometric altitudes does not assure obstacle clearance. The Flight Management System does not provide an automatic approach barometric altitude temperature compensation function.

Electronics - Flight Management System - Barometric Altitude Temperature Compensation A201609

The following is applicable upon installation of FMCS Software update U14 and later versions with BARO VNAV option enabled:

For hot temperature automatic altitude compensation, do not use VNAV in the event of a go-around or missed approach.
Automatic temperature compensation is not authorized using QFE.

Electronics - Flight Management Computer System (FMCS) - FMCS Calculated Takeoff Speeds A163052

When operating at an alternate forward center of gravity limit for takeoff, disregard FMCS calculated takeoff speeds, if displayed. Determine takeoff speeds from Airplane Flight Manual - Digital Performance Information (AFM-DPI).

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Electronics - Global Landing Unit (GLU) A201322

(Required by AD 2020-03-20)

Operations that require less than 0.3 RNP (For example, 0.1, 0.11, 0.15, etc.) in the region identified below are prohibited with GLU-2100 OPS software number COL4D-0087-0002, COL4E-0087-0001, COL48-0087-0700, or COL49-0087-0701 installed.

Exception: Anchorage (PANC) approach procedures that allow less than RNP 0.3 are authorized provided the instructions outlined in the Electronics – Global Landing Unit Section of Normal Procedures Chapter are followed.

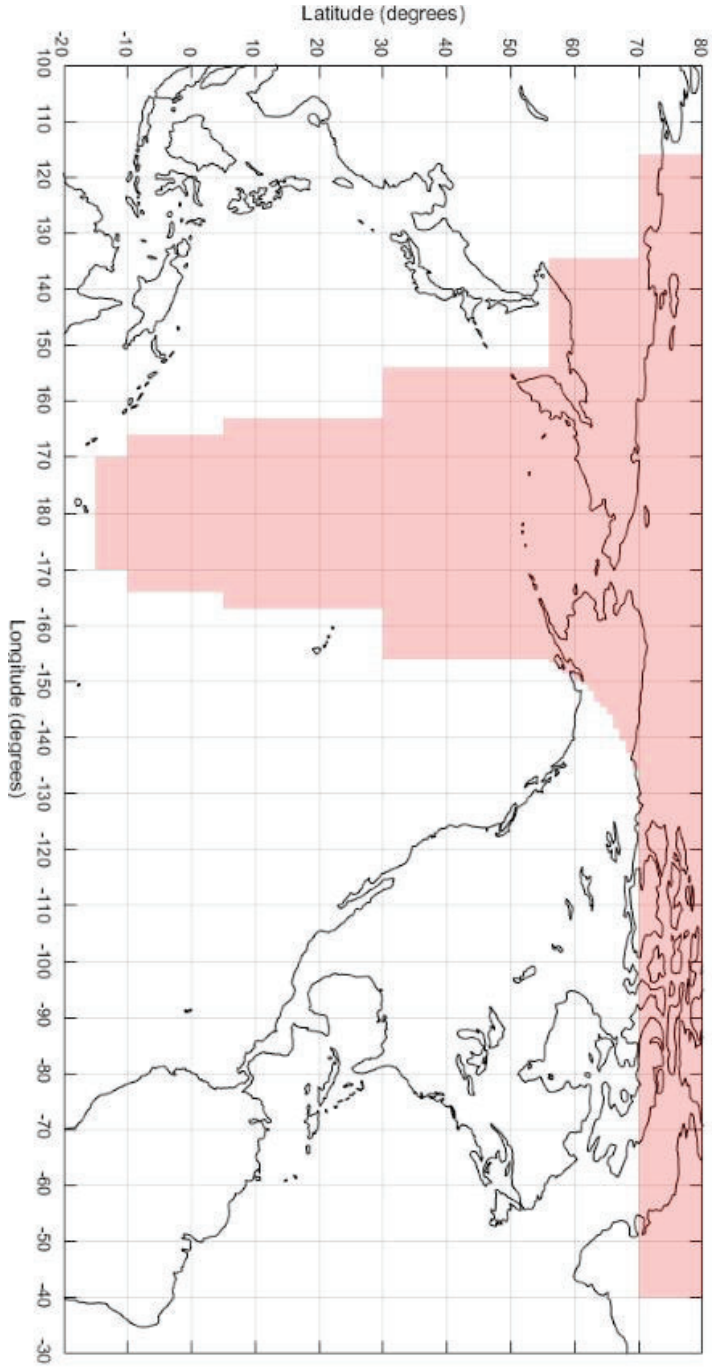
Note: Currently, Fairbanks (PAFA) and Anchorage (PANC) are the only airports in the region with an RNP approach that requires better than 0.3 nmi performance.

Region bounded by the following coordinates:

Latitude Range (degrees)	Longitude Range (degrees)
80 N to 70 N	40 E to 40 W
70 N to 69 N	134.5 E to 134.38 W
69 N to 68 N	134.5 E to 137.28 W
68 N to 67 N	134.5 E to 139.50 W
67 N to 66 N	134.5 E to 141.58 W
66 N to 65 N	134.5 E to 144.23 W
65 N to 64 N	134.5 E to 145.48 W
64 N to 63 N	134.5 E to 146.44 W
63 N to 62 N	134.5 E to 148.33 W
62 N to 61 N	134.5 E to 149.50 W
61 N to 60 N	134.5 E to 150.35 W
60 N to 59 N	134.5 E to 151.00 W
59 N to 58 N	134.5 E to 151.40 W
58 N to 57 N	134.5 E to 152.62 W
57 N to 56 N	134.5 E to 153.42 W
56 N to 30 N	154 E to 154 W
30 N to 5 N	163 E to 163 W
5 N to 10 S	166 E to 166 W
10 S to 15 S	170 E to 170 W

Continued on next page

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Electronics - Ground Proximity Warning System (GPWS) - Look Ahead Terrain Alerting A163052

Look-Ahead Terrain Alerting

Do not use the terrain display for navigation.

The use of look-ahead terrain alerting and terrain display functions are prohibited within 15 NM of takeoff, approach or landing at an airport or runway not contained in the GPWS terrain database. Refer to Honeywell Document 060-4267-000 for airports and runways contained in the installed GPWS terrain database.

Electronics - Ground Proximity Warning System (GPWS) - Runway Awareness and Advisory System (RAAS) A163052

Runway Awareness and Advisory System - If Installed

Do not use Runway Awareness and Advisory System (RAAS) alerting or voice annunciations to determine real-time runway suitability for airplane performance. RAAS runway length alerting is set by airline policy and does not consider real-time factors such as weather, runway conditions, or Notices to Airmen (NOTAMS).

Electronics - Ground Proximity Warning System (GPWS) - Runway Situation Awareness Tool (RSAT) A163052

Applicable to ASN(s) : 42831, 42834, 42835, 42836, 43361, 62904, 62907, 62911, 62923, 62926, 63971, 63972, 63973, 64992, 66350

Overrun Warning System (ORW)

Do not use the Overrun Warning alerting system for landing performance planning or as a landing distance assessment replacement.

The Overrun Warning alerting system does not replace the requirement to conduct an en route landing distance performance assessment prior to landing. The lack of an overrun alert does not guarantee the airplane can stop prior to reaching the runway end.

If the In-Air Overrun Warning (“OVERRUN, GO-AROUND”) occurs during approach, execute an immediate go-around.

Inhibit the overrun alert when:

Landing altitude is above 10,000 ft pressure altitude; or
Landing OAT on the ground is below -40C or greater than 50C; or
Gross Weight is greater than Maximum Landing Weight

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Electronics - Inertial Reference System (IRS) A201489

All flight operations based on magnetic heading or magnetic track angle are prohibited in geographic areas where the installed IRS MagVar table errors are greater than 5 degrees.*

For “Fail-Operational Autopilot” the following applies:

All autopilot/flight director ILS (and GLS if installed) approach and landing operations that use magnetic north referenced courses or bearings are prohibited in geographic areas where the installed IRS MagVar table errors are greater than 3 degrees.*

*See Operational Data/Inertial Reference System (IRS) section for procedures to determine the geographic areas and magnitude of MagVar errors for the specific MagVar table installed in the IRS and if any of these limitations apply.

Electronics - Integrated Standby Flight Display A170119

The following applies to airplanes equipped with an Integrated Standby Flight Display (ISFD) earlier than Boeing Part Number 231A120-4:

The ISFD must be initialized and alignment complete prior to moving the aircraft.

Electronics - QFE Selection A170255

The use of VNAV or LNAV with QFE selected is prohibited.

The use of the Vertical Situation Display during QFE operation is prohibited.

Electronics - Single Generator Operation A163052

On the ground, limit one generator (engine driven) operation to a maximum of 75 KVA (215 amps).

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Extended Range Operations A201489

Applicable to ASN(s) : 42825, 42826, 42827, 42828, 42829, 42830, 42831, 42832, 42833, 42834, 42835, 42836, 43361, 62897, 62903, 62904, 62907, 62911, 62922, 62923, 62926, 62928, 62931, 63564, 63970, 63971, 63972, 63973, 64992, 64993, 66350

The following is applicable only to airplanes with Extended Range Operations type design approval, per Operational Data chapter of this document.

Perform an operational check of the fuel crossfeed valve prior to each extended range operations flight. This check may be accomplished on the ground by maintenance personnel or the flight crew as part of the preflight procedure for the specific extended range operations flight.

Flap Operation A163052

Do not extend flaps above 20,000 feet pressure altitude.

Flight Controls A180103

Avoid rapid and large alternating control inputs, especially in combination with large changes in pitch, roll or yaw (e.g. large side slip angles) as they may result in structural failure at any speed, including below V_A.

Flight Deck Security Door A163052

Reinforced Cockpit Door

Operational check of the Flight Deck Access System must be accomplished according to approved procedures once each flight day.

Flight Maneuvering Load Acceleration Limits A163052

Flaps Up	+2.5g to -1.0g
Flaps Down	+2.0g to 0.0g

Note:
With flaps 30 and 40, positive limits vary linearly from +2.0g at maximum landing weight to +1.5g at maximum takeoff weight.

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Fuel System - Center Tank Fuel Pumps A163052

Intentional dry running of a center tank fuel pump (low pressure light illuminated) is prohibited.

Fuel System - Lateral Fuel Imbalance A180643

The lateral fuel imbalance between main wing tanks must be scheduled to be zero. Fuel imbalance between main wing tanks for taxi, takeoff, flight or landing must not exceed 1,000 pounds (453 kilograms).

Fuel System - Loading A180643

MODEL AIRPLANE	FUEL DENSITY
737-8/-8200/-9	6.3 to 7.1 LB/gallon (0.7549 to 0.8507 KG/liter)

Fuel tanks may be loaded individually, simultaneously, or in any sequence. Load main tanks equally to the desired fuel quantity or until full. Main wing tanks must be scheduled to be full if the center wing tank contains more than 1,000 pounds (453 kilograms) of fuel. With 1,000 pounds (453 kilograms) or less of center wing tank fuel, partial main wing tank fuel may be loaded provided the effects of balance have been considered. Fuel must always be used in accordance with **Fuel System - Usage** in this Chapter.

Fuel System - Usable Tank Quantities A180207

TANK	VOLUME (U.S. Gallons)	MAXIMUM WEIGHT (Pounds)	MAXIMUM WEIGHT (Kilograms)
No. 1 or No. 2	1,273	9,038	4,099
Center	4,274	30,345	13,764

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Fuel System - Usage A163052

With no center tank fuel, use main tank-to-engine fuel feed with all operable main tank boost pumps on, center tank boost pumps off and the crossfeed valve closed [1].

With center tank fuel, use center tank fuel for all operations with all operable boost pumps on and the crossfeed valve closed [1] until center tank fuel is depleted (however, a maximum of 1,000 pounds (453 kilograms) may be retained in the center wing tank provided the effects of balance have been considered). Then continue flight using main tank-to-engine fuel feed with all operable main tank boost pumps on and the crossfeed valve closed [1].

[1] The crossfeed valve is open for minimum fuel operation, and may be opened to correct fuel imbalance.

Holding in Icing Conditions A163052

Holding in icing conditions with flaps extended is prohibited.

Kind of Airplane Operation A163052

The airplane is approved for the following kinds of flight and operation, both day and night, when the required equipment is installed and approved in accordance with the applicable 14 CFR:

Visual (VFR)
Instrument (IFR)
Icing Conditions
Extended Over-Water*

*Operation over water at a horizontal distance of more than 50 nautical miles from the nearest shoreline.

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Landing on Wet Skid-Resistant Runways A180643

Applicable to ASN(s) : 42825, 42826, 42827, 42828, 42829, 42830, 42831, 42832, 42833, 42834, 42835, 42836, 43361, 62897, 62903, 62904, 62907, 62911, 62922, 62923, 62926, 62928, 62931, 63564, 63970, 63971, 63972, 63973, 64992, 64993, 66350

The following paragraph is applicable to the landing performance provided by selecting WET SK-R in AFM-DPI.

The landing performance provided for the 737-8/-8200/-9 flaps 30 and 40 operations has been evaluated in accordance with AC121.195(d)-1A and found suitable for runways constructed and maintained to meet the Friction Level Classification for Runway Pavement Surface defined in AC 150/5320-12C dated 3/18/97 or its equivalent. This finding does not constitute operational approval to base the landing performance requirements on these distances.

The inclusion of this data in the AFM does not constitute operational approval to base the landing performance requirements on these distances. Where such operational approval has not been obtained, the performance information applicable to a smooth, hard-surfaced runway must be used.

The data is not applicable to operations with Antiskid Inoperative.

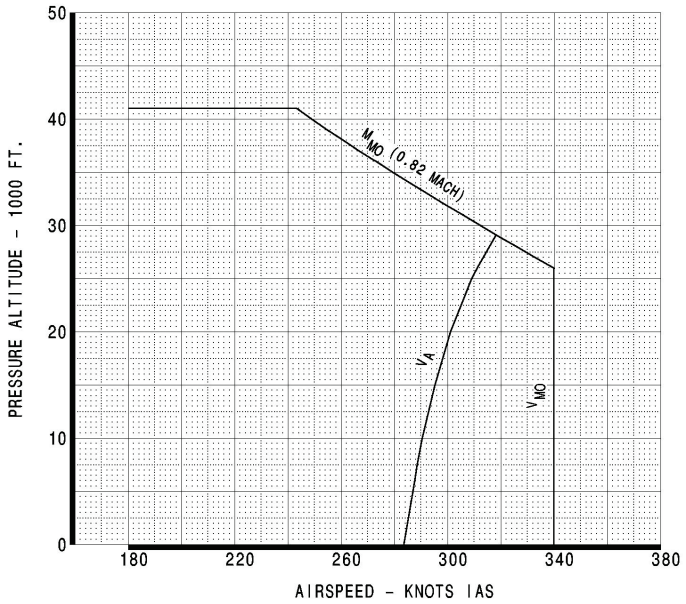
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Maximum Airspeed Limits A180643

FLAP PLACARD SPEEDS	
FLAP POSITION	V_{FE} - KNOTS IAS
1	250*
5	250*
10	210
15	200
25	190
30	175
40	166

*DO NOT EXCEED 230 KNOTS IAS WHEN FLAPS ARE EXTENDED USING THE ALTERNATE (STANDBY) SYSTEM.

- MAXIMUM LANDING GEAR RETRACTING SPEED: V_{LO} = 235 KNOTS IAS
- MAXIMUM LANDING GEAR EXTENDING SPEED: V_{LO} = 270 KNOTS IAS, M_{LO} = 0.82 MACH
- MAXIMUM LANDING GEAR EXTENDED SPEEDS: V_{LE} = 320 KNOTS IAS, M_{LE} = 0.82 MACH



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Maximum Operating Limit Speed (VMO MMO) A191045

The maximum operating limit speed shall not be deliberately exceeded in any regime of flight.

Notes:

1. All instrument markings and placards in the airplane are shown as indicated (IAS, M) values and are not corrected for instrument error. V_{MO} is indicated by the upper barber pole on the speed tape.

2. V_A is defined as the speed above which maneuvers involving full application of rudder, ailerons or elevator, or maneuvers involving angles of attack near stall, should be avoided.

Operational Limits A201495

Runway slope -- ±2%

Maximum Takeoff and Landing Tailwind Component -- 15 knots.

Note: The capability of this airplane has been satisfactorily demonstrated for takeoff and manual landing with a tailwind component up to 15 knots. This finding does not constitute operational approval to conduct takeoffs or landings with a tailwind component greater than 10 knots.

Maximum Operating Altitude -- 41,000 feet pressure altitude.

Maximum Takeoff and Landing Altitude -- 8,400 feet pressure altitude.

Takeoff, Landing, and Enroute Operational Limits - use Environmental Envelope chart, see Performance chapter in Performance appendix.

Maximum Flight Operating Latitude:

Maximum flight operating latitude -- 82 degrees North and 82 degrees South, except for the region between 80 degrees West and 170 degrees West longitude, the maximum flight operating latitude is 73 degrees North, and the region between 120 degrees East and 160 degrees East longitude, the maximum flight operating latitude is 60 degrees South.

Performance Configuration A163052

The airplane configuration as presented under Performance Configuration, Performance Chapter, must be observed.

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Required Crew A163052

Required flight crew: Two Pilots

Required Navigation Performance (RNP) A211975

The following is applicable only to airplanes with U14.1 Operational Program Software (OPS) installed and Flight Management Computer P/N 10-62225-004 installed in either FMC-L or FMC-R:

Required Navigation Performance - Authorization Required (RNP AR)
RNP AR operations with minima below 0.3nm are prohibited.

Speedbrake Deployment A170119

In flight, do not extend the speed brake lever beyond the FLIGHT detent. Do not use speed brakes below 1000 feet above the surface.

In flight, do not extend the speedbrake lever beyond the ARMED detent with flaps 40 selected.

Window Heat Inoperative A170119

Airspeed is limited to 250 knots IAS at altitudes below 10,000 feet with number 1 or number 2 window heat inoperative.

Wing Thermal Anti-Ice System (Ground Operation) A163052

Do not use wing anti-ice as a substitute for ground de-icing/anti- icing and inspection procedures which are necessary to comply with operating rules.

Do not operate wing anti-ice on the ground when the OAT is above 10°C (50°F).

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General A201615

The operating procedures contained in this chapter have been developed and recommended by the manufacturer and approved by the appropriate regulatory agencies for use in the operation of this aircraft. These procedures are provided as guidance and should not be construed as prohibiting the development of equivalent approved procedures.

Procedures in this chapter are peculiar to the operation of the Model 737 airplane and are presented in abbreviated form. These procedures as well as additional procedures and information are found in the 737 Flight Crew Operating Manual for the operation of this aircraft.

This chapter contains "non-normal" procedures which arise out of a system malfunction or failure, and/or which involve the use of special systems or the alternative use of normal systems to protect crew and passengers from harm, and to maintain the airworthiness of the airplane. Emergency procedures requiring immediate crew awareness are designated by (E).

This chapter also contains "normal procedure", designated by (N), where an action or sequence of actions or prohibition of incorrect actions, which if not observed, could result in an adverse effect on the airworthiness of the airplane or the safety of the passengers and crew.

Procedures contained herein assume:

(a) When able, aural warnings are silenced.

(b) Oxygen masks and smoke goggles (if installed) are donned when cabin altitude is excessive or ambient air is contaminated, and normal oxygen is selected if ambient air is not contaminated.

(c) Landing at nearest suitable airport is accomplished in the event of fire or persistent smoke.

(d) All flight crew members go on 100% oxygen whenever a hand-held fire extinguisher is to be discharged in the flight deck area, or when required because of smoke accumulation.

Definitions
(*Required by AD 2020-24-02)*
Recall items are minimum immediate actions items.
Reference items are accomplished after Recall items have been accomplished.

Air Conditioning - No Engine Bleed Landing with APU Bleed Air (N) A163052

(Applicable when making a no engine bleed landing with the APU providing bleed air. Wing Thermal Anti-Ice (WTAI) must be positioned OFF.)

When below 10,000 feet position right PACK switch to AUTO, ISOLATION VALVE switch to CLOSE, left PACK switch to AUTO, Engine No. 1 Bleed air switch to OFF, APU BLEED air switch to ON and Engine No. 2 Bleed air switch to OFF.

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Air Conditioning - No Engine Bleed Landing without APU Bleed Air (N) A163052

(Applicable only when there is no engine bleed during landing and the APU is not providing bleed air)

When below 10,000 feet and starting the turn to final approach, position Engine BLEED air switches to OFF.

Air Conditioning - No Engine Bleed Takeoff (N) A163052

(Applicable only when there is no engine bleed during takeoff roll and the APU is not providing bleed air.)

After engine thrust is reduced from takeoff to climb, position Engine No. 2 BLEED air switch to ON. When CABIN rate of CLIMB indicator stabilizes Position Engine No. 1 BLEED air switch to ON and ISOLATION VALVE switch to AUTO.

(Applicable when making a no engine bleed takeoff with the APU providing bleed air. Wing Thermal Anti-Ice (WTAI) must be positioned OFF.)

After engine thrust is reduced from takeoff to climb, position Engine No. 2 BLEED air switch to ON and APU BLEED air switch to OFF. When CABIN rate of CLIMB indicator stabilizes Position Engine No. 1 BLEED air switch to ON and ISOLATION VALVE switch to AUTO.

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Airspeed Unreliable (E) A211940

(Alternative Method of Compliance (AMOC) to AD 2020-24-02)
Airspeed or Mach indications are suspected to be unreliable:

Recall:
If autopilot is engaged, disengage. If autothrottle is engaged, disengage. Set both F/D switches to off.
Set the following gear up pitch attitude and thrust:
Flaps extended: 10° and 80% N1
Flaps up: 4° and 75% N1

Reference:
PROBE HEAT switches check on.
The following indications are reliable: attitude, N1, ground speed, and radio altitude.
Notes:
1. Stick shaker, overspeed warning and airspeed low alerts may sound erroneously or simultaneously.
2. The flight path vector and pitch limit indicator may be unreliable on the PFD and HUD (as installed).
3. If the AOA indicator option is installed, the stick shaker indicator may be unreliable. AOA digital readout, analog needle, and approach reference band may be unreliable if the airspeed unreliable condition is caused by erroneous AOA.

Attempt to determine a reliable airspeed indication.

If a reliable airspeed indication can be determined:

Use the reliable airspeed indication for the remainder of the flight. If only the standby airspeed indication is reliable do not use autopilot, autothrottle, or flight directors. If the captain's or first officer's airspeed indication is reliable, turn on the flight director switch on the reliable side. If needed, engage autopilot on the reliable side. Do not use autothrottle.
Note: Autopilot may not engage or may disengage automatically.

If a reliable airspeed indication cannot initially be determined:

Using performance tables from an approved source, set the pitch attitude and thrust setting for the current airplane configuration and phase of flight. When in trim and stabilized, compare the captain, first officer, and standby airspeed indicators with the airspeed shown in the table. An airspeed indication that differs by more than 20 knots or 0.03 Mach from the airspeed shown in the table should be considered unreliable. If only the standby airspeed indication is reliable, do not use autopilot, autothrottle, or flight directors. If the captain's or first officer's airspeed indication is reliable, turn on the flight director switch on the reliable side, and autopilot if needed. Do not use autothrottle.
Note: Autopilot may not engage or may disengage automatically.

If a reliable airspeed indication cannot be determined:

Using the performance tables from an approved source, set pitch attitude and thrust setting for the airplane configuration and phase of flight as needed. Reference an approved source for landing distances.
Notes:
1. Maintain visual conditions if possible.
2. Establish landing configuration early.

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Section Airspeed Unreliable (E), continued from previous page A211940

3. Radio altitude reference is available below 2500 feet.
4. Use electronic and visual glideslope indicators, where available, for approach and landing.

Attempt to determine a reliable altitude indication.

Use the most reliable altitude indication for the remainder of the flight. If the captain's or first officer's altitude indication is reliable:
The airplane may not meet RVSM requirements. Set transponder to reliable side and select traffic alerts only mode.

If captain's and first officer's altitude indications are both unreliable:
Turn off transponder altitude reporting.
Note: Airplane does not meet RVSM requirements.

In addition to the normal descent, approach and landing checklists, complete the following deferred items:

For approach, only set the BARO minimums on the reliable PFD. Remove the BARO minimums from the unreliable PFD.
Note: If BARO minimums are set only on the First Officer's PFD, DA/MDA aural callouts are not provided.
Use the performance tables from an approved source to determine the go-around pitch attitude and thrust setting.

The following is applicable to airplanes equipped with split axis flight director:

> In the event of a go-around if either the Captain's or First Officer's airspeed indication is reliable, when TO/GA is pushed, the flight director pitch bar may automatically be removed. An AFDS pitch mode change, such as LVL CHG, restores the flight director pitch bar.
> **Note:** Only use flight director guidance on the reliable PFD.
>
> In the event of a go-around and the standby airspeed indication is the only reliable airspeed, do not use TO/GA.

The following is applicable to airplanes equipped with integrated cue flight director:

> In the event of a go-around if either the Captain's or First Officer's airspeed indication is reliable, when TO/GA is pushed, the flight director may automatically be removed. An AFDS pitch mode change, such as LVL CHG, restores the flight director.
> **Note:** Only use flight director guidance on the reliable PFD.
>
> In the event of a go-around and the standby airspeed indication is the only reliable airspeed, do not use TO/GA.

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ALT Disagree A201615

(Required by AD 2020-24-02)
The ALT DISAGREE alert is displayed on the captain's and first officer's altitude tape on the PFD when the indications disagree.
If the IAS DISAGREE alert is also shown on the speed tape of the PFD, accomplish the Airspeed Unreliable procedure.
If the IAS DISAGREE is not shown, check all altimeters are set to correct barometric setting.
If the ALT DISAGREE alert remains, do not use the flight path vector, and if a reliable altitude is determined, use the transponder for the reliable side.
If a reliable altitude is not determined, set the transponder to not transmit altitude.

In addition to the normal descent, approach and landing checklists, complete the following deferred items:

For approach, only set the BARO minimums on the reliable PFD. Remove the BARO minimums from the unreliable PFD.
Note: If BARO minimums are set only on the First Officer's PFD, DA/MDA aural callouts are not provided.
Establish landing configuration early.
Radio altitude reference is available below 2,500 ft.
Use electronic and visual glideslope indicators where available for approach and landing.

Antiskid Inoperative A170184

Takeoff with antiskid inoperative is allowed only on dry runways.

Autobrakes are not allowed with antiskid inoperative.

AOA Disagree A201615

(Required by AD 2020-24-02)
When AOA DISAGREE appears on the PFD, this indicates the left and right angle of attack vanes disagree.
Accomplish the Airspeed Unreliable procedure.

Automatic Speed Brakes A163052

If a fault in the Automatic Speed Brakes system is indicated, do not arm the automatic speed brakes. Use manual speed brake deployment as required.

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Automatic Temperature Compensation A201609

The following is applicable upon installation of FMCS Software update U14 and later versions with TEMP COMP option enabled:

When using temperature compensation inform ATC, failure to do so may lead to loss of separation between airplanes.

FMC temperature compensation applies corrections to all FMC waypoint altitudes from the START WAYPOINT including the missed approach waypoints.

Verify all FMC temperature compensated waypoint altitudes comply with ATC clearance.

For hot temperature:

- When applying temperature corrections to the final approach segment change the START WAYPOINT to the FAF. In the event of a go-around or missed approach, observe charted altitude restrictions and do not use VNAV.
- Temperature compensation can be applied to other segments of the approach with ATC approval.

A temperature altitude correction is not needed while under ATC radar vectors or when maintaining an ATC assigned flight level (FL).

For cold temperature correction, MDA/DA settings should be set at the corrected minimum altitudes for the approach.

If the corrected indicated altitude to be flown is between 100 foot increments, set the MCP altitude to the closest 100 foot increment above the corrected indicated altitude to be flown.

Note: Regulatory authorities may have other requirements for temperature altitude corrections.

Automatic Wheel Brakes (N) A163052

Monitor system landing performance and apply manual brakes if the AUTO BRAKE DISARM annunciator illuminates or deceleration is not adequate. Use reverse thrust as desired.

When armed, the Rejected Takeoff (RTO) feature of the automatic wheel brake system will apply full hydraulic system pressure to the brakes, provided both forward thrust levers are in the idle position and airplane ground speed is more than 90 knots.

Prior to landing select automatic braking, if desired. Select the appropriate setting for the landing runway length available and stopping distance desired.

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Auxiliary Power Unit (APU) (N) A163052

APU in-flight starts have been successfully demonstrated to 41,000 feet pressure altitude. APU in-flight starts may be attempted at any altitude.

Except for engine start do not open APU bleed air valve when LH engine bleed air valve is open, or when isolation and RH engine bleed air valves are open.

With ground air supply connected and isolation valve open do not open APU bleed air valve.

Braking with Accumulator Pressure Only A163052

With brake pressure from the accumulator only, apply steady brake pressure without modulation to avoid accumulator depletion. Do not attempt to taxi after stopping.

Cabin Unpressurized Flight A170322

During unpressurized flight, do not operate more than one air-conditioning pack on. Manually drive the cabin pressure outflow valve to the 100% open position.

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Cabin Altitude Warning or Rapid Depressurization A191182

If Cabin Altitude warning light illuminates or the intermittent warning horn sounds in flight above 10,000 ft MSL:

Recall:

OXYGEN MASKS AND REGULATORS	ON, 100%
CREW COMMUNICATIONS	ESTABLISH

Reference:

PRESSURIZATION MODE SELECTOR	MANUAL
OUTFLOW VALVE SWITCH	CLOSE*

***Note**: Hold toward close until the valve is fully closed.

If Cabin Altitude is uncontrollable:

EMERGENCY DESCENT	INITIATE
PASSENGER OXYGEN SWITCH	ON
THRUST LEVERS	CLOSE
SPEED BRAKES	FLIGHT DETENT
TARGET SPEED	VMO/MMO

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Cabin Pressurization (N) A170249

Operate no more than one air-conditioning pack from one engine.

For normal operations, the pressurization mode selector should be in AUTO prior to takeoff.

Normal Automatic Operating Differential Pressure

Below 28,000 feet selected flight altitude	7.45 ± 0.1 psi
Above 28,000 feet selected flight altitude and below 37,000 feet selected flight altitude	7.80 ± 0.1 psi
Above 37,000 feet selected flight altitude	8.35 ± 0.1 psi

When, because of a malfunction during flight, the differential pressures corresponding to the selected flight level cannot be controlled, the pressurization system may be allowed to operate for the remainder of the flight on the relief valves without affecting the safety of the airplane.

Cockpit Window Failure A163052

When arcing or cracking is discovered, remove heat from that window and, if necessary, reduce the cabin pressure differential at a rate consistent with passenger comfort.

For number 1 and 2 windows, descend to 14,000 feet or minimum safe altitude, whichever is higher, with a cracked inner pane. Set the LAND ALT on the pressurization panel to 9,000 feet. There are no restrictions when an outer pane alone is cracked.

For unheated number 3 windows, there are no restrictions when an inner or outer pane alone is cracked. If both inner and outer panes are cracked, descend to 9,000 feet or the minimum safe altitude, which ever is higher, and depressurize the airplane at a rate consistent with passenger comfort.

For heated number 3 windows, descend to 14,000 feet or minimum safe altitude, whichever is higher, with a cracked inner pane. Set the LAND ALT on the pressurization panel to 9,000 feet. There are no restrictions when an outer pane alone is cracked.

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Ditching (E) A163052

Select the GEAR INHIBIT and TERR INHIBIT switches (if installed) to INHIBIT, turn OFF engine air bleeds and APU, and manually close the cabin outflow valve.

For ditching approach, gear is up with landing flaps and speed V_{REF} with rate of descent approximately 200 to 300 feet per minute.

To ensure no cabin differential pressure, open cockpit windows immediately after ditching.

Do not open aft entry or aft service doors.

Electronics - Global Landing Unit (GLU) A201322

(Required by AD 2020-03-20)

To conduct an approach procedure with GLU-2100 OPS software number COL4D-0087-0002, COL4E-0087-0001, COL48-0087-0700, or COL49-0087-0701, installed at Anchorage (PANC) with less than 0.3 RNP, accomplish the following prior to dispatch in accordance with AC 90-101A:

> Perform a RNP GPS prediction to ensure the predicted availability of GPS Horizontal Integrity Limit (HIL) is less than MAX HIL for the planned operation time frame at Anchorage (PANC).
>
> MAX HIL = 1.8 (RNP – 0.0726 nm) for LNAV with A/P engaged
> MAX HIL = 1.8 (RNP – 0.0926 nm) for LNAV with F/D

Engine - Engine Overheat A163052

Autothrottle (If engaged)	DISENGAGE
Thrust Lever	CLOSE

If the ENG OVERHEAT light remains illuminated, accomplish the ENGINE FIRE, SEVERE ENGINE DAMAGE OR SEPARATION procedure.

If the ENG OVERHEAT light extinguishes, operate the engine at reduced thrust to keep the light extinguished.

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Engine - Fire Severe Damage or Separation (E) A163052

If engine fire, severe engine damage, or engine separation is indicated, accomplish the steps below. This condition is recognized by the following: the fire warning bell rings, illumination of the FIRE WARN lights and an Engine Fire Warning Switch; or illumination of an ENG OVERHEAT light which does not extinguish when the thrust lever is closed.

Autothrottle (If engaged)	DISENGAGE
Thrust Lever	CLOSE
Start Lever	CUTOFF
Engine Fire Warning Switch	PULL

If Engine Fire Warning Switch or ENG OVERHEAT light remains illuminated:

Engine Fire Warning Switch	ROTATE TO L OR R

If after 30 seconds, the Engine Fire Warning Switch or ENG OVERHEAT light remains illuminated:

Engine Fire Warning Switch	ROTATE TO REMAINING BOTTLE

If high airframe vibration occurs and continues after engine is shut down:
Without delay, reduce airspeed and descend to a safe altitude which results in an acceptable vibration level. If high vibration returns and further airspeed reduction and descent are not practicable, increasing airspeed may reduce vibration.

Isolation Valve Switch	CLOSE
APU Bleed Air Switch	OFF
APU (if available)	START & ON BUS
Fuel	BALANCE
Transponder Mode Selector (Option with TCAS)	TA

Land at the nearest suitable airport.

If wing anti-ice is required:

Pack Switch (Affected side)	OFF
Isolation Valve Switch	AUTO

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Engine - High Oil Temperature A163052

If oil temperature is in the amber band, retard thrust lever until engine oil temperature is within normal operating range.

If oil temperature is in the amber band for 15 minutes or reaches the red line, shut down the engine if flight conditions permit.

Engine - High Vibration A170249

If VIB levels are 4.0 units accompanied by airframe vibrations, accomplish the following:

If not in icing conditions:

Disengage autothrottle and reduce thrust on affected engine to maintain VIB below 4.0 units.

Note: If the VIB indication does not decrease when the thrust lever is retarded, check other engine indications on the affected engine. If other engine indications are normal, run the engine at reduced thrust.

If in icing conditions (increase fan vibration due to fan ice accumulation):

1. Reduce thrust on one engine at a time to 45% N1. After approximately 5 seconds, advance thrust lever to a minimum of 80% N1 while monitoring engine vibrations.
2. Reduce thrust as needed for flight conditions.
3. Wait 15 seconds. This allows engine vibration level to stabilize.
4. If vibration does not decrease accomplish the procedure for "if not in icing conditions".
5. If above steps successfully remove fan ice and vibration is less than 4.0 units after thrust is reduced, repeat at approximately 15 minutes intervals or as needed.
6. If vibration decreases, indicating ice removal, resume normal operation.

Note: Engine vibration may indicate above the normal operating range up to the maximum display value during ice shedding, however, this will have no adverse effect on the engine.

Engine - Low Oil Pressure A163052

If oil pressure is at or below red line, shut down the engine.

Engine - Oil Filter Bypass A163052

If the oil filter is bypassed and normal operation can be reestablished by reducing thrust, operate at the reduced thrust.

If oil continues to bypass, shut down the engine if flight conditions permit.

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Equipment Cooling Off A163052

Illumination of the Equipment Cooling Supply or Exhaust OFF light indicates a loss of airflow from the selected cooling fan. Turn Equipment Cooling Supply/Exhaust Switch to Alternate.

No further action is necessary in flight if a single equipment cooling OFF light does not extinguish.

If both the Supply and Exhaust equipment cooling OFF lights do not extinguish, plan to land at nearest suitable airport.

Fire - APU (E) A163052

APU FIRE WARNING
RECALL:

APU Fire Warning Switch	PULL AND ROTATE
APU Switch	OFF

REFERENCE:

Master Fire Warning	RESET

If APU fire switch remains illuminated, land at nearest suitable airport.

Fire - Cargo Fire (E) A191036

CARGO FIRE ARM SWITCH (FWD/AFT)	ARMED
CARGO FIRE DISCHARGE SWITCH	PUSH
Push and hold for 1 second (ensures bottle discharge).	

Note: DISCH light may require approximately 30 seconds to illuminate.

RECIRC FAN SWITCH(ES)	OFF
OPERATING PACK(S) SWITCH(ES)	HIGH
LANDING ALTITUDE SELECTOR	8000 Feet
CAB/UTIL SWITCH	OFF

Land at the nearest suitable airfield. At top of descent, set the landing altitude to the landing field elevation.

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Fire - Cargo Fire Detector Fault A163052

Illumination of the detector fault light indicates the selected detector loop(s) have failed.

The selected fire detection loop(s) in one or both cargo compartments are inoperative.

Fire - Wheel Well (E) A163052

Landing Gear (Maximum 270 KIAS/.82M).	DOWN

If the landing gear must be retracted for airplane performance, leave the landing gear extended for 20 minutes after the WHEEL WELL FIRE warning light has extinguished.

If the WHEEL WELL FIRE warning light stays on, land at nearest suitable airport.

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Flaps Anomalies A180643

All Flaps-Up Landing

Land at as low a gross weight as possible.
Maintain flaps-up maneuvering speed until on final approach.
Limit bank angle to 15 degrees below flaps-up maneuver speed.
Approach and land at V_{REF} flaps 40 plus 55 knots.

Leading Edge Flaps Transit

While the transit light in general illuminates for leading edge surfaces not in commanded position cases, it will also illuminate during normal L.E. extension/retraction operation.

Illumination of the leading edge flaps transit light indicates one or more leading edge devices are not in the correct position. Limit airspeed to:

230 kts for multiple surfaces out of position or if roll is encountered.
300 kts for a single surface out of position and if no roll is encountered.

Plan to make a flaps 15 landing at V_{REF} 15 plus 15 knots. Limit bank angle to 15 degrees below flaps-up maneuvering speed.

Trailing Edge Flap Asymmetry

After a trailing edge flap asymmetry shut-off, do not attempt to override flap asymmetry protection. Place flap lever in nearest detent corresponding to the smallest actual flap position, and land using the following speeds:

Flaps are 15 or greater, set V_{REF} for the smallest flap position.
Flaps are 1 or greater, and less than 15, set V_{REF} 40 + 30 knots (Land at as low a gross weight as possible).
Flaps less than 1, use trailing edge flaps-up landing procedure.

Trailing Edge Flap Disagree

After positioning the flap handle, if the flap position indicators are not in agreement with the flap handle, and there is no asymmetry indicated, land using the following:
Indicated flap position is 30 or greater, and less than 40, land using existing flaps and set V_{REF} 30.
Indicated flap position is 15 or greater, and less than 30, land using existing flaps and set V_{REF} 15.
Indicated flap position is less than 15, plan to extend flaps to 15 using alternate flap extension and set V_{REF} 15.

Trailing Edge Flaps-Up Landing

Land at as low a gross weight as possible.
Maintain flaps-up maneuver speed until on final approach.
Limit bank angle to 15 degrees below flaps-up maneuver speed.
Approach and land at V_{REF} flaps 40 plus 40 knots.

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Section Flaps Anomalies, continued from previous page

Use alternate system (momentarily) to extend leading edge devices at 230 knots or lower.

Fuel Quantity Low A170415

With less than 2,000* pounds (907 kilograms) indicated fuel load in either wing tank, open the crossfeed valve and turn all fuel pump switches ON. Apply thrust changes slowly and smoothly. Avoid high nose-up attitudes.

* 1,300 pounds (590 kilograms) upon incorporation of approved Boeing Service Bulletin or production equivalent.

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Fuel System (N) A180643

Nominal Tank Capacity

	USABLE FUEL		
TANK	U.S. GALLONS	IMP GALLONS	LITERS
#1 OR #2	1,273	1,059	4,819
CENTER	4,274	3,558	16,179

Lateral Fuel Imbalance

The lateral fuel imbalance between main wing tanks must be scheduled to be zero. Fuel imbalance between main wing tanks for taxi, takeoff, flight or landing must not exceed 1,000 pounds (453 kilograms).

Fuel Loading

Fuel tanks may be loaded individually, simultaneously, or in any sequence. Load main tanks equally to the desired fuel quantity or until full. Main wing tanks must be scheduled to be full if the center wing tank contains more than 1,000 pounds (453 kilograms) of fuel. With 1,000 pounds (453 kilograms) or less of center wing tank fuel, partial main wing tank fuel may be loaded provided the effects of balance have been considered. Fuel must always be used in accordance with **Normal Fuel Usage**, below.

Normal Fuel Usage

The crossfeed valve is normally closed [1].

All boost pumps are ON for all normal operations (center tank pumps ON only when fuel is available).

To equalize main wing tank fuel, open crossfeed valve and turn off both boost pumps in the main wing tank with the least amount of fuel until fuel is equalized.

With no center tank fuel, use main tank-to-engine fuel feed with all operable main tank boost pumps on, center tank boost pumps off and the crossfeed valve closed [1].

With center tank fuel, use center tank fuel for all operations with all operable boost pumps on and the crossfeed valve closed [1] until center tank fuel is depleted (however, a maximum of 1,000 pounds (453 kilograms) may be retained in the center wing tank provided the effects of balance have been considered). Then continue flight using main tank-to-engine fuel feed with all operable main tank boost pumps on, center tank boost pumps off and the crossfeed valve closed [1].

[1] The crossfeed valve is open for minimum fuel operation, and may be opened to correct fuel imbalance.

Center Tank Fuel Pumps

Center tank fuel pumps must not be “ON” unless personnel are available in the flight deck to monitor low pressure lights.

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Section Fuel System (N), continued from previous page

For ground operation, center tank fuel pump switches must not be positioned “ON” unless the center tank fuel quantity exceeds 1,000 pounds (453 kilograms), except when defueling or transferring fuel. Upon positioning the center tank fuel pump switches “ON” verify momentary illumination of each center tank fuel pump low pressure light.

For ground and flight operations, the corresponding center tank fuel pump switch must be positioned “OFF” when a center tank fuel pump low pressure light illuminates [1].

Both center tank fuel pump switches must be positioned “OFF” when the first center tank fuel pump low pressure light illuminates if the center tank is empty.

[1] When established in a level flight attitude, both center tank pump switches should be positioned “ON” again if the center tank contains usable fuel.

Defueling and Fuel Transfer

When transferring fuel or defueling center or main tanks, the fuel pump low pressure lights must be monitored and the fuel pumps positioned to “OFF” at the first indication of the fuel pump low pressure [1].

Defueling the main tanks with passengers on board is prohibited if the main tank fuel pumps are powered [2].

Defueling the center tank with passengers on board is prohibited if the center tank fuel pumps are powered and the auto-shutoff system is inhibited [2].

[1] Prior to transferring fuel or defueling, conduct a lamp test of the respective fuel pump low pressure lights.

[2] Fuel may be transferred from tank to tank or the aircraft maybe defueled with passengers on board, provided fuel quantity in the tank from which fuel is being taken is maintained at or above 2,000 pounds (907 kilograms).

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Go-Around (N) A170344

When go-around is initiated the autothrottle system (if engaged) will advance the thrust levers automatically. Flaps and landing gear must be controlled manually.

An Autothrottle, Flight Director and/or Dual Autopilot go-around may be initiated below a radio altitude of 2,000 feet by pressing a go-around switch.

When the decision is made to abort the approach, actuate a go-around switch and assure rotation to go-around attitude. Verify thrust lever movement to achieve a nominal rate of climb* and retract flaps to flaps 15.**

After a positive rate of climb has been established, retract landing gear. Climb to a safe altitude, accelerate and retract remaining flaps according to takeoff flap retraction speed schedule. Monitor rate of climb, attitude, and airspeed.

* Full go-around thrust may be obtained, after engine spool-up, by reactivating a go-around switch.

** In windshear, the recommended procedure is to delay flap and gear retraction until windshear is no longer a factor.

Ground Proximity Warning System A163052

When a ground proximity warning occurs, aggressively pull up, apply engine thrust and climb until the warning ceases or clear of terrain.

When a ground proximity caution occurs, correct the flight path/configuration so that the alert ceases.

The Ground Proximity Warning System (GPWS) Mode 4 function may be deactivated to prevent nuisance warnings when following approved procedures which specify landing with flaps not in normal landing position or landing gear up.

Hydraulic Pump Overheat or Low Pressure A163052

If overheat or low pressure of any hydraulic pump is indicated, turn the affected pump off.

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Hydraulic System A163052

If either hydraulic system is lost: Switch failed system FLT CONTROL to STBY RUD.

Loss of System A

If system A is lost, it will be necessary to use the standby system for left engine thrust reverser. Flight spoilers (two on each wing), nose wheel steering, alternate brakes and Autopilot channel "A" are inoperative. Landing gear must be extended manually.

Loss of System B

If system B is lost, use alternate operation for trailing edge flaps. Selection of standby rudder automatically activates the standby system for leading edge device extension and right engine thrust reverser operation. The flight spoilers (two on each wing), autobrake system and Autopilot channel "B" are inoperative. Approach and landing with flaps 15 at V_{REF} flaps 15 is recommended.

IAS Disagree A201615

(Required by AD 2020-24-02)
When IAS DISAGREE appears on the PFD, this indicates the captain's and first officer's airspeed indicators disagree. Accomplish the Airspeed Unreliable procedure.

Inflight Engine Fuel Leak A163052

If an inflight fuel leak is suspected, turn off the center wing tank fuel pumps and close the crossfeed valve. Watch for any unusual decrease in fuel tank quantity and/or a fuel imbalance to determine if there is an engine fuel leak.

If an engine fuel leak is confirmed (either visually or by flight deck indications), shut down the affected engine to stop the leak and retain the remaining fuel. After shutdown of the affected engine, resume normal fuel management procedures. All remaining fuel can be used for the operating engine. Land at the nearest suitable airport.

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IRS Fault A163052

Applicable to ASN(s) : 42825, 42826, 42827, 42828, 42829, 42830, 42833, 42835, 43361, 62897, 62903, 62904, 62907, 62911, 62922, 62923, 62926, 62928, 62931, 63564, 63970, 63971, 63972, 63973, 64993, 66350

Position instrument source selector switch(es) to operating IRS. Do not engage either autopilot.

IRS Fault A170462

Applicable to ASN(s) : 42831, 42832, 42834, 42836, 64992

Position instrument source selector switch(es) to operating IRS. Do not engage the autopilot for single channel approaches.

With the failure of one IRS, the autopilot on the side opposite the failed IRS may be engaged for cruise operations, however, the yaw damper will disengage 60 seconds after the autopilot is engaged and cannot be re-engaged.

Jammed Flight Controls A170249

In the event of a jammed rudder, elevator or aileron, do not hesitate to apply additional force to maintain control of the airplane. Do not turn off any flight control switches. Stabilizer or rudder trim may be used to offload control forces. Do not make abrupt thrust or speedbrake changes. Limit bank angle to 15 degrees. Make approach and landing with flaps 15 at V_{REF} flaps 15 plus 10 knots. If pitch control is jammed or restricted, set the Elevator Jam Landing Assist to ON after the flaps have been selected to 1 or greater for the approach.

Landing Configuration A163052

Condition: A steady warning horn sounds in flight.
Assure proper airplane landing configuration.

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Landing Gear Lever Unable To Place In Up Position After Takeoff A191043

If landing gear retraction is necessary:

Landing Gear Lever	DOWN

If the takeoff configuration warning remains silent after the flaps are fully retracted:

Note: This condition indicates a failure of the landing gear latch solenoid.

Landing Gear Lever LOCK OVRD switch	PUSH & HOLD
Landing Gear Lever	UP

If the takeoff configuration warning sounds when the flaps are fully retracted:

Land at the nearest suitable airport.

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Loss of Both Engine Driven Generators A163052

All TRANSFER BUS OFF, SOURCE OFF, and GEN OFF BUS lights illuminated indicate the loss of both engine driven generators.

Note: With main tank fuel pumps inoperative above 30,000 feet, thrust deterioration or engine flameout may occur.

Engine Generator Switches	ON

If only one SOURCE OFF light extinguishes:

APU (if available)	START & ON BUS

If both SOURCE OFF lights remain illuminated:

Bus Transfer Switch	OFF
Electric Hydraulic Pump Switches	OFF
APU (if available)	START & ON BUSES
Bus Transfer Switch	AUTO
Electric Hydraulic Pump Switches	ON (ONE AT A TIME)
RCCB Remote C/B (STBY Power Control Unit).	RESET*

***Note**: This step is not required for airplanes equipped with a single (one) standby battery. This step is not required for airplanes equipped with dual (two) standby batteries if the Standby Power Control Unit (SPCU) is supplier part number S281A601-3001 or later.

If both primary attitude displays are inoperative:

IRS Mode Selector Switches	ATT
Maintain wings level, constant speed flight until attitude displays recover (approximately 30 seconds).	
Magnetic Heading	ENTER

If both SOURCE OFF lights remain illuminated:

Avoid icing conditions.

Switch the following switches to the OFF position on the P5-13 Power Panel: CAB/UTIL and IFE/PASS SEAT.

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Note: Flight in icing conditions may result in erroneous flight instrument indications.

Land at the nearest suitable airport.

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Loss of Thrust On Both Engines (E) A191078

Engine Start Switches (both)	FLT
Engine Start Levers (both)	CUTOFF

EGT Decreasing:

Engine Start Levers (both)	IDLE DETENT

If EGT reaches the redline limit or no increase in EGT within 30 seconds:

Engine Start Lever (affected engine)	CUTOFF, then IDLE DETENT

If EGT again reaches the redline limit or no increase in EGT within 30 seconds, repeat.

Note: Engines may accelerate to idle very slowly, especially at high altitudes or in heavy precipitation. Slow acceleration may be incorrectly interpreted as a hung start or an engine malfunction. If N2 is steadily increasing, and EGT remains within limits, do not interrupt the start.

Airspeed. . . . . .at or above FL270	275 KIAS
Airspeed. . . . . .below FL270	300 KIAS
APU (if available)	START
APU Generator	ON

If neither restart is successful and N2 is above 8.5%

Thrust Levers	CLOSE
Either Engine Start Lever	CUTOFF
Then	IDLE DETENT

Note: Engine may accelerate to idle very slowly. If N2 is steadily increasing, and EGT remains within limits, do not interrupt the start.

If neither restart is successful and N2 is below 8.5%:

Wing Anti-Ice Switch	OFF
Pack Switches	OFF
APU Bleed Air Switch	ON
Ignition Select Switch	BOTH
Either Engine Start Switch	GRD

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Section Loss of Thrust On Both Engines (E), continued from previous page A191078

Note: Engine may accelerate to idle very slowly. If N2 is steadily increasing, and EGT remains within limits, do not interrupt the start.

When engine parameters have stabilized:

APU Bleed Air Switch	OFF
Engine Start Switch	FLT
Thrust Lever	Advance
Engine Generator Switch	ON
Pack Switch	AUTO

Attempt to start the other engine.

If neither IRS attitude display recovers after a transfer bus is restored:

IRS Mode Selector Switches	ATT
Maintain wings level, constant speed flight until attitude displays recover (approximately 30 seconds)	
Magnetic Heading	ENTER
APU	AS REQUIRED

If APU Not Available, place the following switches to the OFF position on the P5-13 Power Pane: CAB/UTIL and IFE/PASS SEAT.

Mach Trim System A163052

If the Mach trim system malfunctions, or becomes inoperative during flight, do not exceed 280 KIAS/.82 Mach.

Manual Flight Control A163052

Larger control forces should be anticipated and larger amounts of control movements are required. Select standby rudder, turn yaw damper on and turn system A and B hydraulic pump switches off. Plan for manual gear and alternate flap extension. Make approach and landing with flaps 15 at V_{REF} flaps 15. The crosswind capability of the airplane will be greatly reduced.

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Manual Landing Gear Extension A163052

Extend landing gear manually as follows:

T-handles (3)	PULL

Wait 15 seconds after the last manual gear extension handle is pulled:

Gear lever	DOWN
Gear down and locked lights	CHECK GREEN

One Engine Inoperative Landing A163052

Use flaps 15 and V_{REF} 15 + 5 knots for landing approach. Land using flaps 15. Use flaps 1 for missed approach/go-around.

One Generator Operation A163052

The limit for one generator operation in the air, is 75 KVA (215 Amps).

If the APU is not available, the Cabin/Utility and IFE/Passenger Seat switches on the P5-13 panel will need to be set to the OFF position to limit the total load to below 75 KVA (215 Amps).

On the ground, limit one generator operation (engine driven) to a maximum of 75 KVA (215 amps).

Operation in Icing Conditions - Wing Anti-Ice System (N) A163052

In flight, the primary method of operating the wing anti-ice system is to operate it as a de-icing system. Ice accumulation on the cockpit window frames, windshield center post, below the windshield wiper blade or on the windshield wiper arm may be used as an indication of structural Icing Conditions and the need to turn on the wing anti-ice system.

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Operation on Standby Electrical Power A180643

When operating on standby power only, a minimum of 30 minutes of electrical power is provided from a single battery system (the dual battery system, if installed, provides 60 minutes of electrical power). DC power is supplied to the hot battery bus, switched hot battery bus, the battery bus and the standby DC bus. 115-volt AC power is supplied from the static inverter to the standby AC bus and the 28-volt AC standby bus is supplied from a transformer off the 115-volt AC standby bus.

Services provided for include: Flight instruments, including Captain's attitude/heading system, airspeed, Mach, barometric altitude, vertical speed, #1 navigation and communication radios, #1 ADF, #1 stall warning, standby altimeter/airspeed, and standby attitude with ILS/GLS* deviation bars. The Captain's outboard and inboard display unit, left display processing computer, EFIS control panel are active. As an optional feature, one or more of the following additional items may be provided for: #1 ATC, #1 DME, GPS, #1 FMC/CDU.

***Note**: GLS is an optional feature and may or may not be installed.

The Integrated Standby Flight Display (ISFD) System includes a dedicated battery that provides an additional 150 minutes minimum of ISFD operation following the loss of standby power (main battery depletion). Services provided for include:

APU and engine generator controls and indicators.

Electronic engine control, engine and APU starting, ignition, fuel shutoff and crossfeed valves and fuel quantity indicators, engine N1 and EGT indicators, primary engine display on the left inboard display unit, and hydraulic fluid shutoff valves.

Takeoff warning, landing gear indication and warning, and antiskid INOP lights.

Master fire warning, engine and APU fire detection, warning and extinguishing systems.

Thrust reversers, inboard antiskid system and parking brake light.

Oxygen indicators and valve.

Air conditioning overheat indicators, pack valves and manual pressurization control.

Standby flood and compass lights, master caution and integral lights for standby flight instruments, digital clock, control cabin white dome lights, and forward air stair standby controls and actuator.

Passenger address, and flight and service interphone.

Pitot probes, the total air temperature probe and the alpha vanes are electrically heated. Static ports are not heated. Probe/vane heat is not provided for on standby power. The captain's pitot probe may be heated on standby power (as an optional feature), however, the CAPT PITOT light does not illuminate for a failure.

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Section Operation on Standby Electrical Power, continued from previous page

Emergency area lights, floor proximity lights and exit signs, which are powered by battery packs independent of the main battery.

Passenger Evacuation (E) A163052

In the event passengers must be rapidly evacuated from the airplane, reduction of personal injury and damage may be enhanced by early notification of the evacuation. Set parking brake, select Flaps 40, speedbrake lever down, open outflow valve (as required), shut down engines, and discharge engine and APU fire bottles (as required) prior to leaving the airplane.

Rejected Takeoff (RTO) A163052

Simultaneously close the thrust levers (disengage the autothrottles, if required) and apply maximum manual wheel brakes or verify operation of RTO autobrakes.

If RTO autobrake is selected, monitor system performance and apply manual wheel brakes if the AUTO BRAKE DISARM light illuminates or deceleration is not adequate.

Raise SPEED BRAKE lever.

Apply maximum reverse thrust consistent with conditions.
Continue maximum braking until certain the airplane will stop on the runway.

Reverse Thrust (N) A163052

When on the ground with forward thrust levers at idle, pull reverse thrust levers aft to the interlock position, then to the reverse thrust detent. As the airplane decelerates below approximately 60 knots, modulate power to idle detent by taxi speed. Position levers to forward thrust after engine deceleration to idle.

Reverser Limited

On the ground, a fault has occurred in the engine thrust reverser system.

In air, a fault in the engine thrust reverser system limits reverse thrust. Engine thrust reverser will not deploy or reverse thrust will be limited to idle.

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Runaway Stabilizer (E) A201615

(Required by AD 2020-24-02)
If uncommanded stabilizer movement occurs continuously or in a manner not appropriate for flight conditions:

Recall:
Firmly hold control column. Disengage autopilot if engaged. Disengage autothrottle if engaged. Use the control column and thrust levers to control airplane pitch attitude and airspeed. Use main electric stabilizer trim to reduce control column forces.

If the runaway stops after autopilot is disengaged, do not re-engage autopilot or autothrottle; end of procedure.

If the runaway continues after autopilot is disengaged, place both STAB TRIM cutout switches to CUTOUT.

If the runaway continues, grasp and hold stabilizer trim wheel.

Reference:
Trim the stabilizer manually.
Notes:
1. A two-pilot effort may be used to correct an out of trim condition.
2. Reducing airspeed reduces airloads on the stabilizer which can reduce the effort needed to manually trim. Anticipate trim requirements. Do not re-engage autopilot or autothrottle.

In addition to the normal descent, approach and landing checklists, complete the following deferred item:

Establish landing configuration and in-trim condition early on final approach.

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Severe Turbulent Air Penetration (N) A163052

The maximum degree of turbulence encountered at the pilot's station during certification flight tests was evaluated as moderate.

Flight through severe turbulence should be avoided, if possible. When flying at 30,000 feet or higher, it is not advisable to avoid a turbulent area by climbing over it, unless it is obvious that it can be overflown well in the clear. For turbulence of the same intensity, greater buffet margins are achieved by flying the recommended speeds at reduced altitudes.

Selection of the autopilot Control Wheel Steering (CWS) is recommended for operation in severe turbulence. Do not use Altitude Hold (ALT HOLD) mode.

The recommended procedures for flight in severe turbulence are:

1. Air Speed

Target air speed should be approximately 280 KIAS or 0.76 Mach, whichever is lower. Severe turbulence will cause large and often rapid variations in indicated air speed. DO NOT CHASE THE AIR SPEED.

2. Yaw Damper – ENGAGED

3. Autopilot – OPTIONAL

If Autopilot Engaged:

Use CWS position, do not use ALT HOLD mode.

4. Autothrottle – DISENGAGE

5. Attitude

Maintain wings level and the desired pitch attitude. Use attitude indicator as the primary instrument. In extreme drafts, large attitude changes may occur. DO NOT USE SUDDEN LARGE CONTROL INPUTS. After establishing the trim setting for penetration speed, DO NOT CHANGE STABILIZER TRIM.

6. Altitude

Allow altitude to vary. Large altitude variations are possible in severe turbulence. Sacrifice altitude in order to maintain the desired attitude and air speed. DO NOT CHASE ALTITUDE.

7. Thrust

Engine ignition should be on. Make an initial thrust setting for the target air speed. CHANGE THRUST ONLY IN CASE OF EXTREME AIR SPEED VARIATION. The FMC cruise page displays N1 target values for turbulence.

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Smoke Fire or Fumes (E) A180643

Oxygen Masks and Regulators	On, 100%
Smoke Goggles	On
Crew and Cabin Communications	Establish
Bus Transfer Switch	OFF
Cabin/Utility Switch	OFF
IFE/Passenger Seat Switch	OFF
Recirculation Fan Switch(es)	OFF
Gasper Fan Switch (if installed)	OFF
APU Bleed Air Switch	OFF

Do the SMOKE OR FUMES REMOVAL procedure if needed.

If the source of the smoke, fire or fumes is both obvious and can be extinguished quickly, isolate and extinguish the source. If possible, remove electrical power from the affected equipment.

If the source is visually confirmed to be extinguished and the smoke or fumes are decreasing, continue flight at the captain's discretion.

If the source is not visually confirmed to be extinguished continue with the following steps:

Select the Equipment Cooling Supply/Exhaust Switches to ALTERNATE.

Initiate a diversion to the nearest suitable airport.

If time permits prior to landing, isolate each side of the air conditioning system one at a time.

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Smoke or Fumes Removal (E) A163052

Do this procedure only when directed by the SMOKE, FIRE OR FUMES Procedure.

Flight Deck Door	Close

If pack(s) are AUTO:

L and R PACK Switches	High

Do not turn on any PACK switch that was turned off by the Smoke, Fire or Fumes procedure.

Landing Altitude	10,000 FT
Engine No. 1 and No. 2 Bleed Air Switches	Verify ON
Engine Thrust	Maximum Practical N1(Minimum 45%)
Flight Deck Air Conditioning and Gasper Outlets	Open

CAUTION: Do not open any flight deck window. Keep the flight deck door closed.

If smoke or fumes are uncontrollable:

Airplane Altitude	Lowest Safe Altitude, or 10,000 FT, whichever is higher

At 14,000 feet or below:

Pressurization Mode Selector	Man
Outflow Valve Switch	Open*

***Note**: Hold toward open until the valve is fully open.

Return to the Smoke, Fire or Fumes procedure

If pack(s) are OFF and smoke or fumes source is confirmed to be in the flight deck:

Normal Holding Airspeed	Establish
First Officer's Sliding Window	Open

CAUTION: Window should not be opened unless the source is confirmed to be originating in the flight deck.

Return to the Smoke, Fire or Fumes procedure

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Speed Brakes (N) A163052

The automatic function will raise all speed brakes (spoilers) from the DOWN position during an aborted takeoff (above 60 knots) upon actuation of reverse thrust lever(s).

Do not allow high sink rates to develop when using speed brakes during approach.

For automatic speed brake extension on landing, place speed brake lever in ARM detent prior to touchdown. If automatic operation is delayed, either on rejected takeoff or on landing, manually extend speed brakes. Manual extension of the speed brakes is required if the antiskid is inoperative.

Speed Trim Fail A201615

(Required by AD 2020-24-02)
The Speed Trim function and MCAS function are inoperative.
Continue normal operation.
Note: The Speed Trim System will not provide stabilizer trim inputs when deviating from a trimmed airspeed.

Stabilizer Out of Trim A201615

(Required by AD 2020-24-02)
The STAB OUT OF TRIM light illuminates for the following conditions:
On the ground: A partial failure of a Flight Control Computer.
In-flight: the autopilot does not set the stabilizer trim correctly.

If on ground, do not takeoff. End of procedure.

In flight, during large changes in trim requirements, the STAB OUT OF TRIM light may illuminate momentarily. If the stabilizer is trimming, continue normal operation; end of procedure.

In flight, if the stabilizer is not trimming, hold control column firmly. Disengage autopilot. Disengage autothrottle if engaged. Use main electric stabilizer trim as needed.

If the stabilizer responds to electric trim inputs, do not re-engage the autopilot or autothrottle; end of procedure.

If the stabilizer does not respond to electric trim inputs, accomplish the Stabilizer Trim Inoperative procedure.

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Stabilizer Trim Inoperative A211940

(Alternative Method of Compliance (AMOC) to AD 2020-24-02)
Loss of electric trim through the main electric stabilizer trim switches, or when directed by the Stabilizer Out of Trim procedure.

If autothrottle is engaged, disengage. Do not re-engage the autothrottle.

Place both STAB TRIM cutout switches to CUTOUT. The autopilot is not available. Trim stabilizer manually. A two-pilot effort may be used and will not cause system damage.
Notes:
1. Reducing airspeed reduces airloads on the stabilizer which can reduce the effort needed to manually trim.
2. If the failure could be due to ice accumulation, descend to a warmer temperature and attempt again to trim manually.

If the stabilizer can be trimmed manually, anticipate trim requirements. If the stabilizer cannot be trimmed manually, expect higher than normal elevator forces during approach and landing. The thrust reduction at flare will cause a nose down pitch.

Plan a flaps 15 landing. Set Vref 15+10 knots.
Note: The maximum wind additive should not exceed 5 knots. Check the non-normal landing distance tables in an approved source.

In addition to the normal descent, approach and landing checklists, complete the following deferred items:

Review the normal go-around procedure. During a go-around, advance thrust to go-around smoothly and slowly to avoid excessive pitch-up.
Establish landing configuration early on final approach.

Standby Rudder On A163052

If the standby rudder is ON and both flight control low pressure lights are extinguished, avoid large or abrupt rudder control pedal inputs.

Takeoff (N) A163052

Guidance material for takeoff procedures and flap retraction speed schedules are shown in Chapter 4, Performance.

Window Heat (N) A170119

Except as indicated in Window Heat Inoperative section under Certificate Limitations chapter, window heat must be ON to the No. 1 (FWD) and No. 2 (side) windows for all normal flight operations, and must be turned on to these windows 10 minutes prior to takeoff.

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Wing Thermal Anti-Ice Failure A163052

In flight, if either or both valves fail in the closed position in icing conditions, exit the icing conditions. This is indicated by either or both L VALVE or R VALVE lights continuously illuminating amber with the Wing Anti-Ice switch ON. Turn the switch OFF and avoid icing conditions.

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General A180643

This chapter includes information peculiar to the operation of the Boeing 737 about systems that may assist an operator in complying with operational regulations, statements of demonstrations during certification, statements required by Advisory Circulars in effect at the time of certification, or other information deemed necessary to aid in proper operation of the airplane

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Air Traffic Control Radar Beacon System A180643

Mode S Transponder (If Installed)

Mode A, C and S functions with respect to TCAS have been demonstrated for proper operation.

Selecting Standby mode on the ATC/TCAS control panel will disable all of the following functions:

1. All ATC transponder modes (Modes A/C/S)
2. ADS-B Out
3. TCAS II

Elementary Surveillance - Version 0 ATC Transponder (If Installed)

Elementary Surveillance functions have been demonstrated for proper operation.

Elementary Surveillance - Version 2 ATC Transponder (If Installed)

The installed transponder system is able to respond to interrogations in Mode A, C and S (Mode S Elementary Surveillance).

Enhanced Surveillance - Version 0 ATC Transponder (If Installed)

Enhanced Surveillance functions have been demonstrated for proper operation.

Enhanced Surveillance - Version 2 ATC Transponder (If Installed)

The installed transponder system is able to respond to interrogations in Mode A, C and S (Mode S Enhanced Surveillance).

Version 2 ATC Transponder Failure Modes (If Installed)

ADS-B Out Transponder failure is indicated by the following:

Transponder (TCAS) control panel “XPNDR FAIL” or “ATC FAIL” amber light illuminated on the P8 Aft Aisle stand and “TCAS OFF” on Navigation Displays.

ADS-B Out function failure (loss of position source from the Global Positioning System (GPS)) is indicated by the following:

Transponder (TCAS) control panel “XPNDR FAIL” or “ATC FAIL” amber light illuminated on the P8 aft aisle stand, FMC scratch pad message “GPS-L INVALID” or “GPS-R INVALID” text displayed on MCDU and illumination of the associated amber “FMC” lights on P1-3 and P3-1.

Extended Squitter/ADS-B Out - Version 0 (If Installed)

The following is applicable only if the extended squitter is installed and the Multi Mode Receivers (MMRs) are interfaced with the ATC Transponders.

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Extended Squitter transmission has been demonstrated for proper operation per EASA AMC 20-24 for broadcast of ADS-B related position information, with the following exceptions.

1. The extended squitter transmission system does not take into account the system's uncompensated latency into its transmitted horizontal quality indicator value.

2. The extended squitter transmission system does not base the transmitted horizontal quality indicator solely on the integrity information from the horizontal position source. The horizontal quality indicator value may be encoded based on the horizontal position source's accuracy quality information.

3. The extended squitter transmission system does not have the capability to transmit the discrete emergency codes (7500, 7600, 7700). Instead the extended squitter system will transmit a generic emergency indicator when any emergency code is dialed into the control panel.

Extended Squitter/ADS-B Out Version 2 (If Installed)

The installed ADS-B OUT system has shown to meet the equipment requirements of 14 CFR 91.227 (1090 MHz Extended Squitter ADS-B OUT).

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Aircraft Communications Addressing and Reporting System (ACARS) A232344

Applicable to ASN(s) : 42831, 42832, 42833, 42835, 62897, 62903, 62904, 62907, 62911, 62922, 62923, 62926, 62928, 62931, 63971, 63972, 63973, 64992, 66350

The following is applicable when using (a) ACARS Controller Pilot Datalink Communications (CPDLC) or (b) ACARS Protective Mode - Controller Pilot Datalink Communications (PM-CPDLC):

The FAA has approved the datalink system and applications to the criteria contained in AC 20-140/20-140A.

The aircraft ATC datalink system has been demonstrated to comply with the applicable safety and performance requirements of EUROCAE ED-120, and the interoperability requirements of ED110.

For installations other than PM-CPDLC Rockwell Collins Communication Management Unit (CMU) part number 822-1239-151 or Honeywell CMU part number 965-0758-006: The aircrew shall ascertain the accuracy and validity of every altimeter setting obtained via CPDLC datalink before using the setting. CPDLC shall not be the only means of verifying altimeter settings.

The Aircraft ATC Datalink System is intended for the following datalink services:

Datalink Initiation Capability (DLIC) enabling initial contact between the aircraft and the ATC unit that supports data communications, to accurately determine the identity of the aircraft, and to ensure compatibility of aircraft equipage with ATC. It is a prerequisite to any other operational datalink services.

ATC Communication Management (ACM) providing the necessary information to the aircraft to enable transfer of frequencies for both voice and data communications, either within the same sector, between two sectors or between two ATC centers. ATC Clearances (ACL) enabling uplink of a set of clearance and information messages and downlink of pilot responses and requests.

ATC Microphone Check (AMC) enabling the controller to send a message to datalink equipped aircraft (of appropriate interoperability) to request a stuck microphone check.

Departure Clearance (DCL) enabling the request and the delivery of departure information and clearance.

Downstream Clearance (DSC) enabling the request and the delivery of clearance with a downstream ATC Center (i.e. oceanic clearance)

D-ATIS enabling the request and delivery of ATIS via datalink.

Aircraft Communications Addressing and Reporting System (ACARS) A222195

Applicable to ASN(s) : 42825, 42826, 42827, 42828, 42829, 42830, 42831, 42832, 42833, 42834, 42835, 42836, 43361, 62897, 62903, 62904, 62907, 62911, 62922, 62923, 62926, 62928, 62931, 63564, 63970, 63971, 63972, 63973, 64992, 64993, 66350

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Section Aircraft Communications Addressing and Reporting System (ACARS), continued from previous page A222195

Installation of the ACARS system does not include operational approval for use of the ACARS system. The Operator must receive approval from their Civil Aviation Authority (CAA), in accordance with DO-296 or equivalent, prior to use of ACARS.

The following is applicable if ACARS Airline Operation Control (AOC) Datalink is installed:

When using the ACARS Airline Operation Control (AOC) Datalink, the following procedures are one means which may be used to verify Pre-Departure Clearance, Digital-Automatic Terminal Information Service, Oceanic Clearances, Weight and Balance and Takeoff Data messages transmitted over ACARS:

Pre-Departure Clearance:
The flight crew shall manually verify (compare) the filed flight plan versus the digital pre-departure clearance and shall initiate voice contact with Air Traffic Control if any question/confusion exists between the filed flight plan and the digital pre-departure clearance.

Digital-Automatic Terminal Information Service:
The flight crew shall verify that the D-ATIS altimeter setting numeric value and alpha value are identical. If the D-ATIS altimeter setting numeric value and alpha value are different, the flight crew must not accept the D-ATIS altimeter setting.

Oceanic Clearances:
The flight crew shall manually verify (compare) the filed flight plan versus the digital oceanic clearance and initiate voice contact with Air Traffic Control if any questions/confusion exists between the filed flight plan and the digital oceanic clearance.

Weight and Balance:
The flight crew shall verify that the Weight and Balance numeric and alphabetic values are identical. If the Weight and Balance numeric and alphabetic values are different, the flight crew must not accept the Weight and Balance data.

Takeoff Data:
The flight crew shall verify that the Takeoff Data numeric and alphabetic values are identical. If the Takeoff Data numeric and alphabetic values are different, the flight crew must not accept the Takeoff Data message.

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Electrical and Electronics A163052

Air Data Systems

The following systems (if installed) receive airspeed, altitude, angle of attack and/or total air temperature information from the air data system:

Flight Management System (autopilot, flight director, altitude alerting, auto throttle and flight management computer system)
Altitude reporting
Cabin Pressurization
Max Display System for primary flight instruments (altimeters, Mach/airspeed and vertical speed indicators)
Ground Proximity Warning System
Windshear System
Inertial Navigation Reference System
Mach Trim
Speed Trim
Overspeed Warning Switches
Stall Management Yaw Damper
VMO indication
Weather Radar
Rudder and Rudder Trim Control System
Fuel Systems
Spoiler Control Electronic
ATC Transponder
High Lift System
Air Conditioning System
Flight Controls Computer
Propulsion
Cabin Pressure Control System
Elevator and Tab Control Systems
Environmental Control Systems
Bleed Air System

Lack of this information may cause faulty operation of these systems.

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Extended Operations A180764

Applicable to ASN(s) : 42825, 42826, 42827, 42828, 42829, 42830, 42835, 62897, 62903, 62904, 62907, 62911, 62922, 62923, 62926, 62928, 62931, 63971, 63972, 63973

The type design reliability and performance of this airplane/engine combination has been evaluated in accordance with 14 CFR 25.1535/EASA Certification Specification (CS) 25.1535 and Acceptable Means of Compliance (AMC) 20-6, and found suitable for up to 180 minutes extended operations (ETOPS) when the configuration, maintenance and procedure standards contained in the 737 MAX ETOPS Configuration, Maintenance, and Procedures (CMP) document D044A032, are met. The actual maximum approved diversion time for this airplane may be less based on its most limiting system time capability. This finding does not constitute approval to conduct ETOPS.

System Time Capabilities:

The Class C cargo fire suppression system is limited to 195 minutes.

No other airplane systems are time limited.

The APU must be started and running prior to reaching a point farther than 60 minutes from an adequate alternate airport and kept running for the duration of the ETOPS flight.

Extended Operations A191022

Applicable to ASN(s) : 42831, 42832, 42833, 42834, 42836, 43361, 63564, 63970, 64992, 64993, 66350

The type design reliability and performance of this airplane/engine combination has been evaluated in accordance with 14 CFR 25.1535/ EASA Certification Specification (CS) 25.1535 and Acceptable Means of Compliance (AMC) 20-6, and found suitable for up to 180 minutes extended operations (ETOPS) when the configuration, maintenance and procedure standards contained in the 737 MAX ETOPS Configuration, Maintenance, and Procedures (CMP) document D044A032, are met. The actual maximum approved diversion time for this airplane may be less based on its most limiting system time capability. This finding does not constitute approval to conduct ETOPS.

System Time Capabilities:

The Class C cargo fire suppression system is limited to 195 minutes.

No other airplane systems are time limited.

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Flight Deck Auxiliary Power A180643

The Flight Deck 110V AC, 400 Hz Auxiliary Power outlet has been demonstrated to a maximum normal power level of 150 watts.

The two Personal Electronic Device (PED) 110V AC, 60 Hz outlets (if installed) are capable of a maximum normal power level of 550 watts between both of them.

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Flight Management Computer System (FMCS) A180376

FMCS Software update U13 and later versions.

The FMCS has been demonstrated to meet the requirements of FAA Advisory Circular (AC) 20-138D for a multi-sensor area navigation system when operated with Global Position System (GPS) or radio updating. When operated in this configuration, the FMCS may be used for RNP and RNAV enroute operations, terminal area operations and instrument approach navigation (excluding ILS, LOC, LOC-BC, LDA, SDF, GLS, and MLS).

The FMCS has been demonstrated to meet the requirements of FAA AC 20-138D for Baro-VNAV vertical navigation (Baro-VNAV) for enroute operations, terminal area operation and instrument approaches (excluding GLS & ILS glide slope approach procedures).

The FMCS has been shown to meet the requirements of FAA AC 90-101A Appendix 2, AC 120-29A Appendix 2, AC 20-138D, EASA AMC 20-26 and AMC 20-27 for RNP and RNAV instrument approaches (excluding ILS, GLS glide slope approach procedures) using LNAV, VNAV, or FAC, G/P guidance coupled with the autopilot or flight director and using the ND map with the Primary Flight Display (PFD) lateral and vertical deviations. EASA AMC 20-26 requires selection of the NPS option. FAA AC 90-101A does not require selection of the NPS option.

The FMCS has been shown to meet the requirements of FAA Advisory Circular (AC) 25-15 for long range navigation with the following equipment operational at departure:

(1) Two FMC, and
(2) Two CDU, and
(3) Two IRU in NAV mode.

The FMCS has been shown to meet the requirements for one of two independent navigation systems for RNAV operations (per AC 90-96A, AC 90-100A, EASA AMC 20-26, AMC 20-27) with the following equipment operational at departure (unless other appropriate procedures are used):

(1) One FMC, and
(2) One CDU, and
(3) One IRU in NAV mode;
(4) Either One Global Positioning System (GPS); or
One Distance Measuring Equipment (DME), if authorized for intend RNAV operation/ procedure;
(5) One VHF Omni-directional radio (VOR); or
One Automatic Direction Finder (ADF) radio (if required by the route or operation).

The FMCS has been shown to meet the requirements for two of two independent navigation systems for RNP and RNAV operations (per AC 90-96A, AC 90-100A, AC 90-101A Appendix 2, AC 90-105A, AC 120-29A Appendix 2, JAA TGL-10, EASA AMC 20-26, AMC 20-27) with the following equipment operational at departure (unless other appropriate procedures are used):

(1) Two FMC, and
(2) Two CDU, and
(3) Two IRU in NAV mode.

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(4) Two sensors capable of complying with the RNP. Example sensors are GPS, DME, VOR/DME, if authorized for intend RNP operation/procedure.

For a flight operation without a RNP, the FMCS ND Map has been demonstrated for enroute and terminal area flight operations and instrument approach navigation (excluding ILS, LOC, LOC-BC, LDA, SDF, GLS and MLS).

For flight operations in airspace designated for RNP, without NPS option selected, the demonstrated RNP is as follows:

Minimum Demonstrated RNP Flight Operations Versus Mode of Flight		
Mode of Flight	**FMC GPS Enabled**	**FMC GPS Not Installed/ Enabled**
LNAV with A/P Engaged	0.10 NM	0.10 NM
LNAV with F/D	0.10 NM	0.20 NM
Manual Control with Map display only	0.64 NM	0.72 NM

When using the MAP for manual control for RNP operations of 2.00 or less, down to the minimum demonstrated RNP, using a MAP scale of 10nm or less has been shown to assure acceptable lateral path track performance.

For flight operations in airspace designated for RNP, with NPS option selected the demonstrated RNP is as follows:

Minimum Demonstrated RNP Flight Operations Versus Mode of Flight		
Mode of Flight	**FMC GPS Enabled**	**FMC GPS Not Installed/ Enabled**
LNAV with A/P Engaged	0.10 NM	0.21 NM
LNAV with F/D	0.10 NM	0.22 NM

The demonstrated RNP capabilities are predicated upon the assumptions, definitions, requirements, and analysis in FAA approved Boeing Document D6-39067-3, "RNP Capability of FMC Equipped 737, Generation 3", Revision New or later FAA approved version.

Caution: Boeing Vertical RNP has been demonstrated only for situational awareness with respect to a barometric vertical path and does not provide performance benefit on the final approach segment beyond what barometric VNAV provides. Boeing Vertical RNP is not based on ICAO or any other internationally harmonized standard or operational concept that defines accuracy and containment requirements.

RNP flight operations are subject to navaid coverage and/or GPS satellite availability (for configurations with GPS installed) for the selected route.

For configurations with GPS installed, GPS updating must be disabled for approach operation when operating outside the United States National Airspace, if the FMC database and charts are not referenced to WGS-84 reference datum, unless other appropriate procedures are used.

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Fuel management and range calculation values presented by the FMCS have not been evaluated by the FAA.

These FMCS demonstrations do not constitute operational approval.

The FAA has approved the data link system and applications to the criteria contained in AC 20-140. This approval was based on the safety and interoperability requirements contained in the FAA approved Boeing Documents listed below:

D926A106 "737 Air Traffic Services Systems Requirements and Objectives" for Generation 3, U13 and later versions.

Flight Management Computer System (FMCS) A201609

The following is applicable upon installation of FMCS Software update U14 and later versions with TEMP COMP option enabled:

The FMCS Barometric Altitude Temperature Compensation has been demonstrated to meet the requirements of RTCA/DO-236C Appendix H and FAA AC 20-138D, Section 11-1.b to enable baro-VNAV operations outside of the temperature limitations published on approach procedure charts.

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Inertial Reference System (IRS) A180254

Ground alignment of the IRS has been demonstrated to be satisfactory between 78.25 degrees North/South latitudes with a variable alignment period. The ISDU can be used to monitor time remaining in the alignment period.

CONFIGURATION	IRS data in NAV Mode
ADIRU with 2015 Magnetic Variation table activation	In the NAV (navigation) mode the IRS will not provide valid magnetic heading and track above 82 degrees North and below 82 degrees South latitudes, and in the region above 73 degrees North between 80 degrees West and 170 degrees West longitude, and the region below 60 degrees South between 120 degrees East and 160 degrees East.

Compliance with the design requirements for magnetic heading and magnetic track angle accuracy within the above boundaries must be maintained for the continued airworthiness of the airplane as these accuracies degrade over time relative to the earth's ever changing geomagnetic reference field. To maintain continuous world-wide compliance, new Magnetic Variation (MagVar) tables are developed and certified for production and retrofit into the IRS (ADIRU) every 10 years. Failure to maintain the installed IRS MagVar table in compliance can result in operational limitations (see Certificate Limitations/Electronics-Inertial Reference System (IRS) section of this AFM). Information for determining which specific Epoch MagVar table is installed in the IRS (ADIRU), and geographic maps which show the location and magnitude of the specific MagVar table errors relative to current and projected National Oceanic and Atmospheric Administration (NOAA) World Magnetic Model (WMM) data is provided in Boeing document D6-85262, "Guidance for Airplane Flight Operations Using Inertial Reference Systems with Magnetic Variation Tables", revision A or later.

During flight, the IRS navigation maybe automatically recovered by align-in-motion if GNSS data is available.

During flight, the IRS attitude may be automatically recovered.

Practical Operational Flight Envelope (POFE) A201490

The Practical Operational Flight Envelope describes the boundaries where compliance to the controllability and maneuverability requirements of 14 CFR Part 25 are met with a single failure of any Stability Augmentation, Automatic, or Power-Operated System.

For Speed Trim System Inoperative, the controllability and maneuverability requirements of 14 CFR Part 25.672(c)(2) are met Flaps Up at load factors up to +1.8g or prior to initial buffet, at speeds up to Vmo/Mmo. Beyond this envelope, column forces may be slightly lower than normal.

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Reduced Vertical Separation Minimum (RVSM) A163052

The type design reliability and performance of this airplane/engine combination has been evaluated in accordance with FAA Memorandum 91-RVSM, dated 14 March 1994, "Interim Guidance for Approval of Aircraft for Reduced Vertical Separation Minima (RVSM) Flight" and is qualified for operation in RVSM airspace. This finding does not constitute approval to conduct reduced vertical separation minimum operations.

Traffic Alert and Collision Avoidance System (TCAS) A211679

(If installed)

Immediate compliance with a TCAS II/ACAS resolution advisory (RA) is necessary unless the pilot flying considers it unsafe to do so, or unless there is an obvious TCAS II/ACAS system failure.

CAUTION: Once an RA has been issued, safe separation could be compromised if current vertical speed is changed, except as necessary to comply with the RA. This is because TCAS II/ACAS-to-TCAS II/ACAS coordination may be in progress with the intruder aircraft, and any change in vertical speed that does not comply with the RA may negate the effectiveness of the other aircraft's compliance with the RA.

Note: The consequences of not following an RA may result in additional RAs in which aural alert and visual annunciations may not agree with each other.

The pilot should not initiate evasive maneuvers using information from the traffic display only. The pilot may maneuver, based on a traffic advisory (TA) displayed on the ND, if the traffic is also visually acquired and it is considered safe to do so. However, while climbing or descending, a modest change in vertical speed based on traffic display information is not considered evasive maneuvering.

Following a TCAS II/ACAS "clear of conflict" advisory, the pilot should expeditiously return to the applicable ATC clearance unless otherwise directed by ATC.

All RA and TA aural messages are inhibited at a radio altitude of less than 1100 feet above ground level (AGL) climbing, and less than 900 feet AGL descending.

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Weather Radar A163052

The weather mode should be used for evaluating weather conditions and should be the basis for all decisions regarding maneuvering the airplane to avoid weather. Turbulence information presented by the weather radar may be used as additional information to aid in decisions to maneuver the airplane to avoid weather related threats.

Clear air turbulence (CAT) conditions are not detectable with the weather radar system.

The WXR system provides variable gain allowing the gain to be adjusted below the calibrated level (CAL or AUTO). When below the calibrated gain level, weather returns may be shown at reduced levels or even removed from the display. When not performing temporary weather analysis, the gain selection should be in calibrated position.

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Authorized Alternate Performance Capability A232381

Applicable to ASN(s) : 42825, 42826, 42827, 42828, 42829, 42830, 42831, 42832, 42833, 42834, 42835, 42836, 43361, 62897, 62903, 62904, 62907, 62911, 62922, 62923, 62926, 62928, 62931, 63564, 63970, 63971, 63972, 63973, 64992, 64993, 66350

Although additional alternate performance may be available in AFM-DPI, only the alternate performance indicated below is authorized for use.

Alternate EEC Mode
Ten Minute Takeoff Thrust
Gear Down Dispatch
Forward CG Limit
Wet Skid-Resistant Landing (WSRL)

Automatic Wheel Brakes A163052

The automatic wheel brake system is designed to improve the consistency and smoothness of the landing roll-out by modulating brake pressure to achieve a selected deceleration rate. Use of the system is optional and does not relieve the pilot of the responsibility to assure a safe stop within the available runway length.

The automatic wheel brake system is triggered at touchdown by air/ground signals and main gear wheel spin-up. Therefore, a delay in lowering the nose gear will not delay application of the brakes. Maximum pedal braking achieves a higher deceleration rate (shorter stopping distance) than the MAX setting, provided sufficient runway friction is available. The use of reverse thrust is demonstrated to have no adverse effects on autobrake stopping distances.

Stopping distances that can be achieved on wet runways depend on the friction characteristics, or slipperiness, of the particular runway. The wet runway stopping distance that is achieved for a given setting will be the longer of (1) the distance produced by the fixed deceleration of the setting or (2) the distance produced by the runway friction available. A higher setting (MAX rather than 3, or 3 rather than 2) will produce shorter stopping distances on wet runways if friction characteristics are such to permit the higher deceleration. The use of reverse thrust will decrease wet runway stopping distance when the deceleration is limited by runway friction characteristics. Extremely wet and slippery conditions can result in longer distances than indicated.

Note: Autobrakes are not allowed with antiskid inoperative.

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Buffet Onset Characteristics A170073

Buffet onset occurs when the airflow starts to separate from the wing. This characteristic is a function of angle of attack and Mach number.

From the Normal Cruise Altitude Limits Due to Buffet charts, it is possible to determine the altitude, low-speed, and high-speed limits that will still provide maneuver margins of either 0.3 g or 0.2 g before buffet onset occurs.

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Definitions - Airspeeds A180643

All airspeeds and Mach values in AFM-DPI are based on the primary airspeed system and assume zero instrument error.

Static source position error, ΔV_p, is the error due to the location of the static port within the airplane pressure field.

Indicated Airspeed, V_I, IAS- Airspeed reading, as displayed in the airplane.

Calibrated Airspeed, V_C, CAS - Indicated airspeed corrected for static source position error ($V_C = V_I + \Delta V_p$).

Equivalent Airspeed, V_e, EAS - Calibrated airspeed corrected for compressibility ($V_e = V_C - \Delta V_C$).

True Airspeed, V_T, TAS - Equivalent airspeed corrected for atmospheric density effects ($V_T = V_e/\sqrt{\sigma}$).

True Mach Number, M - Machmeter reading, as installed in the airplane, corrected for static source position error

Critical Engine Failure Speed, V_{EF} - The speed at which, if the most critical engine fails, the engine failure is recognized at V_1, one second later.

Takeoff Decision Speed, V_1 - The speed from which a decision to continue the takeoff, following a failure of the critical engine at V_{EF}, results in a takeoff distance to a height of 35 feet at V_2 speed that will not exceed the usable takeoff distance and the speed at which the first action is taken to bring the airplane to a full stop that will not exceed the accelerate-stop distance available. V_1 must not be less than the minimum V_1 for control on the ground, $V_{1\,(MCG)}$, or greater than the rotation speed, V_R, or greater than the maximum brake energy speed, V_{MBE}.

Rotation Speed, V_R- The speed at which rotation from the three-point attitude to the takeoff attitude is initiated. The scheduled rotation speed must be equal to or greater than 1.05 V_{MCA} and V_1.

Maximum Brake Energy Speed, V_{MBE} - The highest takeoff decision speed from which the airplane may be brought to a stop without exceeding the maximum energy absorption capability of the brakes. Maximum Brake Energy Speed is compared in takeoff planning to V_1 speed.

Takeoff Safety Speed, V_2 - The target speed to be attained at the 35 foot height, assuming recognition of an engine failure at or after V_1 . This speed must be at least 1.13 times the reference stall speed in the takeoff configuration.

Air Minimum Control Speed, V_{MCA} - The minimum flight speed at which the airplane is controllable with a maximum of 5 degrees bank when the critical engine suddenly becomes inoperative with the remaining engine at takeoff thrust.

Continued on next page

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Section Definitions - Airspeeds, continued from previous page A180643

Minimum V_1 for Control on the Ground, $V_{1\,(MCG)}$ - The minimum takeoff decision speed, V_1, at which, when the critical engine suddenly becomes inoperative at V_{EF} with the remaining engine at takeoff thrust, it is possible to control the airplane with primary aerodynamic controls alone and continue the takeoff.

Minimum Control Speed for Landing, V_{MCL} - The minimum control speed with the airplane configured for approach at which the airplane is controllable with a maximum of 5 degrees bank when the critical engine suddenly becomes inoperative with the remaining engine at go-around thrust.

Landing Speed - The minimum speed at the 50 foot height in a normal landing. This speed must be at least 1.23 times the reference stall speed in the landing configuration.

Reference Landing Speed, V_{REF} - This speed is equal to the landing speed in the flaps 40 landing configuration.

Design Maneuvering Speed, V_A - The speed above which full application of rudder, ailerons or elevator, or maneuvers involving angles of attack near stall, should be avoided.

Definitions - Brake Assembly - Category A180643

Brake part numbers in the same category for the same model airplane produce equal braking performance. This does not allow interchangeability of the brakes themselves.

MODEL	CATEGORY	BOEING PART NO. [1]
737-8/-8200	B	S277A016-312, -510

[1] Part Number:

S277A016-510 - Messier-Bugatti Carbon Brakes

S277A016-312 - Goodrich Carbon Brakes

Definitions - Icing Conditions A163052

Icing conditions exist when the OAT on the ground and for takeoff, or TAT inflight, is 10 degrees C (50 degrees F) or below and visible moisture in any form is present (such as clouds, fog with visibility of one mile or less, rain, snow, sleet and ice crystals).

Icing conditions also exist when the OAT on the ground and for takeoff is 10 degrees C (50 degrees F) or below when operating on ramps, taxiways or runways where surface snow, ice, standing water or slush may be ingested by the engines or freeze on engines, nacelles or engine sensor probes.

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Definitions - Takeoff Flight Path A170249

The takeoff flight path begins 35 feet above the takeoff surface at the end of the takeoff distance and extends to a point where the airplane is at least 1500 feet above the takeoff surface and has achieved the enroute configuration and final climb speed.

Climb Gradient - The ratio, expressed as a percentage, of the change in geometric height divided by the horizontal distance traveled in a given time. Gross gradient is the actual calculated performance of the airplane under specified conditions, while Net gradient is the gross gradient reduced by an increment specified in the regulations.

Gross Height - The geometric height attained at any point in the takeoff flight path using gross climb performance. Gross height is used for calculating actual pressure altitude at which obstacle clearance procedures and wing flap retraction are initiated, and level-off height scheduled.

Net Height - The geometric height attained at any point in the takeoff flight path using net climb performance. Net height is used to determine the net flight path which must clear any obstacle by at least 35 feet to comply with the regulations.

Reference Zero - A point on the runway or clearway plane at the end of the takeoff distance and 35 feet below the flight path to which the height and distance coordinates of other points in the takeoff flight path are referred.

First Segment - Extends from the end of the takeoff distance to the point where the landing gear is assumed to be fully retracted, using takeoff thrust and takeoff flaps at a constant V_2 speed.

Second Segment - Extends from the gear up point to a gross height of at least 400 feet, using takeoff thrust and takeoff flaps at a constant V_2 speed.

Third Segment - The horizontal distance required to accelerate, at constant altitude using takeoff thrust, to the final climb speed while retracting flaps in accordance with the recommended speed schedule.

Maximum Level-Off Height - The maximum height at which the third segment can be completed before the time limit on the use of takeoff thrust expires.

Final Takeoff Segment - Extends from the end of the third segment to a gross height of at least 1500 feet, with flaps-up, maximum continuous thrust and at final climb speed.

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Definitions - Takeoff Path A163052

The takeoff path assumes failure of the most critical engine at V_{EF} and extends from a standing start to a point where the airplane is at least 1500 feet above the takeoff surface and has achieved the enroute configuration and final climb speed.

The takeoff path is divided into segments representing distinct changes in airplane configuration, airspeed, and engine thrust as defined below. Some of these definitions are abbreviated versions of those appearing in 14 of CFR Parts 1 and 25; the appropriate regulations should be consulted for complete details.

Clearway - An obstacle-free area beyond the takeoff runway which can be used as a part of the takeoff distance available.

Stopway - An area beyond the takeoff runway capable of supporting the airplane in an aborted takeoff which can be used as part of the accelerate-stop distance available. To be applicable for use with this manual, the surface characteristics of the stopway must not differ substantially from those of the runway.

Takeoff Distance Required - The greater of:

1. The distance to takeoff and climb to a height of 35 feet (dry runway[1]) with a failure of the critical engine at V_{EF};

or

2. 115 percent of the distance to takeoff and climb to a height of 35 feet with all engines operating.

Takeoff Run Required - The greater of:

1. The distance to takeoff and climb to a point equidistant between lift off and the 35 foot height point (dry runway[2]) with a failure of the critical engine at V_{EF};

or

2. 115 percent of the distance to takeoff and climb to a point equidistant between lift off and the 35 foot height point with all engines operating.

Accelerate-Stop Distance Required - The greater of:

1. The sum of the distances required to accelerate with all engines operating and come to a complete stop assuming a critical engine failure at V_{EF}

or

2. The sum of the distances required to accelerate with all engines operating and come to a complete stop with all engines still operating.

Notes:

1. On a wet runway, the height requirement with a failed engine is 15 feet.
2. On a wet runway, the takeoff run required is the distance to takeoff and climb to 15 feet with a failure of the critical engine at V_{EF}.

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Definitions - Temperature A163052

ISA - International Standard Atmosphere, as accepted by the International Civil Aviation Organization.
OAT - Outside Air Temperature - the free air static (ambient) temperature.
SAT - Static Air Temperature - outside air (ambient) temperature as computed from TAT.
TAT - Total Air Temperature - static air temperature plus adiabatic compression (ram) rise as indicated on the Total Air Temperature indicator.

Definitions - Wind Velocity A163052

The actual wind velocity at a 10 meter height reported from the tower and corrected to a headwind or tailwind component parallel to the flight path.

Flap Retraction Speed Schedule A163052

Maximum level-off heights, Third Segment distances and Final Segment climb performance are based upon retracting the wing flaps during Third Segment acceleration using the schedule below. This schedule is recommended for all flap retraction operations.

During acceleration, select flap positions at the following initiation speeds:

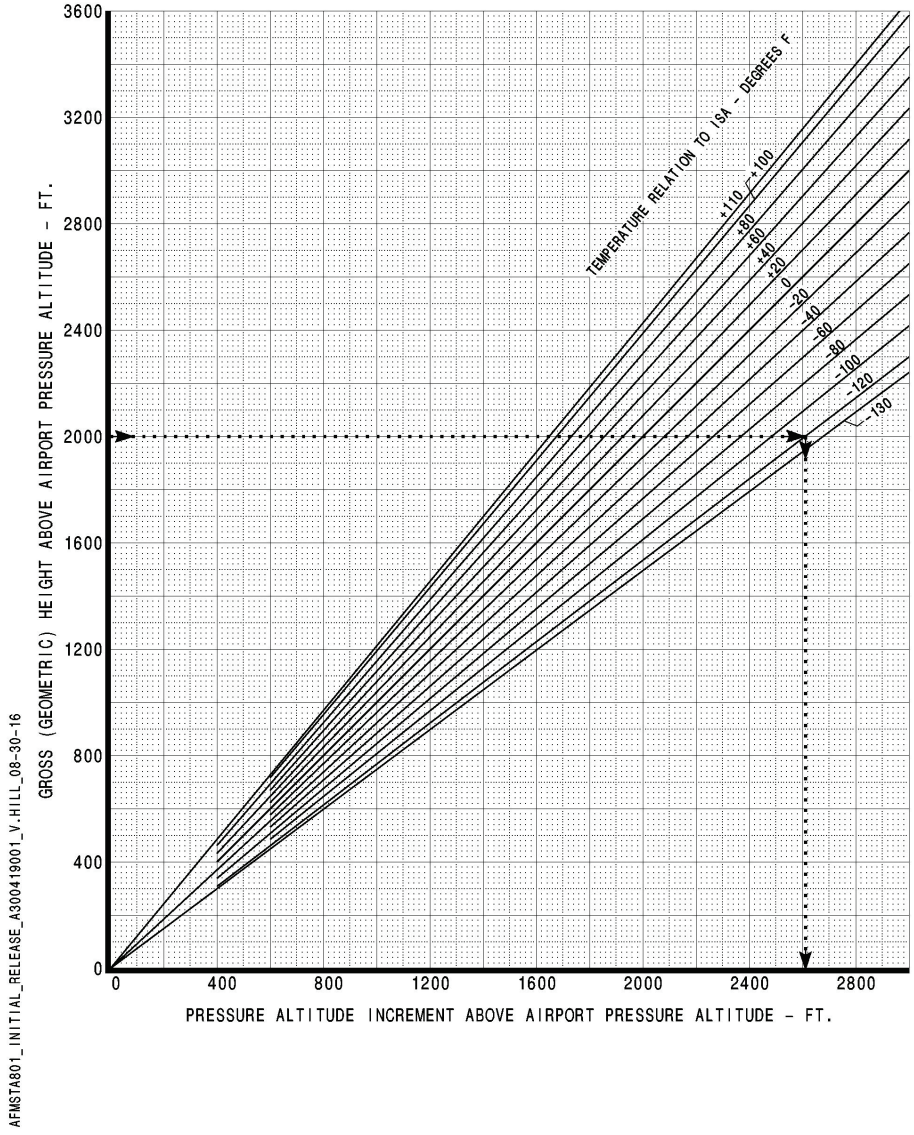
INITIAL FLAP POSITION	SELECT FLAP POSITION	INITIATION SPEED, KNOTS
25	15 5 1 0	V_2 + 15 V_{REF40} + 20 V_{REF40} + 30 V_{REF40} + 50
15	5 1 0	V_2 + 15 V_{REF40} + 30 V_{REF40} + 50
10	5 1 0	V_2 + 15 V_{REF40} + 30 V_{REF40} + 50
5	1 0	V_2 + 15 V_{REF40} + 50
1	0	V_{REF40} + 50
Final Segment Climb Speed:		V_{REF40} + 70

When flaps are being retracted at a constant altitude, begin climbing when the above final segment climb speed is achieved, maintaining takeoff thrust setting until flaps are completely retracted.

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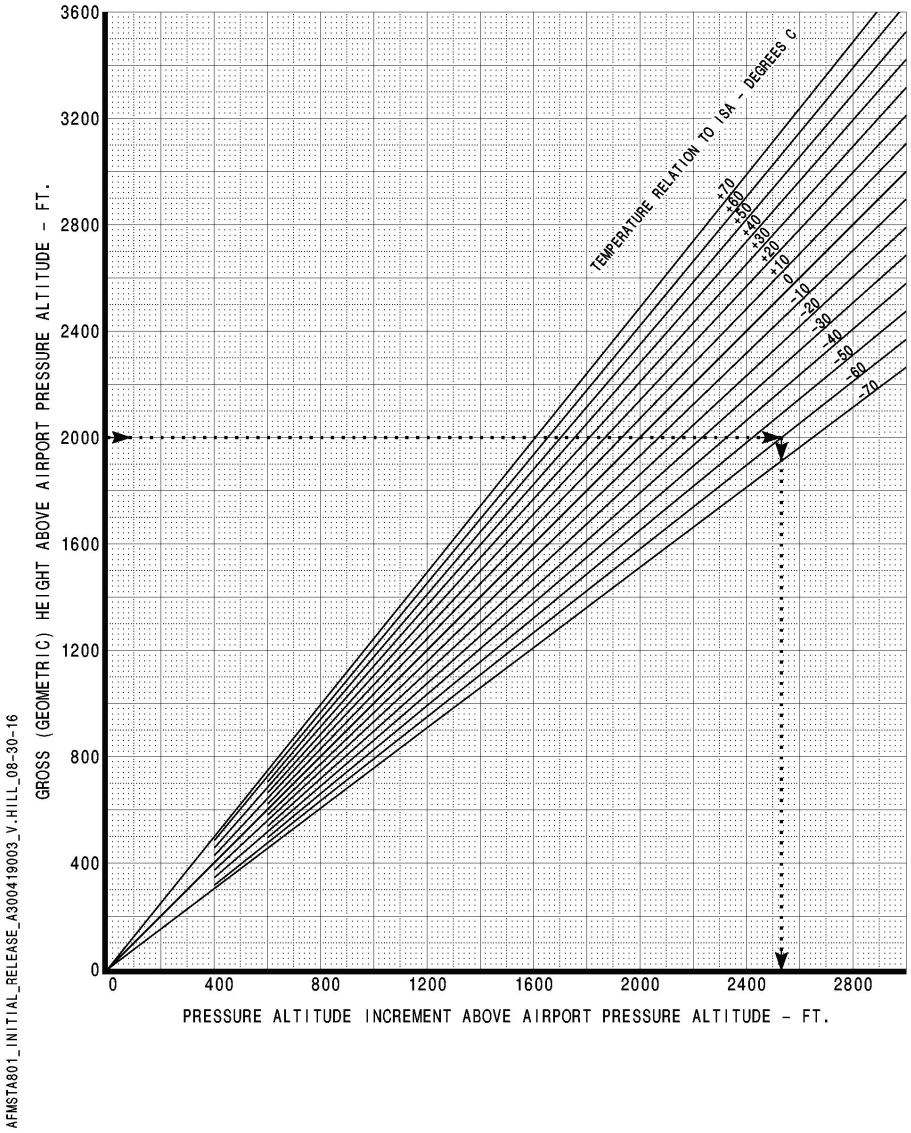
Gross Height - Pressure Altitude Conversion A180643

When ambient temperatures are above or below ISA, the altitude change displayed on a pressure altimeter can differ significantly from the geometric heights calculated for the flight path. This chart relates the geometric height increment to a pressure altitude increment so that flight path procedures can be developed with reference to altimeter indications.



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Maximum Crosswind A163052

The maximum demonstrated crosswind component as reported at a 10 meter height for takeoff is 35 knots and for landing is 32 knots. This component is not considered to be limiting on a dry runway with all engines operating.

Maximum Quick Turnaround Weight A170091

Use the following procedures:

1. For all brake categories: After landing at weights exceeding those determined from the maximum quick turnaround option in AFM-DPI, wait at least the Minimum Wait Time indicated in the table below and check that wheel thermal plugs have not melted before making a subsequent takeoff.

2. For all brake categories: If a Brake Temperature Monitoring System (BTMS) is installed: No sooner than 10 and no later than 15 minutes after parking, check the BRAKE TEMP light. If the BRAKE TEMP light is not on, no ground waiting time is required. If the Brake TEMP light is on, do not dispatch until (a) at least the Minimum Wait Time indicated in the table below has expired after landing, or (b) the BRAKE TEMP light is off and all BTMS readings on the Systems Display are below 3.0 (carbon brake part #S277A016-x). Check that wheel thermal plugs have not melted before making a subsequent takeoff .

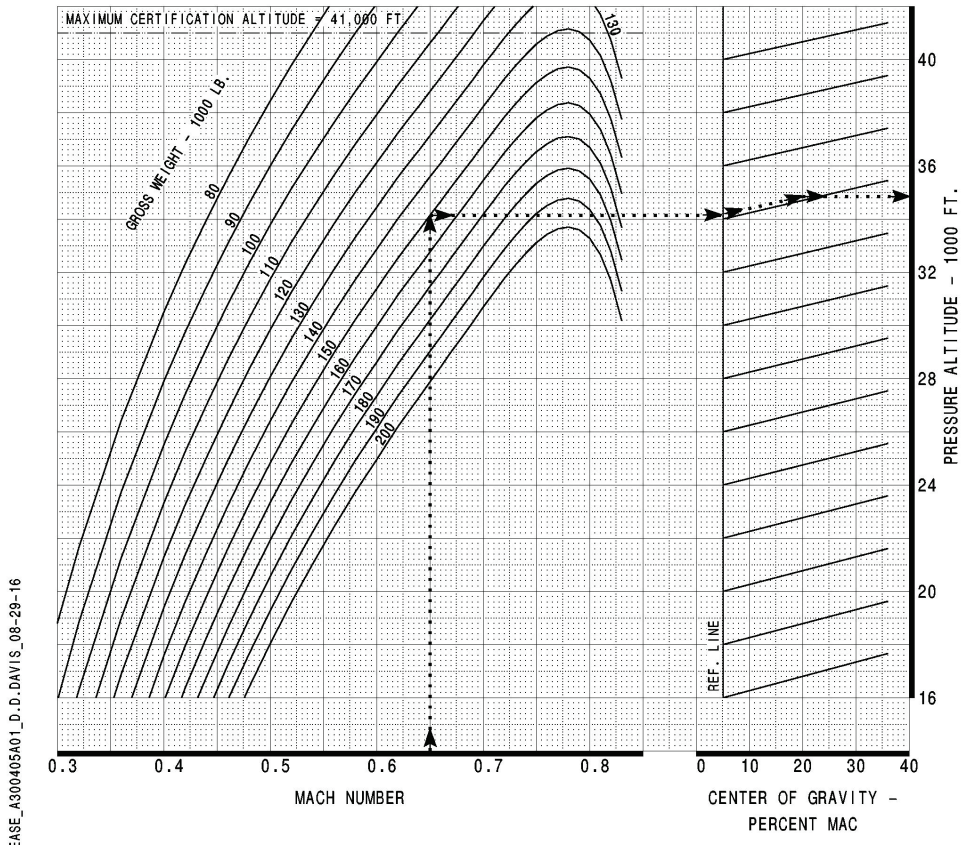
Note: If any (BTMS) brake temperature display digit is blank or indicates 0.0 or 0.1, then this method cannot be used.

MODEL	BRAKE CATEGORIES	BRAKE PART NUMBERS	MINIMUM WAIT TIME (MINUTES)
737-8	B	S277A016-312, -510	48

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Normal Cruise Altitude Limits Due to Buffet A163052

- ALTITUDE CAPABILITY BASED ON 0.3g MARGIN TO INITIAL BUFFET.
- M_{MO} = 0.82 MACH



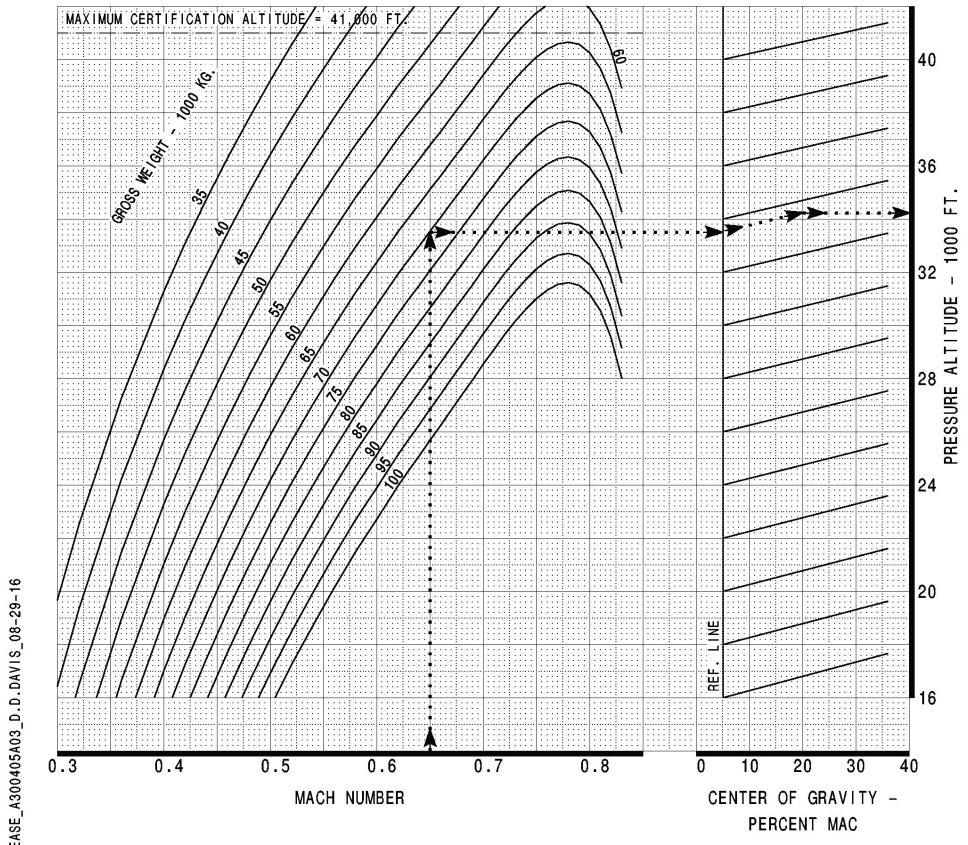
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Section Normal Cruise Altitude Limits Due to Buffet, continued from previous page A163052

- ALTITUDE CAPABILITY BASED ON 0.3g MARGIN TO INITIAL BUFFET.
- M_{MO} = 0.82 MACH

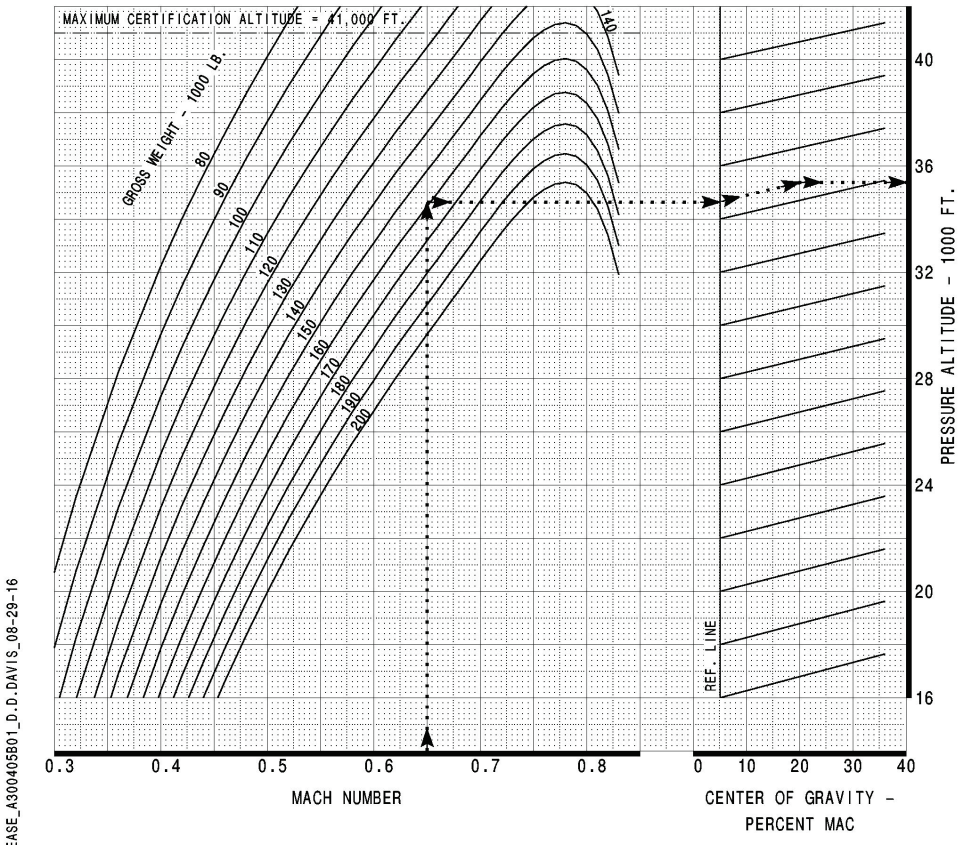


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- ALTITUDE CAPABILITY BASED ON 0.2g MARGIN TO INITIAL BUFFET.
- M_{MO} = 0.82 MACH



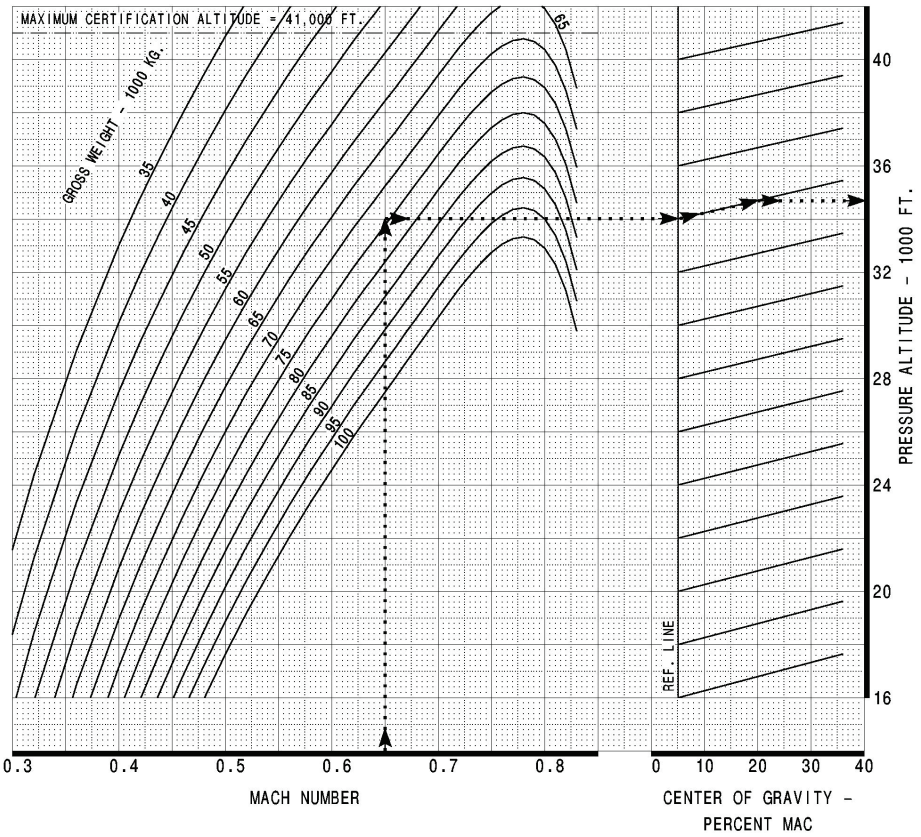
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Section Normal Cruise Altitude Limits Due to Buffet, continued from previous page A163052

- ALTITUDE CAPABILITY BASED ON 0.2g MARGIN TO INITIAL BUFFET.
- M_{MO} = 0.82 MACH



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Performance Conditions and Procedures A163052

The conditions and procedures used in establishing the performance data in this manual are presented for each phase of operation. Airplane configuration and engine thrust settings are shown under Performance Configuration. These procedures include only the essential steps which affect airplane performance, and should be considered as guidance material from which more complete operational procedures can be developed.

Performance Conditions and Procedures - Climb-out A163052

Conditions

Climb gradient and obstacle clearance flight path performance is based on failure of the most critical engine at VEF.

Procedures

Maintain takeoff flap setting and a speed not less than V_2 to the height selected for initiation of flap retraction.

Retract flaps according to the flap retraction speed schedule, in this section.

Follow enroute procedures after all takeoff flight path obstacles have been cleared or after climbing to at least 1500 feet above runway elevation.

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Performance Conditions and Procedures - Landing Field Length A180643

Conditions

The scheduled landing field lengths shown in the AFM-DPI software are based on standard day temperatures on a smooth, level, hard-surfaced runway. Scheduled landing field lengths on a dry runway are maximum performance landing distances from a landing speed of VREF at a 50 feet height to a full stop, multiplied by a factor of 1.67. Scheduled landing field lengths on a non-dry smooth runway are determined by multiplying the dry scheduled landing field lengths by a factor of 1.15. Reverse thrust was not used in establishing these distances.

The unfactored distances from a 50 feet height to touchdown used in the landing distance calculations are based on flight testing conducted with sink rates of approximately 4 to 6 feet per second from a nominal 3 degree glidepath using the applicable procedures defined below. The flight test data were analytically adjusted to 8 feet per second from a 3.5 degree glidepath consistent with AC 25-7C methodology. Unfactored distances from a 50 feet height to touchdown calculated using this method may not be achieved in typical operations. Scheduled landing field lengths are provided for dispatch purposes only.

Procedures

Make final approach and landing with landing gear down and flaps in the landing position. Initiate the flare when the main gear is approximately 20 feet above the runway. Reduce thrust to idle on all engines after initiating the flare.

When landing with automatic speed brakes armed, the speed brakes are automatically deployed on touchdown; apply maximum wheel braking immediately.

When landing with manual speed brakes, apply full speed brakes and maximum wheel braking two seconds or less after touchdown.

When landing with antiskid inoperative, raise the speed brakes at touchdown. Initiate wheel braking using very light pedal pressure and increase as ground speed decreases.

Performance Conditions and Procedures - Obstacle Clearance A163052

Procedures

When engine failure occurs and the takeoff weight is obstacle limited, maintain V_2 or the speed at engine failure (up to V_2 + 20 knots maximum) up to the gross height required for obstacle clearance.

When the height selected for initiation of flap retraction is reached, initiate flap retraction and accelerate to final takeoff climb speed while maintaining constant altitude and initial takeoff thrust setting.

Continue final takeoff climb to 1500 feet above runway elevation, or to the minimum gross height required for obstacle clearance, at final takeoff climb speed and maximum continuous thrust.

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Performance Conditions and Procedures - Rejected Takeoff A180643

Conditions

Calculated accelerate-stop distances account for demonstrated recognition and reaction times, plus additional time delays.

Distances are valid with autobrakes set to RTO. With autobrakes either disarmed or off, maximum manual braking is assumed.

Reverse thrust is not used in calculating these distances on a dry runway surface, but is used in calculating these distances on a wet runway surface.

Procedures

Initiate rejected takeoffs at or prior to V_1 using the following procedures:

Antiskid Operative

Simultaneously close thrust levers, disengage the autothrottles, and apply maximum manual wheel braking or verify operation of RTO Automatic Wheel Braking.
Raise SPEED BRAKE lever.

Antiskid Inoperative

Close thrust levers and raise the speedbrakes. Initiate wheel braking using very light pedal pressure and increase as ground speed decreases.

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Performance Conditions and Procedures - Takeoff A201614

Conditions

Takeoff field length performance accounts for the greater of: (1) 115 percent of the all-engines operating distance, or (2) the engine-inoperative distance, considering a failure of the critical engine at VEF. These distances are based on three potential runway conditions: (1) Smooth, dry, hard surfaced, (2) Smooth, wet, hard surfaced and (3) Skid-Resistant, wet, hard-surfaced.

Procedures

Performance data is based on a static takeoff power setting procedure. Operationally, a rolling takeoff procedure is recommended. Analysis has shown that the change in field length due to the rolling takeoff procedure is negligible (less than 50 feet) when compared to the standing takeoff.

The recommended power setting procedure is as follows: Set 40% N1 prior to brake release or as the airplane is aligned with the runway. As the airplane accelerates, advance the thrust levers to the takeoff N1 setting. Readjust thrust levers, if necessary, prior to 60 knots to obtain target N1. Do not make further adjustments after 60 knots.

A speed not less than V_2 is assumed to have been achieved by a height of 35 feet. Initiation of landing gear retraction is assumed to occur after a positive rate of climb has been established.

Make a smooth positive rotation to the initial climb attitude. Make minor attitude variations after liftoff to achieve the initial climb speed.

The following statement is not applicable to FMC Software U14 or later:

> When performing a takeoff using full rated takeoff thrust at an airport above a pressure altitude of 8,400 ft, ensure that the portion of the Takeoff Flight Path data based on the use of takeoff thrust is completed by an airplane pressure altitude of 19,000 ft. This can be verified by ensuring that the AFM-DPI flight path points labeled as “Level-Off” and “Start Final Climb” (if present) do not exceed an airplane pressure altitude of 19,000 ft. Use of fixed derate or assumed temperature thrust reduction will result in AFM-DPI performing this check.

Note: Takeoff with antiskid inoperative is allowed only on dry runways.

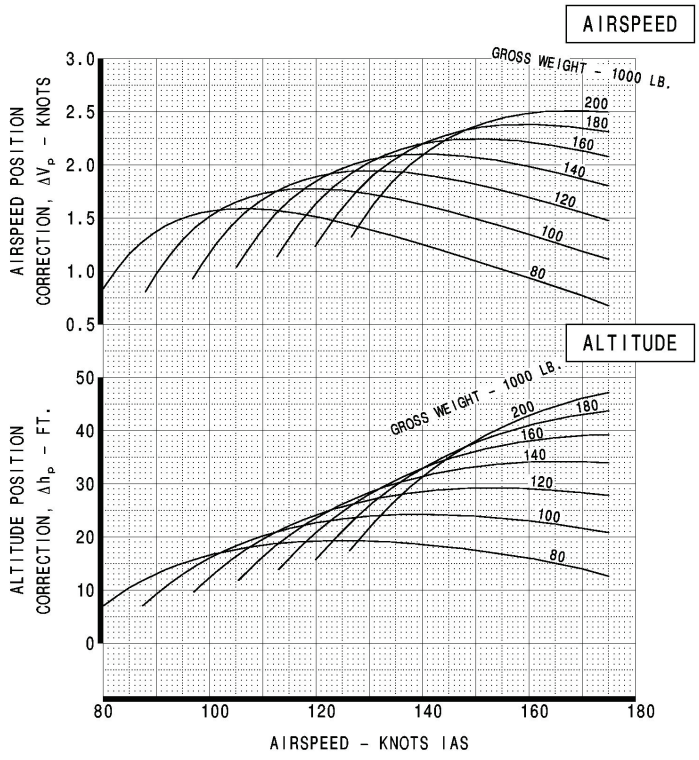
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Position Corrections - Standby System; Flaps 30; Gear Down A180643

737-8/-8200

FLAPS 30 / GEAR DOWN

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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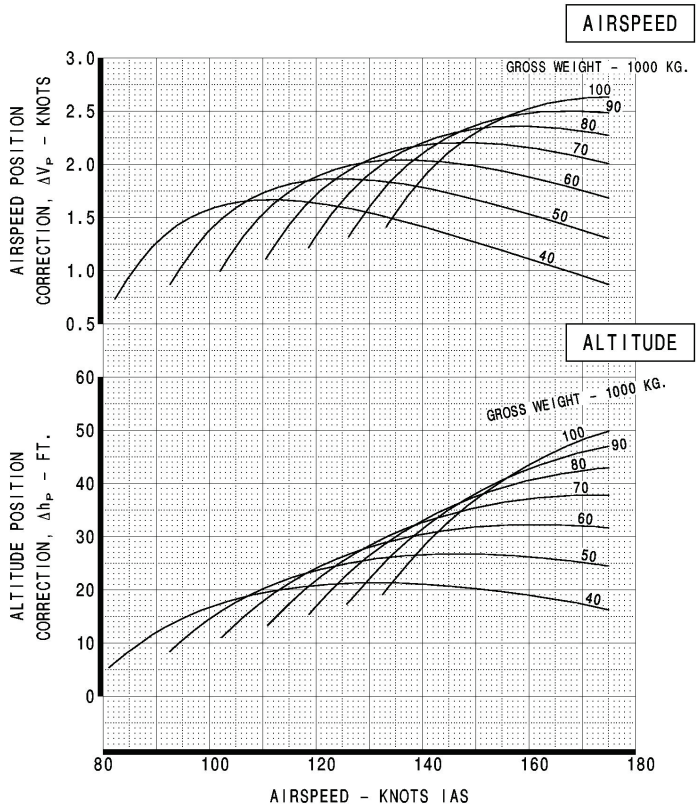
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Section Position Corrections - Standby System; Flaps 30; Gear Down, continued from previous page A180643

737-8/-8200

FLAPS 30 / GEAR DOWN

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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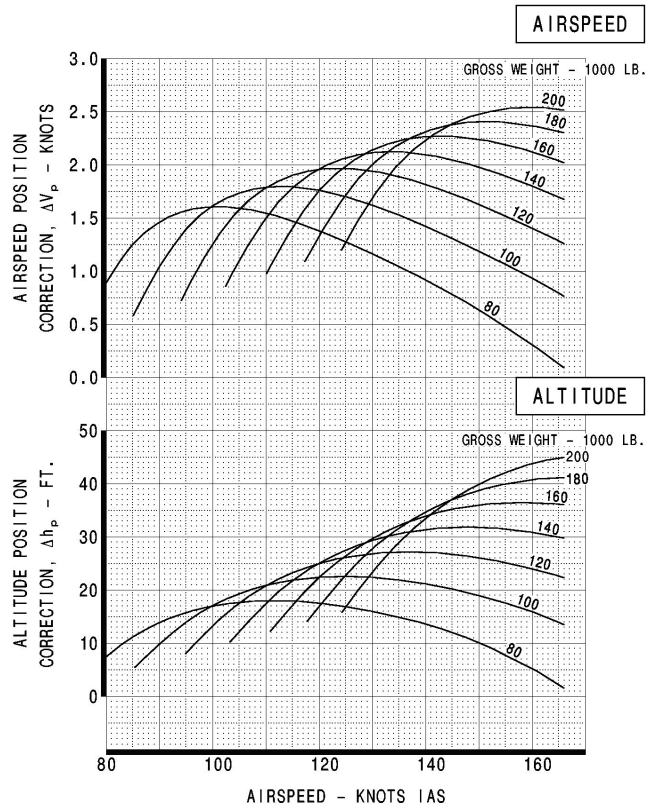
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Position Corrections - Standby System; Flaps 40; Gear Down A180643

737-8/-8200

FLAPS 40 / GEAR DOWN

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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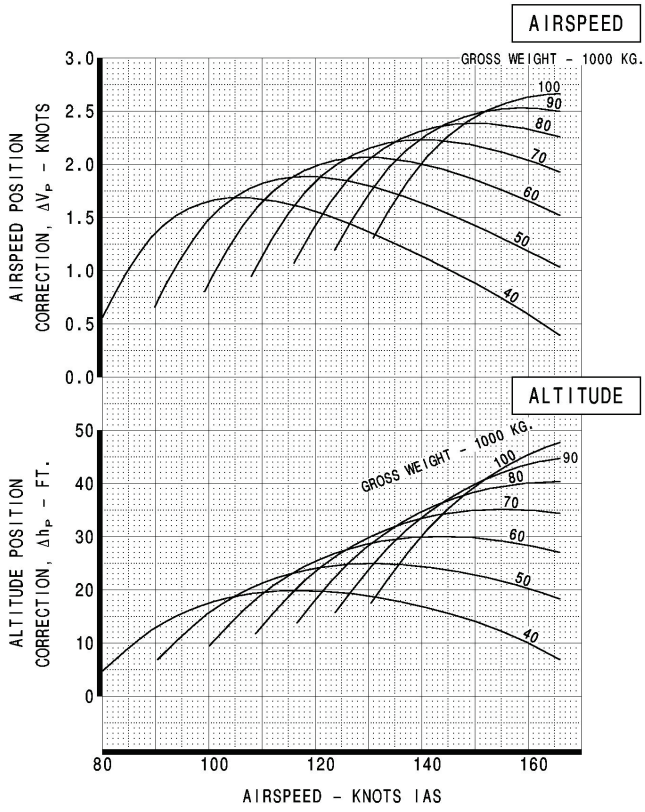
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Section Position Corrections - Standby System; Flaps 40; Gear Down, continued from previous page A180643

737-8/-8200

FLAPS 40 / GEAR DOWN

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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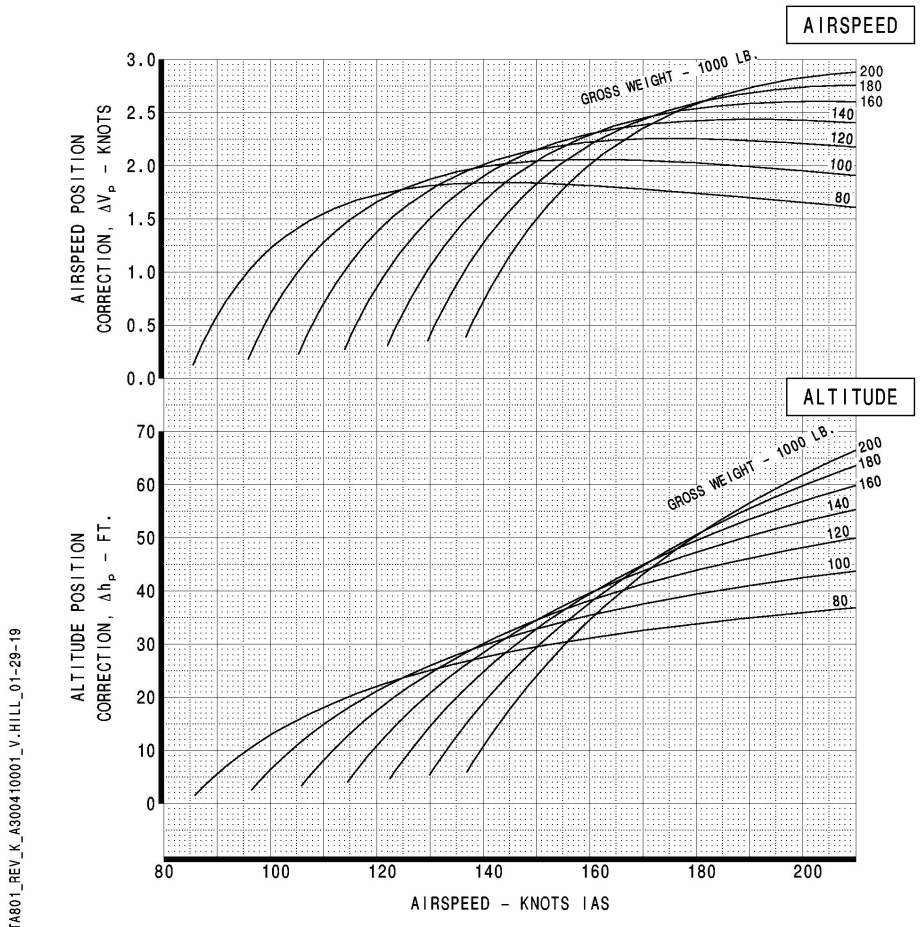
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Position Corrections - Standby System; Flap 10; Gear Up A180643

737-8/-8200

FLAPS 10 / GEAR UP

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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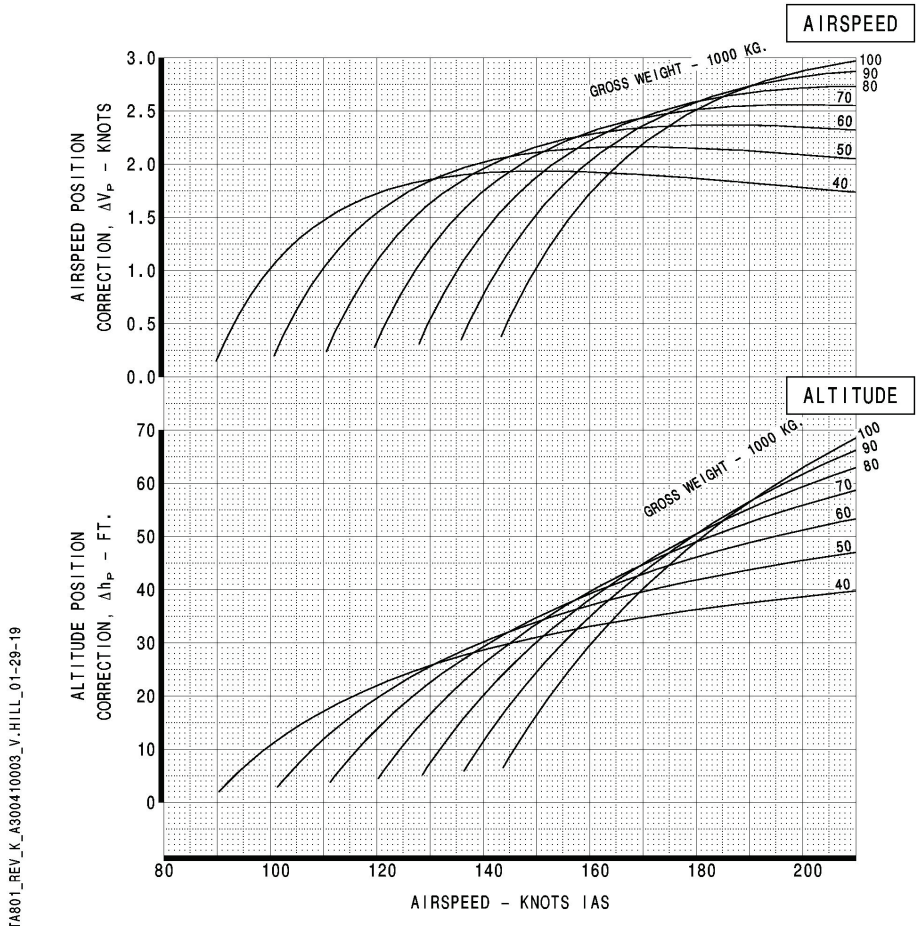
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737-8/-8200

FLAPS 10 / GEAR UP

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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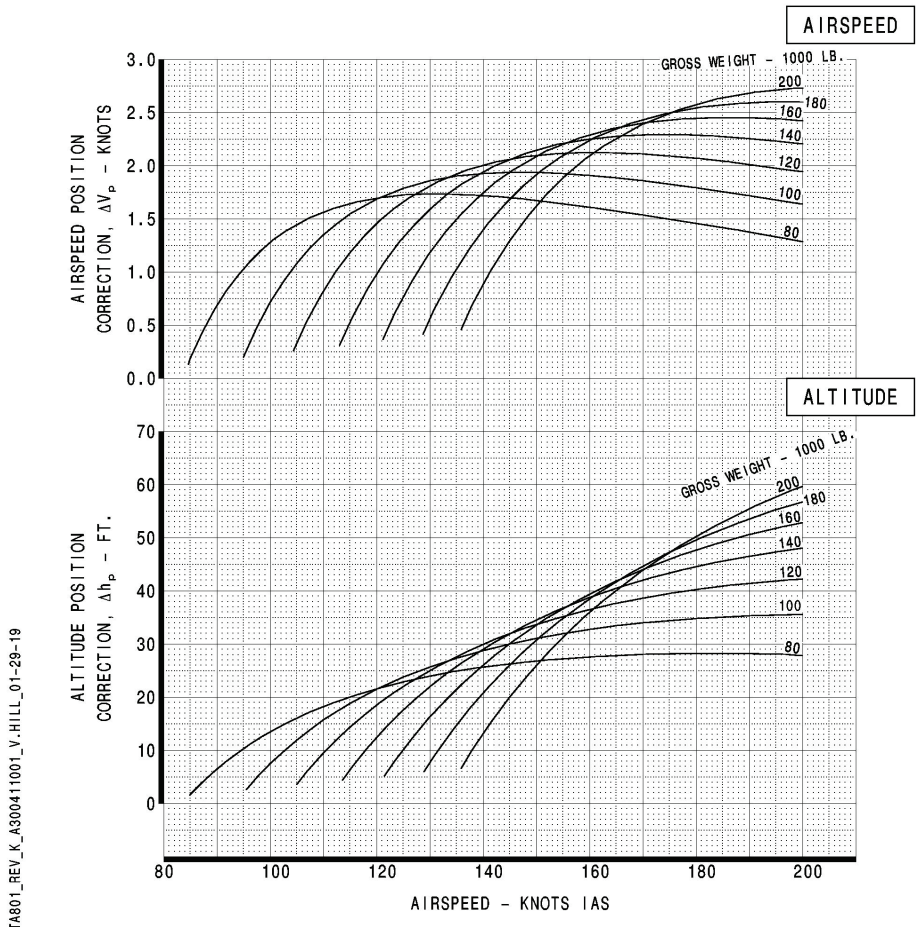
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Position Corrections - Standby System; Flap 15; Gear Up A180643

737-8/-8200

FLAPS 15 / GEAR UP

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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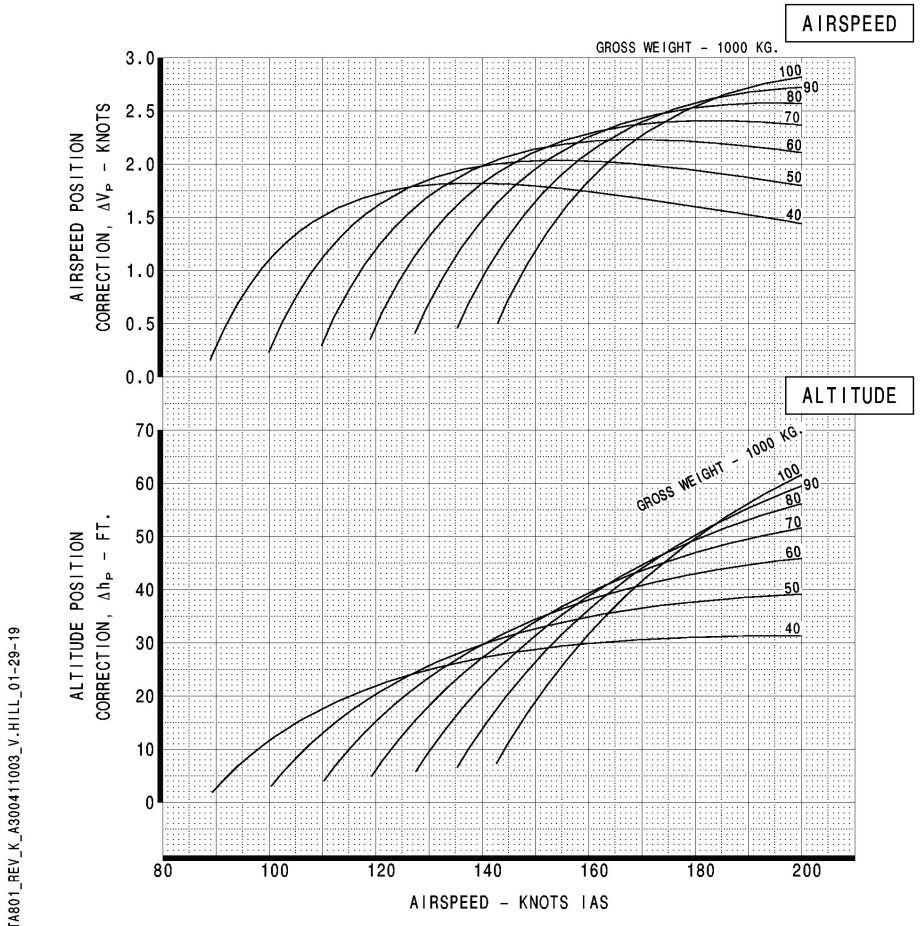
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Section Position Corrections - Standby System; Flap 15; Gear Up, continued from previous page A180643

737-8/-8200

FLAPS 15 / GEAR UP

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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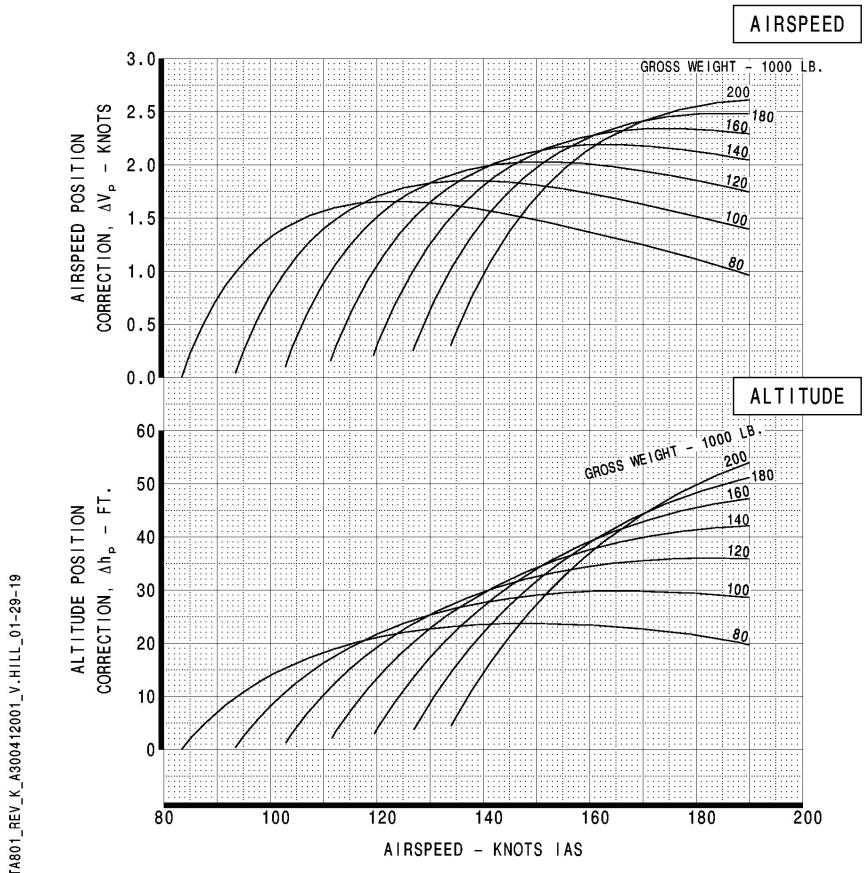
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Position Corrections - Standby System; Flap 25; Gear Up A180643

737-8/-8200

FLAPS 25 / GEAR UP

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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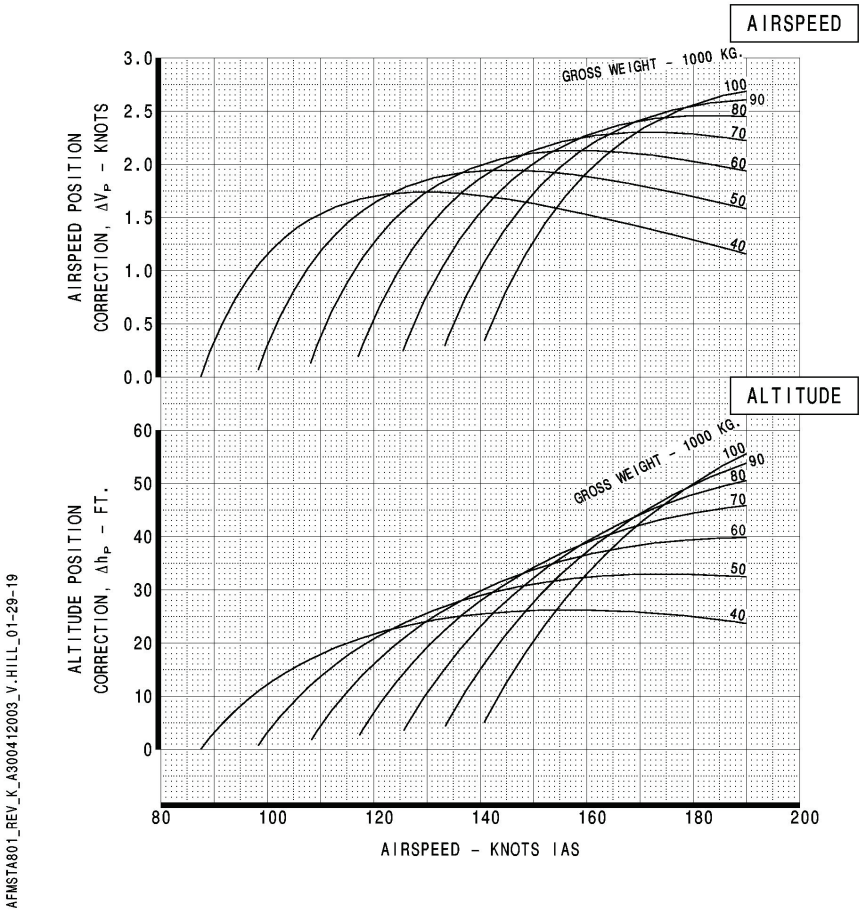
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Section Position Corrections - Standby System; Flap 25; Gear Up, continued from previous page A180643

737-8/-8200

FLAPS 25 / GEAR UP

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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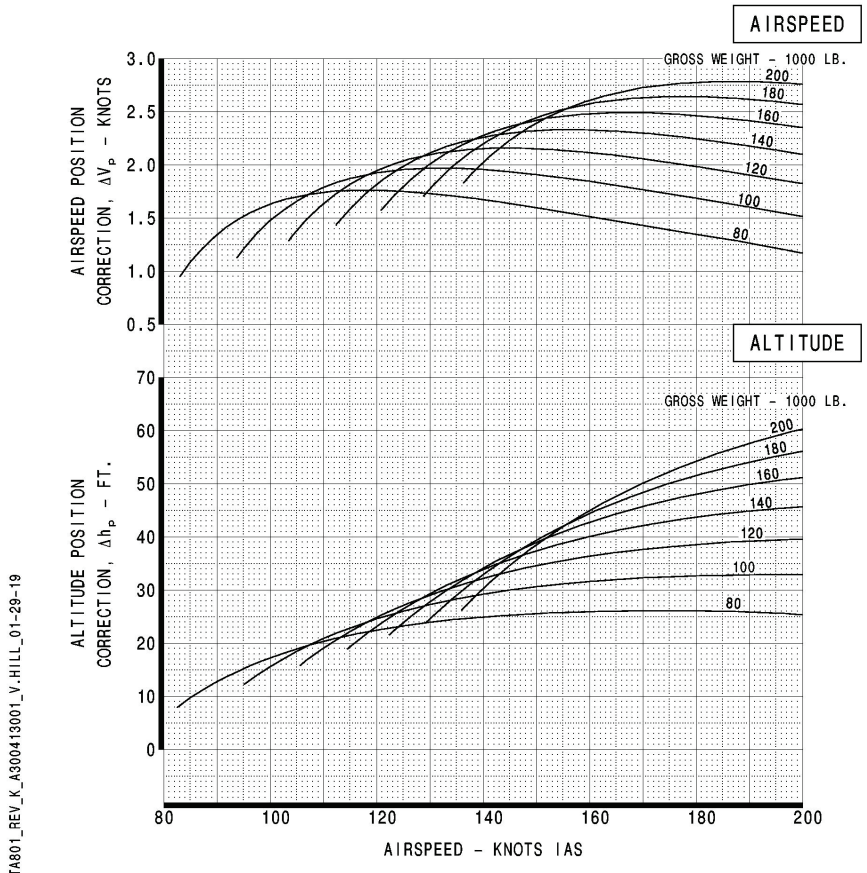
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Position Corrections - Standby System; Flaps 15; Gear Down A180643

737-8/-8200

FLAPS 15 / GEAR DOWN

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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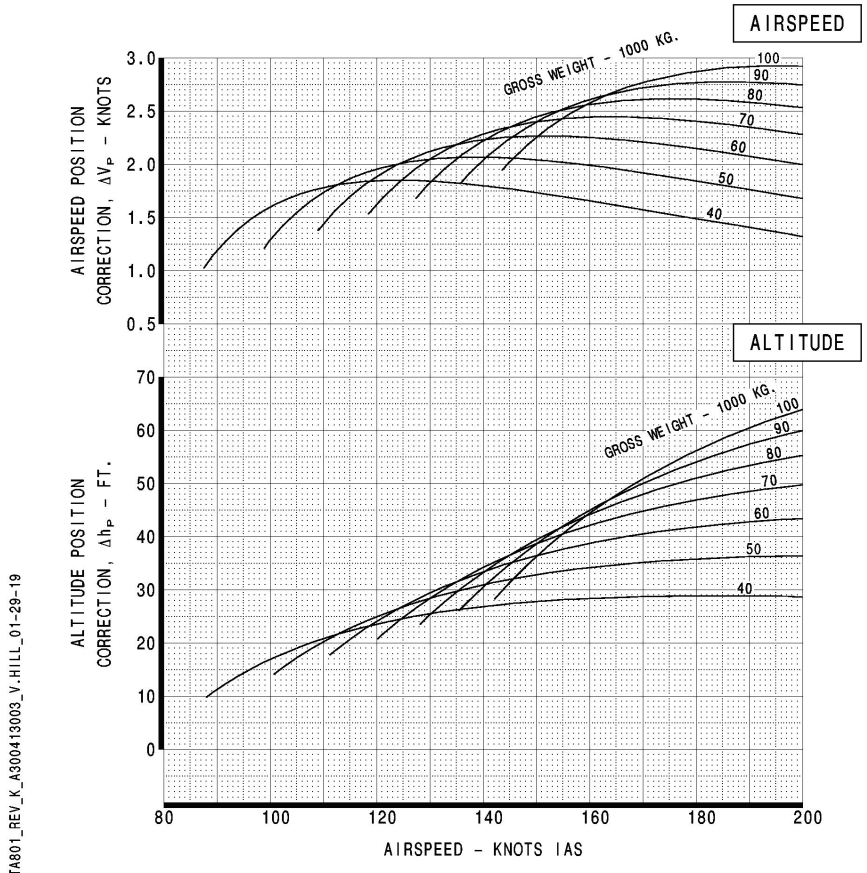
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Section Position Corrections - Standby System; Flaps 15; Gear Down, continued from previous page A180643

737-8/-8200

FLAPS 15 / GEAR DOWN

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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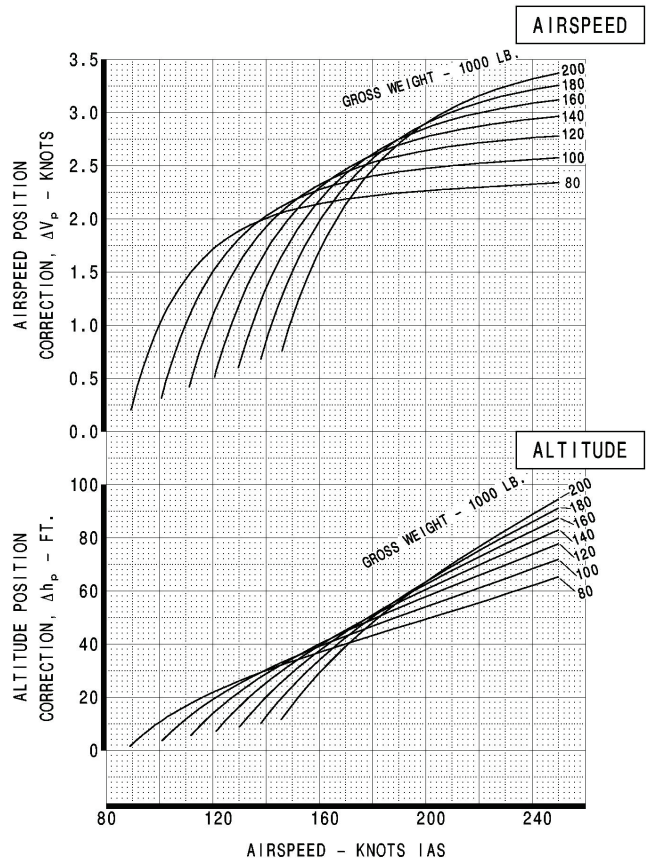
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Position Corrections - Standby System; Flaps 1; Gear Up A180643

737-8/-8200

FLAPS 1 / GEAR UP

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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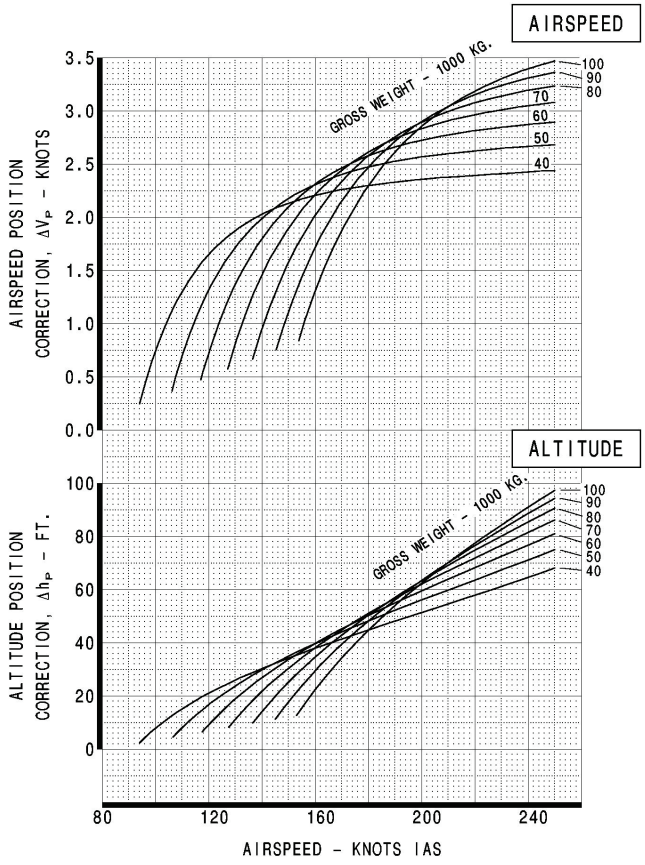
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Section Position Corrections - Standby System; Flaps 1; Gear Up, continued from previous page A180643

737-8/-8200

FLAPS 1 / GEAR UP

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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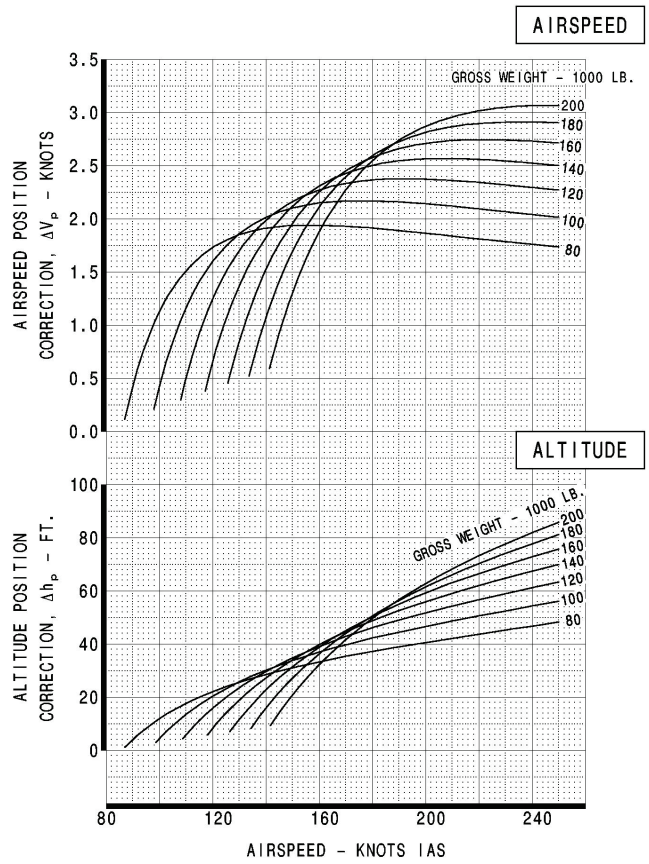
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Position Corrections - Standby System; Flaps 5; Gear Up A180643

737-8/-8200

FLAPS 5 / GEAR UP

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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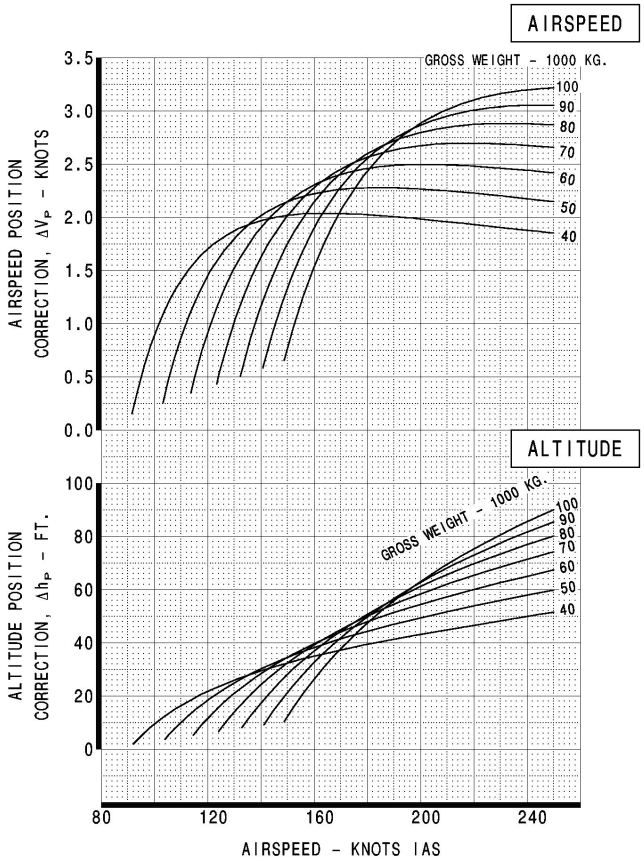
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Section Position Corrections - Standby System; Flaps 5; Gear Up, continued from previous page A180643

737-8/-8200

FLAPS 5 / GEAR UP

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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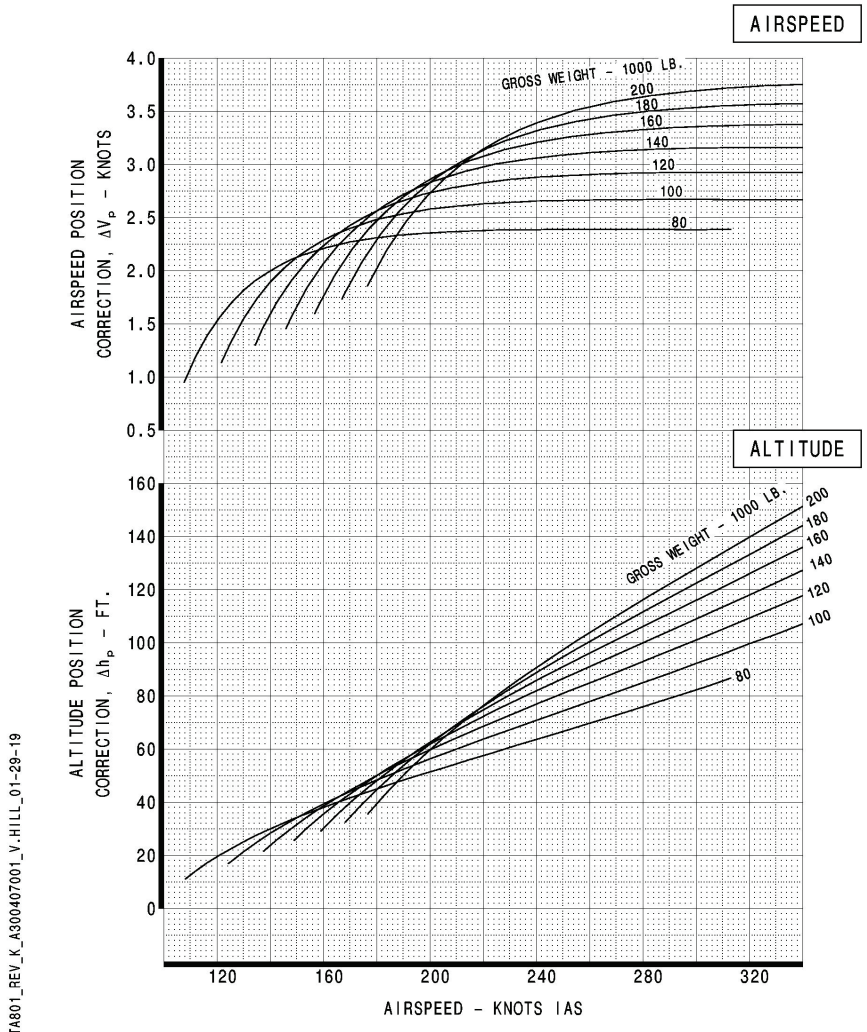
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Position Corrections - Standby System; Flaps Up; Gear Up A180643

737-8/-8200

FLAPS UP / GEAR UP

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$



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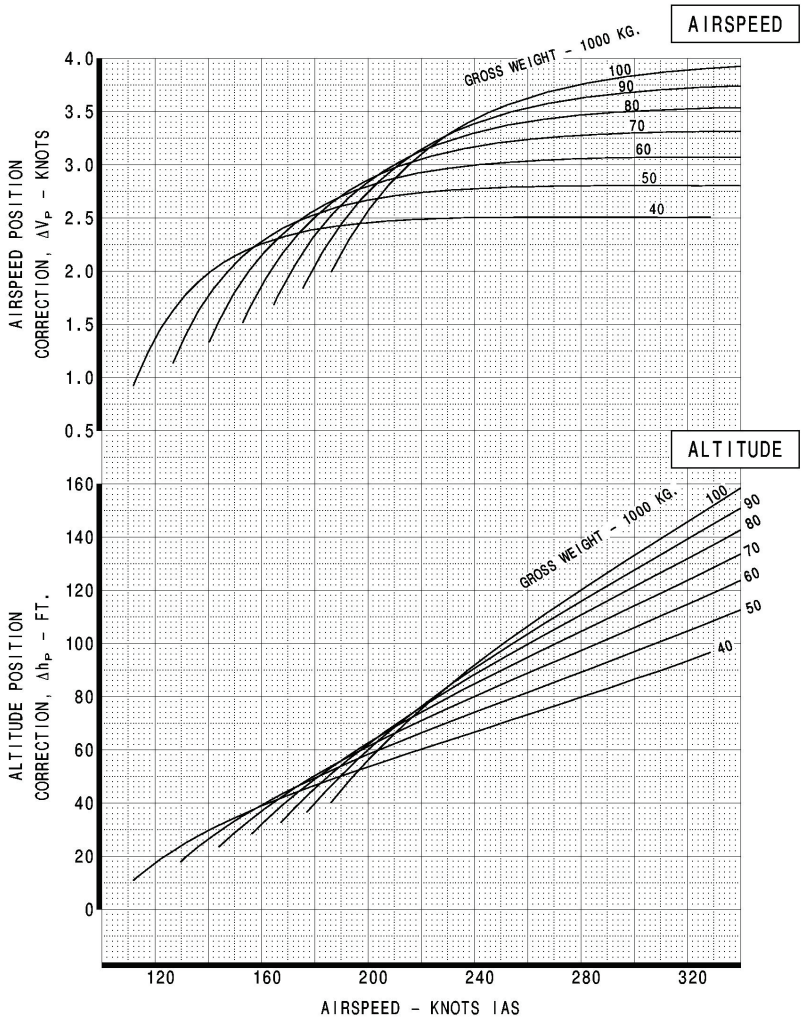
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737-8/-8200

FLAPS UP / GEAR UP

- APPLICABLE TO LOW SPEED OPERATIONS BELOW 17,000 FT. PRESSURE ALTITUDE.
- CALIBRATED AIRSPEED, $V_C = V_I + \Delta V_P$
- TRUE PRESSURE ALTITUDE, $h_P = h_I + \Delta h_P$

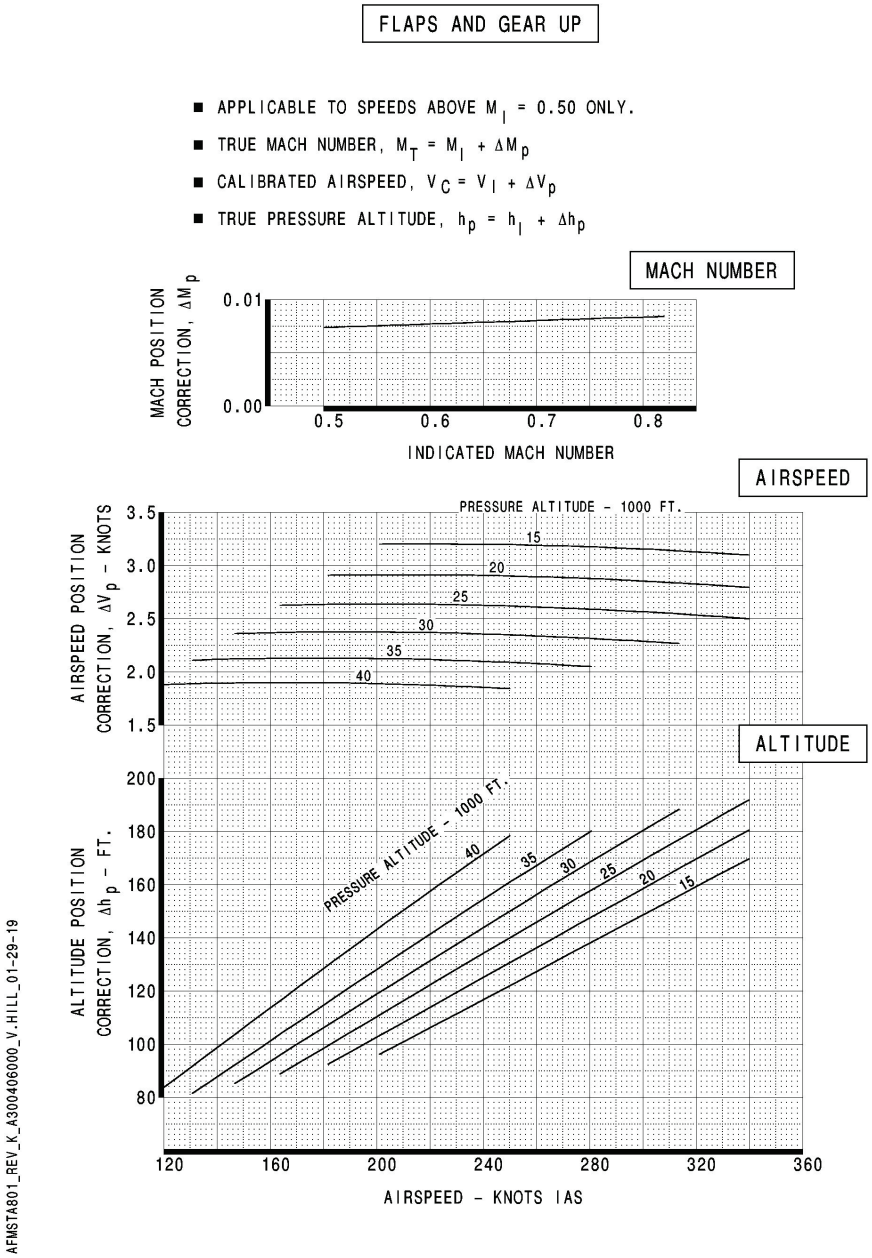


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Position Corrections - Standby System; Flaps Up; Gear Up; High Speed A180643

737-8/-8200



Regulatory Compliance A170119

The information in this Chapter is presented in compliance with the requirements of 14 CFR Part 25.

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Standard Performance Conditions A170229

All performance is based on the following:

1. Approved engine thrust ratings less installation, airbleed, and accessory losses.

2. Trailing edge flap positions as follows:

	Trailing Edge Flaps
Takeoff	1, 5, 10, 15, 25
Enroute	0
Approach	1, 5*, 15
Landing	15, 30, 40
Autoland (all engines)	30, 40

* For operation with Approach trailing edge Flap 5 settings see AFM-DPI Authorized Alternate Performance Capability in the Airplane Flight Manual - Digital Performance Information (AFM-DPI) Description section.

3. Leading edge devices in the appropriate position for trailing edge flap position.

4. Full temperature accountability within the operational limits, except for landing distance, which is based on standard day temperatures.

5. Wind corrections are calculated using not more than 50% of actual headwinds, or not less than 150% of actual tailwinds, in compliance with relevant operating regulations unless otherwise noted. Tower winds are assumed to be measured at 10 meters (32.8 feet).

6. The effect of humidity on engine thrust is based on the following relative humidities: 80 percent, at and below standard temperatures, and 34 percent, at and above standard temperatures plus 28 degrees C (standard temperatures plus 50 degrees F). Between these two temperatures, the relative humidity varies linearly.

7. Aircraft performance adjusted for possible ice accumulations is provided for planning flights into known or anticipated icing conditions.

Takeoff Performance: Accounting for ice accumulation is not required.

Enroute Performance: Accounting for ice accumulation is not required.

Landing Performance:

Climb: Performance for approach climb and landing climb accounting for possible ice accumulation is provided in AFM-DPI by selecting “Inflight Icing”. Use the approach climb and landing climb “Inflight Icing” performance when operating in icing conditions during any part of the flight and the forecast landing temperature is below 10 degrees C.

Speeds and Landing Distances: No speed or landing distance adjustments are necessary for the effect of ice accumulation for Flaps 40 or when using Flaps 15 during a go-around from a Flaps 30 approach.

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Section Standard Performance Conditions, continued from previous page A170229

Flaps 1 stall speeds are adjusted to account for possible ice accumulation prior to landing. Flaps 15 landing speeds and distances are adjusted to account for possible ice accumulation. When using Flaps 5 as an alternate flap setting during a go-around from a Flaps 30 approach, the Flaps 30 landing speeds and distances are adjusted for possible ice accumulation. Ice accumulation performance information is provided in AFM-DPI by selecting “Inflight Icing”.

Use the adjusted Flaps 1 stall speeds and the Flaps 15 and 30 “Inflight Icing” landing speeds and distances under the following conditions:

a. When operating in icing conditions during any part of the flight and the forecast landing temperature is below 10 degrees C.

b. Anytime after wing thermal anti-ice (WTAI) has been turned on regardless of whether WTAI was subsequently turned off.

c. Anytime cowl thermal anti-ice (CTAI) is on during approach.

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Revision Highlights

Revision 4 to D631A002-AP3GIE dated 11/30/2022

The purpose of this revision is to add 737-8200 data.

The following changes comprise this revision:

Chapter - Front Matter

Title (Revised)
Revised to add the 737-8200.

Revision Approval (Revised)
This section uniquely identifies the approval authority and the reference number for this revision.

Log of Sections (Revised)
Revised to reflect sections changed in this revision.

Chapter - Certificate Limitations

Electronics - Autopilot (AP) Flight Director (FD) System (Revised)
Revised to add the 737-8200.

Chapter - Operational Data

Autopilot Flight Director System (Revised)
Revised to add the 737-8200.

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Title **A211727**

APPENDIX AP3GIE

AUTOPILOT FLIGHT DIRECTOR SYSTEM

CATEGORY IIIA AUTOLAND OPERATIONS
GNSS LANDING SYSTEM (GLS) ACTIVATION
INTERGRATED APPROACH NAVIGATION (IAN)

EASA

MODEL 737-8/-8200/-9 SERIES

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Revision Approval A211727

This Revision Approval may only be used in conjunction with a Log of Sections that refers to Revision Approval Number A211727 in the revision approval number column for the Revision Approval section.

Approved by:  Date:11/30/2022

Authorized Representative,
Regulatory Administration,
Boeing Commercial Airplanes

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Log of Sections A211727

D631A002-AP3GIE Revision 4 Date: 11/30/2022

	Chapter/Section	Revision Approval Number	Approval Date
	Front Matter		
*	Title	A211727	11/30/2022
*	Revision Approval	A211727	11/30/2022
*	Log of Sections	A211727	11/30/2022
	Introduction	A201492	08/04/2020
	Certificate Limitations		
*	Electronics - Autopilot (AP) Flight Director (FD) System	A211727	11/30/2022
	Operating Procedures		
	Autopilot (AP) Flight Director (FD) System	A170100	04/19/2017
	Operational Data		
*	Autopilot Flight Director System	A211727	11/30/2022

* = Revised Sections

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Introduction A201492

This appendix contains the additional Certificate Limitations and Procedures applicable to airplanes equipped for Category IIIa Autoland operations, GNSS Landing System (GLS) activated, and Integrated Approach Navigation (IAN).

This appendix is used in conjunction with the Certificate Limitations, Operating Procedures, and Operational Data of the basic Airplane Flight Manual. Refer to ASN & Appendix Effectivity page of the basic manual for applicability.

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Electronics - Autopilot (AP) Flight Director (FD) System A211727

1. For single channel operation during approach, the autopilot shall not remain engaged below the Minimum Use Height (MUH).

The following table defines the MUH for single channel autopilot operation as function of aircraft model:

Aircraft Model	MUH (feet AGL)
737-8/-8200	111
737-9	130

2. The autopilot must not be engaged below a minimum engage altitude of 400 feet AGL after takeoff.

3. Use of aileron trim with the autopilot engaged is prohibited.

Automatic Landing

The maximum allowable wind speeds, when conducting a dual channel Cat II or Cat III landing predicated on autoland operations, are:

Wind Component (knots)	**737-8/-8200/-9**
Headwind	25
Tailwind	15*
Crosswind	20

The maximum and minimum glideslope angles are 3.25 degrees and 2.5 degrees respectively.

Autoland capability may only be used with flaps 30 and 40 for both engines operative.

Autoland capability may only be used to runways at or below 8,400 ft field elevation.

Notes:
* Operations in tailwinds greater than 10 knots maybe further restricted, see Operational Limitation for Takeoff and Landing Tailwind Component in the basic AFM.

Low Weather Minima Operations

The autopilot system and associated equipment has been found to meet the airworthiness and performance criteria of:

CS AWO Subpart 1 - Autoland Operation
CS AWO Subpart 2 - Category II Operation
CS AWO Subpart 3 - Category III Operation

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Section Electronics - Autopilot (AP) Flight Director (FD) System, continued from previous page A211727

Note: Compliance with the standards noted above does not constitute approval to conduct Category II or Category III operations. Such authorization must be obtained by the operator from the appropriate authorities.

The airplane is approved for the following types of ILS approach and landing:

DH 200 feet or more (Category I) (Flaps 15 for approach, 30 and 40 for approach and landing).
Manual approach with or without flight director.
Single or, dual channel automatic approach and manual landing.
Dual channel automatic approach and landing.

DH below 200 feet but not less than 100 feet (Category II) (Flaps 30 and 40).
Dual channel automatic approach and manual landing.
Dual channel automatic approach and landing.

DH below 100 feet but not less than 50 feet (Category III) (Flaps 30 and 40).
Dual channel automatic approach and landing.

The airplane is approved for the following types of approaches using IAN FAC and/or G/P guidance:

DA(H) 250 HAT feet or more
Manual approach with flight director.
Single channel automatic approach and manual landing.

The airplane is approved for the following types of GLS approach and landing:

DH 200 feet or more (Category I) (Flaps 15 for approach, 30 and 40 for approach and landing).
Manual approach with or without flight director.
Single or, dual channel automatic approach and manual landing.
Dual channel automatic approach and landing.

These autopilot flight director system limitations do not constitute operational approval.

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Autopilot (AP) Flight Director (FD) System A170100

Approach and Go-Around Alerts

These procedures apply to airplanes equipped with an Autoland Warning Light and are to be used for specified failures during dual channel Category II and IIIa approaches. The procedures assume the actual weather is at the category minimum.

Alert	Approach Below 500 ft. RA
Autoland Warning Light* (If installed)	Execute a manual go-around (In visual conditions, land or execute a go-around)
Flashing red A/P disengage warning light and wailer, or steady red A/P disengage warning light **	Execute a manual go-around (In visual conditions, land or execute a go-around)
ILS Deviation Alert ** GLS Deviation Alert**	Execute a go-around (In visual conditions, land or execute a go-around)
Engine failure	Execute a manual go-around (In visual conditions, land or execude a go-around)

Notes:

* The Autoland Warning Light is a customer selectable airplane configuration option and may or may not be installed on specific airplanes. The Alerts indicated with (**) will activate the Autoland Warning light if it is installed.

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Autopilot Flight Director System A211727

The Autopilot-Flight Director System is used as either a single channel autopilot or flight director for enroute and single channel approaches. Dual autopilot channels provide fail-passive operation for automatic landing and go-around. Dual flight directors provide for takeoff, approach and go-around guidance.

The following flight path control functions for automatic (autopilot) and/or manual (flight director) control of the airplane are provided:

Lateral navigation
Vertical navigation
VOR
Localizer (Front course only)
Approach
Autoland (Dual autopilot only)
Go-around (Dual autopilot and/or flight director only)

The following pilot assist functions for automatic (autopilot) and/or manual (flight director) control of the airplane are provided:

Control Wheel Steering (Autopilot only)
Heading select and hold
Vertical speed select and hold
IAS/Mach select and hold (Elevator control of speed in level change)
Altitude Select/Acquire or Capture and Hold
Takeoff (Dual Flight director only)
Go-around, one engine inoperative (Dual Flight director only)

The Captain's and First Officer's instruments (Display Source, VHF NAV and IRS) must not be on the same source when making approaches to Category II or lower weather minima.

An interlock is provided with the electrical transfer bus sensing circuit to preclude dual-channel autopilot operation on a single source of power. However, the Auxiliary Power Unit generator may be used as an independent power source.

Integrated Approach Navigation (IAN) Glide Path (G/P) guidance is based on use of barometric altitude. Glide path true gradient will vary with ambient temperature. When evaluating an approach procedure for operational approval, the affect of excessive ambient temperature must be considered. For cold day conditions the shallower approach path may affect obstacle clearance, and for hot day conditions the approach path will be steeper than normal.

Offset Approaches: Instrument approaches offset from the runway centerline following vertical path guidance may require expeditious maneuvering to align with the runway when at or near landing minimums. If alignment and touchdown within the runway touchdown zone cannot be assured using normal bank and pitch attitudes, and suitable descent rates, a go-around should be executed.

Demonstrated Conditions

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The system has been demonstrated both with and without yaw damper and autothrottle and with normal landing flaps 30 and 40.

An autopilot minimum engage height (MEH) of 400 feet has been demonstrated to provide satisfactory performance.

The approach speed selected for automatic approaches using autothrottles was V_{REF} + 5 knots (No wind correction was made).

The approach speed selected for autothrottle inoperative was:

737-8/-8200/-9	V_{REF} + 5 knots for calm air conditions and V_{REF} + 1/2 (Headwind) + Full Gust limited to minimum of V_{REF} + 5 and maximum of V_{REF} + 20 for wind conditions.

The automatic landing system has been demonstrated with the following wind conditions:

Wind Component (knots)	**737-8/-8200**	**737-9**
Headwind	35	29
Tailwind	18	19
Crosswind	27	27

Refer to Certificate Limitations chapter for limiting wind conditions.

Satisfactory Automatic Landing System performance has been demonstrated on US type II and type III ILS ground facilities.

Satisfactory GLS Automatic Landing System performance has been demonstrated on a Ground Based Augmentation System (GBAS) facility that is compliant with ICAO Annex 10 Standards and Recommended Practices (Amendment 89).

The demonstrated altitude loss during an automatic go-around initiated below 100 feet AGL is listed below:

GA Altitude (ft AGL)	**Altitude Loss (ft)**	
	737-8/-8200	**737-9**
60 to 100	28	25
50	24	25
40	20	21
30	15	16
20	11	12
10	6	7

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Integrated Approach Navigation (IAN) Final approach Course (FAC) and Glide Path (G/P) guidance have been demonstrated to be satisfactory for:

Manual approach with flight director
Single channel automatic approach and manual landing.
Approaches using Flaps 15, 30, and 40.
Autopilot engaged as low as MUH.
Approaches having a final segment glidepath angle up to 3.77 degrees.

These autopilot flight director system demonstrations do not constitute operational approval.

Minimum Multichannel Engage Altitude for Autoland

On approach for autoland, dual channel operation should be engaged prior to 800 feet AGL. Check FLARE arm annunciation at approximately 500 feet AGL.

AFDS System Configuration

The AFDS equipment listings in this section do not necessarily denote all of the systems and equipment required for the types of operation specified. Applicable regulations and guidance material may prescribe an operational requirement for such additional systems such as autothrottle, or autobrakes. Operators should determine the total systems requirements for each type of operation prior to requesting Operations Specifications authorization.

Demonstrated compliance with the airworthiness performance standards does not constitute approval to conduct operations in lower weather minimums.

Single Engine Approach
The AFDS has demonstrated adequate performance to allow a single engine flaps 15 approach to Cat I minima.

Demonstrated Altitude Loss

The demonstrated altitude loss due to a simulated hard-over autopilot malfunction is:

Level Flight:

Flaps up when recovery was initiated 3 seconds after the recognition point.

Demonstrated Loss - Feet	
737-8/-8200	737-9
259	230

Approach:

(a) With a 1 second time delay between recognition point and initiation of recovery.

Continued on next page

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Demonstrated Loss - Feet	
737-8/-8200	737-9
31	35

(b) When a recovery was initiated without delay after pilot recognition.

Demonstrated Loss - Feet
737-8/-8200/-9
8

Normal Approach Procedures

Autoland

PILOT FLYING	PILOT NOT FLYING
When on localizer intercept heading, arm APP mode and engage second autopilot	Set transponder mode selector to desired TCAS mode.
Verifies mode annunciation.	
At localizer capture verifies proper mode annunciation and sets appropriate heading.	Verifies proper mode annunciation.
At glide slope "alive", announce "GEAR DOWN", "FLAPS 15". Arm speed brake and check green light illuminated.	Position landing gear lever DN, flap lever to the 15 detent. Position engine start switch to CONT.
At glide slope capture, verify proper mode annunciation, check N1 reference bug at the go-around limit and set missed approach altitude	
Announces "FLAPS ___" as required for landing. Set MCP speed selector at V_{REF} + 5 kts.	Position flap lever as directed.
Call “LANDING CHECKLIST". Respond to the LANDING checklist.	Read the LANDING checklist and report "LANDING CHECKLIST COMPLETE".
Monitor approach progress and guard the controls. At 500 feet AGL, verify FLARE is armed. At approximately 50 feet AGL, verify FLARE is engaged. Ensure the autothrottle retards the thrust levers to idle by touchdown.	

Go-Around

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PILOT FLYING	PILOT NOT FLYING
Press TO/GA switch. Call "FLAPS 15". If full GA thrust is required, press TO/GA switch again after reduced GA thrust is established.	Monitor N1 indication. Position flap lever to 15 and monitor flap retraction.
Confirm rotation to go-around attitude and monitor autopilot.	
Verify Mode annunciation.	
When positive rate of climb is indicated, call "GEAR UP" and monitor acceleration.	Verify positive rate of climb. Position landing gear lever UP.
Check flight instrument indications (MCP speed window blanks).	
Call "TUNE RADIOS FOR MISSED APPROACH".	Tune Radios as directed.
Above 400 feet, select appropriate roll mode and verify proper mode annunciation.	Observe mode annunciation.
Retract flaps on flap speed schedule.	Position flap lever as directed and monitor flaps and slats retraction.
Verify airplane levels off at selected altitude and maintains flap maneuvering speed.	
Call "AFTER TAKEOFF CHECKLIST".	Read and respond to the AFTER TAKEOFF checklist and report "AFTER TAKEOFF CHECKLIST COMPLETE".

ILS Autopilot Approach/Autoland (Fail Passive) (Cat.II/IIIa)

The Autopilot System has been shown to meet the applicable airworthiness performance and integrity criteria established for a fail passive system to comply with EASA CS-AWO Subparts 1, 2, and 3 for automatic approach and landing with the following equipment operative and FLARE arm annunciated:

Dual Channel Autopilot engaged
Low Range Radio Altimeter and display for each Pilot
Decision Height (DH) Display for each Pilot
Windshield Wipers for each Pilot
ILS Receiver and display for each Pilot
Flight Mode Annunciator for each Pilot
Two ADIRU's (associated with the engaged autopilots) in NAV mode
Dual Hydraulic Systems
Two sources of electrical power (The APU generator may be used as an independent power source)
Both Engines Operating

ILS Autopilot Approach (Cat. II)

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The Autopilot System has been shown to meet the applicable airworthiness performance and integrity criteria for a Category II operations in accordance with EASA CS-AWO Subpart 2 for automatic approach with the following equipment operative:

Dual channel Autopilot engaged
Low Range Radio Altimeter and display for each Pilot
Decision Height (DH) Display for each Pilot
Windshield Wipers for each Pilot
ILS Receiver and display for each Pilot
Flight Mode Annunciator for each Pilot
Two ADIRU's (associated with the engaged autopilot) in NAV mode
Two sources of electrical power (The APU generator may be used as an independent power source.)
Both Engines Operating

GLS Autopilot Approach (Cat. I)

The Autopilot System has been shown to meet the applicable airworthiness performance and integrity criteria established by the EASA for automatic approach with the following equipment operative:

Single or Dual channel Autopilot engaged
Decision Altitude (Height) Display for each Pilot
GLS Receiver and display for each Pilot
Flight Mode Annunciator for each Pilot
Two ADIRU's (associated with the engaged autopilot) in NAV mode
One or Both Engines Operating

Flight Director (F/D)

The flight director command may be used as supplemental guidance to the primary speed and attitude indications for takeoff, climb and descent to acquire and maintain desired altitudes.

All of the autopilot command modes, except "CWS", are also available on the flight directors. An additional takeoff mode exists for the F/D only. One or both F/D’s may be on for all modes, except during T/O or GA which requires dual F/D ON.

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Revision Highlights

Revision 14 to D631A002-L1B27E dated 10/10/2023

The purpose of this revision is to update Engine Anti-Ice System to add Engine Anti-Ice limitation when not in actual or anticipated icing conditions per AD 2023-15-05.

The following changes comprise this revision:

Chapter - Front Matter

Revision Approval (Revised)

This section uniquely identifies the approval authority and the reference number for this revision.

Log of Sections (Revised)

Revised to reflect sections changed in this revision.

Chapter - Certificate Limitations

Engines - Anti-Ice System (Revised)

Revised Engine Anti-Ice System to add Engine Anti-Ice limitation when not in actual or anticipated icing conditions per AD 2023-15-05.

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Title **A191201**

APPENDIX L1B27E

PERFORMANCE DATA

OPERATION OF LEAP-1B27 THRUST RATING

MODEL 737-8 SERIES

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Revision Approval A232497

This Revision Approval may only be used in conjunction with a Log of Sections that refers to Revision Approval Number A232497 in the revision approval number column for the Revision Approval section.

Approved by:



Flight Analyst E-UM, ODA-300064-NM, for

10/10/2023

Manager, Flight Test & Human Factors Branch, AIR-710
Federal Aviation Administration

Approved Date

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Log of Sections A232497

D631A002-L1B27E Revision 14 Date: 10/10/2023

	Chapter/Section	Revision Approval Number	Approval Date
	Front Matter		
	Title	A191201	06/30/2021
*	Revision Approval	A232497	10/10/2023
*	Log of Sections	A232497	10/10/2023
	Preface	A163806	03/07/2017
	Introduction	A201491	07/10/2020
	Certificate Limitations		
	Certification Limitations Introduction	A163806	03/07/2017
*	Engines - Anti-Ice System	A163806 , A232395	03/07/2017 , 08/29/2023
	Engines - Cold Soak Start	A163806	03/07/2017
	Engines - EGT	A170428	01/10/2018
	Engines - Fuel System	A201521	08/17/2020
	Engines - Ignition	A163806	03/07/2017
	Engines - Limit Display Markings	A163806	03/07/2017
	Engines - Oil System	A170428	01/10/2018
	Engines - Operational Limits	A163806	03/07/2017
	Engines - RPM	A163806	03/07/2017
	Engines - Reverse Thrust	A163806	03/07/2017
	Engines - Thrust	A163806	03/07/2017
	Operational Data		
	Operational Data Introduction	A163806	03/07/2017
	Operation in Icing Conditions - Engine Anti-Ice System	A163806	03/07/2017
	Performance		
	Performance Introduction	A163806	03/07/2017
	Performance Software Serial Number (PSSN)	A191108	01/28/2021
	Airplane Flight Manual - Digital Performance Information (AFM-DPI)	A191108	01/28/2021
	Airplane Flight Manual - Digital Performance Information (AFM-DPI) Release	A163806	03/07/2017
	Airplane Flight Manual Performance Adjustment - Alternate EEC Mode	A170351	12/13/2017
	Alternate Performance Capability	A191201	06/30/2021
	Engine Power Setting	A170428	01/10/2018
	Environmental Envelope	A180043	01/24/2018
	Gradient Reduction for Banked Flight	A163806	03/07/2017

* = Revised Sections

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Log of Sections (Continued) A232497

Chapter/Section	Revision Approval Number	Approval Date
Inflight Restart Envelope	A163806	03/07/2017
Noise - Characteristics	A170387	10/01/2018
Noise - Configuration	A191108 , A191108	01/28/2021 , 01/28/2021
Non-Normal Configuration Landing Distances	A170245	07/11/2017
Performance Configuration	A180522	11/05/2018
Recommended Takeoff Stabilizer Setting	A191005	01/06/2021
Regulatory Compliance	A163806	03/07/2017

* = Revised Sections

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Preface A163806

This Airplane Flight Manual (AFM) appendix complies with the European Aviation Safety Agency (EASA) Certification Specifications (CS). This AFM appendix adds, amends, replaces, and/or supersedes information found in the U.S. Federal Aviation Administration (FAA) approved AFM.

This AFM appendix is approved by the EASA and signed by the FAA on behalf of the EASA.

Each revision to this AFM appendix includes a new Log of Sections listing the effective sections comprising this AFM appendix. Revisions to text material are indicated by vertical bars in the margins.

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Introduction A201491

This appendix provides the additional certificate limitations and performance data to operate the Boeing Model 737-8 airplane equipped with CFM LEAP-1B27 engines.

This performance data appendix is used in conjunction with the Certificate Limitations, Operating Procedures, Operational Data and Performance of the basic Airplane Flight Manual and other applicable appendices. The performance data is determined from the Airplane Flight Manual - Digital Performance Information (AFM-DPI) software identified by the Performance Software Serial Number(s) (PSSN).

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Certification Limitations Introduction A163806

The Certificate Limitations in the basic Airplane Flight Manual are applicable except as amended herein.

Engines - Anti-Ice System A163806

The engine anti-ice system must be ON during all ground and flight operations when Icing Conditions exist or are anticipated, except during climb and cruise when the temperature is below -40°C SAT. Engine anti-ice must be on prior to and during descent in all Icing Conditions, including temperatures below -40°C SAT.

Note: Icing Conditions exist when the OAT on the ground and for takeoff, or TAT inflight is 10°C or below, and visible moisture in any form is present (such as clouds, fog with visibility of one mile or less, rain, snow, sleet and ice crystals).

Icing conditions also exist when the OAT on the ground and for takeoff is 10°C or below when operating on ramps, taxiways or runways where surface snow, ice, standing water, or slush may be ingested by the engines or freeze on engines or nacelles.

Do not rely on airframe visual icing cues or illumination of the ICING light (if installed) to turn engine anti-ice on. Delaying the use of engine anti-ice until ice build up is visible from the cockpit may result in severe engine damage.

Engines - Anti-Ice System A232395

(Required by AD 2023-15-05)

Engines - Anti-Ice System

Do not operate Engine Anti-Ice (EAI) in-flight when not in actual or anticipated icing conditions.

CAUTION: Operation of EAI when not in actual or anticipated icing conditions may result in severe engine inlet damage or failure.

Engines - Cold Soak Start A163806

Minimum oil starting temperature is -40°C.

Tank fuel temperature prior to engine start must not be less than -43°C (-45°F) or 3°C (5°F) above the fuel freezing point temperature, whichever is higher.

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Engines - EGT A170428

Operating Condition	Temperature Limit	Time Limit
Takeoff Transient	1048°C	30 Seconds
Takeoff	1038°C	5 Minutes**
Maximum Continuous	1013°C	No Limit
Ground Starting	753°C	No Limit
Air Starting	883°C	No Limit
High Power Fuel Cut	981°C	-
Quick Windmill Relight	920°C	-

** 10 minutes allowed in the event of the loss of thrust on one engine during takeoff for airplanes authorized to use Ten Minute Takeoff Thrust performance data under the Authorized Alternate Performance in AFM-DPI Section of the basic AFM.

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Engines - Fuel System A201521

Kerosene jet fuels conforming to the Boeing D6-85140-101 document, revision C or later FAA approved revision, “Aviation Fuel and Fuel Additives Properties, Composition and Performance Requirements”, are authorized for unlimited use with this airplane provided the limitations and requirements specified in the AFM are met. Kerosene jet fuels produced to other specifications and having properties meeting or exceeding the minimum requirements defined in the D6-85140-101 document, revision C or later FAA approved revision, are acceptable for use. The engines will operate satisfactorily with any of the approved fuels or any mixture thereof. Examples of kerosene jet fuel specifications that have been shown to meet the fuel minimum performance and specification requirements as described in the D6-85140-101 document, revision C or later FAA approved revision, include the following:

- Jet A, Jet A-1 as specified in ASTM D1655
- Jet A-1 as specified in UK MoD Def-Stan 91-091
- JP-5 as specified in MIL-DTL-5624
- JP-8 as specified in MIL-DTL-83133

The above list is not exhaustive: other fuel specification/ designation (e.g. GOST 10227 [TS-1], GB 6537 [Chinese No. 3 Jet Fuel], etc.) may be used provided the D6-85140-101, revision C or later FAA approved revision, requirements are met.

Fuel specifications are often changed and updated. It is the responsibility of the operator to ensure the fuel and any additive that are put in the fuel meet the requirements specified in the D6-85140-101 document, revision C or later FAA approved revision, and the AFM.

The fuel additives must meet the requirements listed in the Boeing D6-85140-101 document, revision C or later FAA approved revision, and the requirements of this section. A list of tolerated “incidental materials” and respective maximum concentrations allowed is also provided in the same Boeing D6-85140-101 document, revision C or later FAA approved revision.

The use of any Wide Cut Fuel as defined in the D6-85140-101 document, revision C or later FAA approved revision, (e.g. Jet B as specified in ASTM D6615, JP-4 as specified in MIL-DTL-5624) is prohibited.

The maximum tank fuel temperature should not exceed 49°C (120°F).

Tank fuel temperature prior to takeoff and inflight must not be less than -43°C (-45°F) or 3°C (5°F) above the fuel freezing point temperature, whichever is higher. The use of Fuel System Icing Inhibitor additives does not change the minimum fuel tank temperature limit.

(Required by AD 2020-14-09)
Operation of the CFM LEAP-1B series engines with fuel containing Kathon FP 1.5 biocide is prohibited.

Engines - Ignition A163806

Engine Ignition must be on for takeoff, landing, operation in heavy rain and anti-ice operation.

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Engines - Limit Display Markings A163806

Maximum limits are marked by a red line.
Cautionary range is marked by an amber arc.
Normal operating range is marked by a white arc.
Minimum limits are marked by a red line.

Engines - Oil System A170428

Minimum oil pressure is 17.4 psid (120 kpad). If engine oil pressure is in the yellow band with takeoff thrust set, do not takeoff.
Minimum oil starting temperature is -40°C.
Maximum oil temperature limit for continuous operation is 140°C.
Maximum oil temperature is 155°C for 15 minutes.
After starting, oil temperature rise to at least 31°C must be indicated before setting takeoff thrust.

Engines - Operational Limits A163806

Engine Ground Wind

- For crosswinds greater than 43 knots, limit thrust to a setting normally used for taxi.
- Except when setting takeoff thrust on the runway, limit engine thrust to idle for winds greater than 58 knots.

Engines - RPM A163806

The maximum operational limits are:

N1 - Low Pressure Compressor Rotor 104.3%

N2 - High Pressure Compressor Rotor 117.5%

Engines - Reverse Thrust A163806

Use for ground operation only. Intentional selection of reverse thrust in flight is prohibited. Backing the airplane with the reverse thrust is prohibited. Do not attempt go-around after thrust reverser deployment on landing.

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Engines - Thrust A163806

Power setting values for maximum continuous thrust, go-around thrust and full rated takeoff thrust are provided by AFM-DPI for primary and alternate EEC modes. These power setting values are operational limits. Takeoff and go-around thrust ratings are limited to 5 minutes duration or 10 minutes in the event of loss of thrust on one engine.

Derated takeoff thrust: Power setting values for derated takeoff thrust are available from AFM-DPI for primary EEC mode. These power setting values are operational limits. Use of derated takeoff thrust is not allowed for alternate EEC mode operation. As a condition to the use of derated thrust, operators must perform periodic checks to ensure that the engines are capable of producing full takeoff thrust without exceeding any engine operating limits.

Reduced takeoff thrust: Operation at reduced takeoff thrust based on an assumed temperature higher than the actual ambient temperature is permissible if the airplane meets all applicable performance requirements at the planned takeoff weight and reduced thrust setting. The thrust reduction is referenced to the takeoff thrust, which in turn can be either full rated takeoff thrust or derated takeoff thrust. The amount of thrust reduction must not exceed 25 percent of the full-rated takeoff thrust or derated takeoff thrust. The power setting values for reduced takeoff thrust are available from AFM-DPI. At any time during the takeoff operation, the pilot may select takeoff thrust instead of reduced takeoff thrust. Use of reduced thrust procedures is allowed on a wet runway if suitable performance accountability is made for the increased stopping distance on the wet surface. Reduced thrust is not allowed when the runway is contaminated with standing water, ice, slush, or snow. Reduced thrust is not allowed for alternate EEC mode operation. When allowed, use of reduced thrust is at the pilot's discretion. As a condition to the use of reduced thrust procedures, operators must perform periodic checks to ensure that the engines are capable of producing full takeoff thrust without exceeding any engine operating limits.

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